Mathematics

Class 8 (English Medium)

Solved Exercises — Sindh Textbook Board

12 lessons — 1509 questions

Book Structure — At a Glance

1. Operations on Sets

2. Real Numbers

3. Number Systems

4. Financial Arithmetic

5. Polynomials

6. Factorization and Simultaneous Equations

7. Fundamentals of Geometry

8. Practical Geometry

9. Areas and Volumes

10. Demonstrative Geometry

11. Trigonometry

12. Information Handling

1. Operations on Sets

Question: Review Exercise I, Question 1: Write the following sets in tabular form and set-builder form. (i) A = the set of the first five natural numbers

Answer: Tabular form: A = {1, 2, 3, 4, 5};

set-builder form: A = {x | x ∈ N ∧ x ≤ 5}.

Question: Review Exercise I, Question 1: (ii) B = the set of integers between -2 and 3

Answer: Tabular form: B = {-1, 0, 1, 2};

set-builder form: B = {x | x ∈ Z ∧ -2 < x < 3}.

Question: Review Exercise I, Question 1: (iii) C = the set of prime numbers between 10 and 30

Answer: Tabular form: C = {11, 13, 17, 19, 23, 29};

set-builder form: C = {x | x ∈ P ∧ 10 < x < 30}.

Question: Review Exercise I, Question 2: Which of the following sets are finite and which are infinite? (i) A = {1, 3, 4, 5}

Answer: It is a finite set, because its members can be counted (4 members).

Question: Review Exercise I, Question 2: (ii) B = {1, 3, 5, 7, ...}

Answer: It is an infinite set, because the odd numbers never end.

Question: Review Exercise I, Question 2: (iii) C = the set of even numbers between 10 and 30

Answer: It is a finite set: C = {12, 14, 16, 18, 20, 22, 24, 26, 28}, i.e. 9 members.

Question: Review Exercise I, Question 2: (iv) D = the set of all multiples of 10

Answer: It is an infinite set, because the multiples of 10 never end: D = {10, 20, 30, ...}.

Question: Review Exercise I, Question 2: (v) E = {x | x ∈ Z ∧ x < 0}

Answer: It is an infinite set, because the negative integers never end: E = {..., -3, -2, -1}.

Question: Review Exercise I, Question 3: Explain the following sets with examples. (i) Singleton set

Answer: A singleton set is a set with only one member.

Examples: {5}, {a}.

Question: Review Exercise I, Question 3: (ii) Equivalent set

Answer: Equivalent sets are sets with an equal number of members.

Example: {1, 2, 3} and {a, b, c} are equivalent sets, because both have 3 members.

Question: Review Exercise I, Question 3: (iii) Cardinal of a set

Answer: The cardinal of a set is the number of its members.

Example: n({2, 4, 6}) = 3.

Question: Review Exercise I, Question 4: Prove. (i) A ∪ A' = U

Answer: A' contains all members of U that are not in A, so joining A and A' gives all members of U, i.e. A ∪ A' = U.

Question: Review Exercise I, Question 4: (ii) A ∩ A' = ∅

Answer: No member can be in A and outside A at the same time, so A and A' have no common member, i.e. A ∩ A' = ∅.

Question: Review Exercise I, Question 4: (iii) (A')' = A

Answer: The members outside A' are again the members of A, so the complement of A' is (A')' = A.

Question: Exercise 1.1, Question 1: Decide which of the following sets is a subset of the other set. (i) A = {5, 6, 7} and B = {1, 2, 3, ..., 10}

Answer: Yes, A ⊆ B, because every member of A (5, 6, 7) is in B.

Question: Exercise 1.1, Question 1: (ii) C = {2} and D = {2, 3, 5}

Answer: Yes, C ⊆ D, because the only member of C, 2, is in D.

Question: Exercise 1.1, Question 1: (iii) T = {1, 2, 3, ..., 8} and S = {2, 4, 6}

Answer: No, T ⊆ S is false (T has members not in S);

but S ⊆ T is true, because every member of S is in T.

Question: Exercise 1.1, Question 2: Write two proper subsets of each of the

following sets. (i) A = {a, e, i, o, u}

Answer: Two proper subsets: {a} and {a, e}.

Question: Exercise 1.1, Question 2: (ii) B = {x, y}

Answer: Two proper subsets: {x} and ∅.

Question: Exercise 1.1, Question 3: Write one proper and one improper subset of each of the following sets. (i) X = {2, 4}

Answer: Proper subset: {2};

improper subset: {2, 4}.

Question: Exercise 1.1, Question 3: (ii) Y = the set of prime numbers less than 4

Answer: Y = {2, 3};

proper subset: {2};

improper subset: {2, 3}.

Question: Exercise 1.1, Question 4: Write all possible subsets of the following sets. (i) F = {1, 2, 3}

Answer: ∅, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3} — 8 subsets in all.

Question: Exercise 1.1, Question 4: (ii) A = the set of even prime numbers

Answer: A = {2} (2 is the only even prime number);

subsets: ∅, {2}.

Question: Exercise 1.1, Question 4: (iii) The set of even numbers less than 7

Answer: The set = {2, 4, 6};

subsets: ∅, {2}, {4}, {6}, {2, 4}, {2, 6}, {4, 6}, {2, 4, 6} — 8 in all.

Question: Exercise 1.1, Question 4: (iv) D = {x | x ∈ E^+ ∧ x ≤ 6}

Answer: D = {2, 4, 6};

subsets: ∅, {2}, {4}, {6}, {2, 4}, {2, 6}, {4, 6}, {2, 4, 6} — 8 in all.

Question: Exercise 1.1, Question 4: (v) C = {x | x ∈ Z ∧ -2 < x < 2}

Answer: C = {-1, 0, 1};

subsets: ∅, {-1}, {0}, {1}, {-1, 0}, {-1, 1}, {0, 1}, {-1, 0, 1} — 8 in all.

Question: Exercise 1.1, Question 5: Find the power sets of the following. (i) A = {1, 3, 5}

Answer: P(A) = {∅, {1}, {3}, {5}, {1, 3}, {1, 5}, {3, 5}, {1, 3, 5}} — 8 members in all.

Question: Exercise 1.1, Question 5: (ii) B = {a, b, c, d}

Answer: P(B) = {∅, {a}, {b}, {c}, {d}, {a, b}, {a, c}, {a, d}, {b, c}, {b, d}, {c, d}, {a, b, c}, {a, b, d}, {a, c, d}, {b, c, d}, {a, b, c, d}} — 16 members in all.

Question: Exercise 1.1, Question 5: (iii) C = ∅

Answer: P(C) = {∅} — only one member.

Question: Exercise 1.1, Question 6: Write a set that has exactly one proper subset.

Answer: {a} — it has exactly one proper subset: ∅.

Question: Exercise 1.1, Question 7: Write a set that has no proper subset.

Answer: ∅ — the empty set has no proper subset.

Question: Exercise 1.1, Question 8: Write the improper subsets of the following. (i) {x | x ∈ O ∧ x ≤ 7}

Answer: The set = {1, 3, 5, 7};

its improper subset is itself: {1, 3, 5, 7}.

Question: Exercise 1.1, Question 8: (ii) {x | x ∈ P ∧ x ≤ 11}

Answer: The set = {2, 3, 5, 7, 11};

its improper subset is itself: {2, 3, 5, 7, 11}.

Question: Exercise 1.1, Question 9: Given X = {20, 30, 40, 50, 60, 80, 100}, find the number of members in each of the following subsets of X. (i) The set of numbers exactly divisible by 2

Answer: All numbers in X are divisible by 2: {20, 30, 40, 50, 60, 80, 100} — 7 members.

Question: Exercise 1.1, Question 9: (ii) The set of numbers exactly divisible

by 8

Answer: {40, 80} — 2 members.

Question: Exercise 1.1, Question 9: (iii) The set of numbers exactly divisible by 15

Answer: {30, 60} — 2 members.

Question: Exercise 1.1, Question 9: (iv) The set of numbers exactly divisible by 25

Answer: {50, 100} — 2 members.

Question: Exercise 1.1, Question 10: Decide which of the following sets is a subset of the other given set. (i) N ⊆ W

Answer: Yes, it is true: every natural number (1, 2, 3, ...) is also in the set of whole numbers W.

Question: Exercise 1.1, Question 10: (ii) Z ⊆ N

Answer: No, it is false: negative integers (like -1) are not in N.

Question: Exercise 1.1, Question 10: (iii) P ⊆ N

Answer: Yes, it is true: every prime number is a natural number.

Question: Exercise 1.1, Question 10: (iv) E ⊆ Z

Answer: Yes, it is true: every even number is an integer.

Question: Exercise 1.1, Question 10: (v) Q ⊆ O

Answer: No, it is false: rational numbers (like 1/2) are not odd numbers.

Question: Exercise 1.1, Question 10: (vi) Q ⊆ E

Answer: No, it is false: rational numbers (like 1/2) are not even numbers.

Question: Practical Work: Given U = {a, b, c, d, e, f}, A = {a, b, c, d} and B = {c, d, e, f}, prove De Morgan's laws. (i) (A ∪ B)' = A' ∩ B'

Answer: A ∪ B = {a, b, c, d, e, f} = U, so (A ∪ B)' = ∅.

A' = {e, f}, B' = {a, b}, so A' ∩ B' = ∅.

Both sides are equal, so the law is proved.

Question: Practical Work: (ii) (A ∩ B)' = A' ∪ B'

Answer: A ∩ B = {c, d}, so (A ∩ B)' = {a, b, e, f}.

A' ∪ B' = {e, f} ∪ {a, b} = {a, b, e, f}.

Both sides are equal, so the law is proved.

Question: Exercise 1.2, Question 1: Given A = {1, 2, 3, 4, 5}, B = {1, 3, 5, 7} and C = {2, 4, 6, 8}, verify the following. (i) The commutative law of union of sets

Answer: Left side: A ∪ B = {1, 2, 3, 4, 5, 7};

right side: B ∪ A = {1, 3, 5, 7, 2, 4} = {1, 2, 3, 4, 5, 7}.

Both sides are equal, so A ∪ B = B ∪ A is verified.

Question: Exercise 1.2, Question 1: (ii) The associative law of union of sets

Answer: Left side: A ∪ (B ∪ C) = {1, 2, 3, 4, 5} ∪ {1, 2, 3, 4, 5, 6, 7, 8} = {1, 2, 3, 4, 5, 6, 7, 8};

right side: (A ∪ B) ∪ C = {1, 2, 3, 4, 5, 7} ∪ {2, 4, 6, 8} = {1, 2, 3, 4, 5, 6, 7, 8}.

Both sides are equal.

Question: Exercise 1.2, Question 1: (iii) The distributive law of union over intersection of sets

Answer: Left side: A ∪ (B ∩ C) = {1, 2, 3, 4, 5} ∪ ∅ = {1, 2, 3, 4, 5};

right side: (A ∪ B) ∩ (A ∪ C) = {1, 2, 3, 4, 5, 7} ∩ {1, 2, 3, 4, 5, 6, 8} = {1, 2, 3, 4, 5}.

Both sides are equal.

Question: Exercise 1.2, Question 1: (iv) The distributive law of intersection over union of sets

Answer: Left side: A ∩ (B ∪ C) = {1, 2, 3, 4, 5} ∩ {1, 2, 3, 4, 5, 6, 7, 8} = {1, 2, 3, 4, 5};

right side: (A ∩ B) ∪ (A ∩ C) = {1, 3, 5} ∪ {2, 4} = {1, 2, 3, 4, 5}.

Both sides are equal.

Question: Exercise 1.2, Question 2: Given P = {a, b, c, d, e}, Q = {a, e, i, o, u} and R = {c, d, e, i}, verify the following. (i) P ∪ Q = Q ∪ P

Answer: P ∪ Q = {a, b, c, d, e, i, o, u} = Q ∪ P.

Both sides are equal.

Question: Exercise 1.2, Question 2: (ii) Q ∩ P = P ∩ Q

Answer: Q ∩ P = {a, e} = P ∩ Q.

Both sides are equal.

Question: Exercise 1.2, Question 2: (iii) P ∩ (Q ∩ R) = (P ∩ Q) ∩ R

Answer: Left side: P ∩ {e, i} = {e};

right side: {a, e} ∩ R = {e}.

Both sides are equal.

Question: Exercise 1.2, Question 2: (iv) P ∪ (Q ∩ R) = (P ∪ Q) ∩ (P ∪ R)

Answer: Left side: P ∪ {e, i} = {a, b, c, d, e, i};

right side: {a, b, c, d, e, i, o, u} ∩ {a, b, c, d, e, i} = {a, b, c, d, e, i}.

Both sides are equal.

Question: Exercise 1.2, Question 3: Given U = {1, 2, 3, ..., 10}, A = {1, 2, 3, 4, 5} and B = {2, 4, 6, 8, 10}, verify De Morgan's laws. (i) (A ∪ B)' = A' ∩ B'

Answer: A ∪ B = {1, 2, 3, 4, 5, 6, 8, 10}, so (A ∪ B)' = {7, 9}.

A' = {6, 7, 8, 9, 10}, B' = {1, 3, 5, 7, 9}, so A' ∩ B' = {7, 9}.

Both sides are equal.

Question: Exercise 1.2, Question 3: (ii) (A ∩ B)' = A' ∪ B'

Answer: A ∩ B = {2, 4}, so (A ∩ B)' = {1, 3, 5, 6, 7, 8, 9, 10}.

A' ∪ B' = {6, 7, 8, 9, 10} ∪ {1, 3, 5, 7, 9} = {1, 3, 5, 6, 7, 8, 9, 10}.

Both sides are equal.

Question: Exercise 1.2, Question 4: Given S = {x | x ∈ O ∧ x ≤ 11}, Q = {x | x ∈ P ∧ 3 ≤ x ≤ 11} and R = {x | x ∈ E ∧ x ≤ 10}, verify the distributive laws of union and intersection of sets. (i) S ∪ (Q ∩ R) = (S ∪ Q) ∩ (S ∪ R)

Answer: S = {1, 3, 5, 7, 9, 11}, Q = {3, 5, 7, 11}, R = {0, 2, 4, 6, 8, 10}.

Q ∩ R = ∅, so the left side = S ∪ ∅ = {1, 3, 5, 7, 9, 11}.

Right side: (S ∪ Q) ∩ (S ∪ R) = {1, 3, 5, 7, 9, 11} ∩ {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11} = {1, 3, 5, 7, 9, 11}.

Both sides are equal.

Question: Exercise 1.2, Question 4: (ii) S ∩ (Q ∪ R) = (S ∩ Q) ∪ (S ∩ R)

Answer: Q ∪ R = {0, 2, 3, 4, 5, 6, 7, 8, 10, 11}, so the left side: S ∩ (Q ∪ R) = {3, 5, 7, 11}.

Right side: (S ∩ Q) ∪ (S ∩ R) = {3, 5, 7, 11} ∪ ∅ = {3, 5, 7, 11}.

Both sides are equal.

Question: Exercise 1.2, Question 5: Given U = {x | x ∈ N ∧ x ≤ 11}, A = {x | x ∈ E ∧ 2 < x < 10} and B = {x | x ∈ P ∧ 3 < x < 11}, verify De Morgan's laws of sets. (i) (A ∪ B)' = A' ∩ B'

Answer: A = {4, 6, 8}, B = {5, 7}.

A ∪ B = {4, 5, 6, 7, 8}, so (A ∪ B)' = {1, 2, 3, 9, 10, 11}.

A' = {1, 2, 3, 5, 7, 9, 10, 11}, B' = {1, 2, 3, 4, 6, 8, 9, 10, 11}, so A' ∩ B' = {1, 2, 3, 9, 10, 11}.

Both sides are equal.

Question: Exercise 1.2, Question 5: (ii) (A ∩ B)' = A' ∪ B'

Answer: A ∩ B = ∅, so (A ∩ B)' = U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11}.

A' ∪ B' = {1, 2, 3, 5, 7, 9, 10, 11} ∪ {1, 2, 3, 4, 6, 8, 9, 10, 11} = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11}.

Both sides are equal.

Question: Exercise 1.3, Question 1: Given A = {5, 6, 7}, B = {6, 7, 8} and C = {7, 8, 9}, show the sets in a Venn diagram. (i) A ∪ (B ∪ C)

Answer: B ∪ C = {6, 7, 8, 9}, so A ∪ (B ∪ C) = {5, 6, 7, 8, 9}.

In the Venn diagram the combined shaded part of all three circles shows this set.

Question: Exercise 1.3, Question 1: (ii) A ∩ (B ∩ C)

Answer: B ∩ C = {7, 8}, so A ∩ (B ∩ C) = {7}.

In the Venn diagram the common part of all three circles (the middle part) shows this set.

Question: Exercise 1.3, Question 2: Given P = the set of natural numbers less than 15, Q = the set of prime numbers less than 15 and R = the set of odd numbers less than 15, show the following in a Venn diagram. (i) P ∪ (Q ∪ R)

Answer: P = {1, ..., 14}, Q = {2, 3, 5, 7, 11, 13}, R = {1, 3, 5, 7, 9, 11, 13}.

Q ∪ R = {1, 2, 3, 5, 7, 9, 11, 13}, so P ∪ (Q ∪ R) = {1, ..., 14} = P.

In the Venn diagram the whole P circle will be shaded.

Question: Exercise 1.3, Question 2: (ii) P ∩ (Q ∩ R)

Answer: Q ∩ R = {3, 5, 7, 11, 13}, so P ∩ (Q ∩ R) = {3, 5, 7, 11, 13}.

In the Venn diagram the part of the common portion of Q and R that also lies in P will be shaded.

Question: Exercise 1.3, Question 2: (iii) P ∪ (Q ∩ R) = (P ∪ Q) ∩ (P ∪ R)

Answer: Left side: P ∪ {3, 5, 7, 11, 13} = {1, ..., 14} = P.

Right side: P ∩ P = P (because Q and R are both subsets of P, so P ∪ Q = P and P ∪ R = P).

Both sides are equal.

Question: Exercise 1.3, Question 3: Given A = {a, b, c}, B = {b, c, d} and C = {c, d, e}, verify the sets by means of a Venn diagram. (i) A ∩ (B ∪ C) = (A ∩ B) ∪ (A ∩ C)

Answer: Left side: B ∪ C = {b, c, d, e}, so A ∩ (B ∪ C) = {b, c}.

Right side: (A ∩ B) ∪ (A ∩ C) = {b, c} ∪ {c} = {b, c}.

Both sides are equal.

Question: Exercise 1.3, Question 3: (ii) A ∩ (B ∩ C) = (A ∩ B) ∩ C

Answer: Left side: B ∩ C = {c, d}, so A ∩ (B ∩ C) = {c}.

Right side: (A ∩ B) ∩ C = {b, c} ∩ {c, d, e} = {c}.

Both sides are equal.

Question: Exercise 1.3, Question 4: Given D = {x | x ∈ N ∧ x ≤ 10}, E = {x | x ∈ W ∧ x ≤ 8} and F = {x | x ∈ P ∧ x ≤ 11}, prove the following by means of a Venn diagram. (i) D ∪ (E ∩ F) = (D ∪ E) ∩ (D ∪ F)

Answer: D = {1, ..., 10}, E = {0, ..., 8}, F = {2, 3, 5, 7, 11}.

E ∩ F = {2, 3, 5, 7}, so the left side: D ∪ {2, 3, 5, 7} = {1, ..., 10}.

Right side: (D ∪ E) ∩ (D ∪ F) = {0, 1, ..., 10} ∩ {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11} = {1, ..., 10}.

Both sides are equal.

Question: Exercise 1.3, Question 4: (ii) D ∩ (E ∪ F) = (D ∩ E) ∪ (D ∩ F)

Answer: E ∪ F = {0, 1, 2, 3, 4, 5, 6, 7, 8, 11}, so the left side: D ∩ (E ∪ F) = {1, 2, 3, 4, 5, 6, 7, 8}.

Right side: (D ∩ E) ∪ (D ∩ F) = {1, ..., 8} ∪ {2, 3, 5, 7} = {1, 2, 3, 4, 5, 6, 7, 8}.

Both sides are equal.

Question: Review Exercise 1, Question 1: Circle the correct answer. (i) Which of the following sets is a subset of W?

Answer: (b) N — because every natural number is also a whole number.

Question: Review Exercise 1, Question 1: (ii) The total number of subsets of A = {a, e, i, o, u} is:

Answer: (c) 32 — because A has 5 members and $2^5 = 32$.

Question: Review Exercise 1, Question 1: (iii) For any three sets A, B and C, A ∪ (B ∩ C) is equal to:

Answer: (c) (A ∪ B) ∩ (A ∪ C) — the distributive law of union over intersection.

Question: Review Exercise 1, Question 1: (iv) In a Venn diagram the

universal set is usually shown as:

Answer: (b) a rectangle.

Question: Review Exercise 1, Question 2: Given A = {x, y, z}, find P(A).

Answer: P(A) = {∅, {x}, {y}, {z}, {x, y}, {x, z}, {y, z}, {x, y, z}} — 8 members in all.

Question: Review Exercise 1, Question 3: Given A = {1, 2, 3}, B = {2, 4, 6} and C = {x | x ∈ W ∧ x ≤ 4}, verify the following. (i) (A ∩ B) ∩ C = A ∩ (B ∩ C)

Answer: Left side: (A ∩ B) ∩ C = {2} ∩ {0, 1, 2, 3, 4} = {2};

right side: A ∩ (B ∩ C) = {1, 2, 3} ∩ {2, 4} = {2}.

Both sides are equal.

Question: Review Exercise 1, Question 3: (ii) (A ∪ B) ∪ C = A ∪ (B ∪ C)

Answer: Left side: {1, 2, 3, 4, 6} ∪ {0, 1, 2, 3, 4} = {0, 1, 2, 3, 4, 6};

right side: {1, 2, 3} ∪ {0, 1, 2, 3, 4, 6} = {0, 1, 2, 3, 4, 6}.

Both sides are equal.

Question: Review Exercise 1, Question 3: (iii) A ∩ (B ∪ C) = (A ∩ B) ∪ (A ∩ C)

Answer: Left side: A ∩ {0, 1, 2, 3, 4, 6} = {1, 2, 3};

right side: {2} ∪ {1, 2, 3} = {1, 2, 3}.

Both sides are equal.

Question: Review Exercise 1, Question 3: (iv) (A ∩ B) ∪ C = (A ∪ C) ∩ (B ∪ C)

Answer: Left side: {2} ∪ {0, 1, 2, 3, 4} = {0, 1, 2, 3, 4};

right side: {0, 1, 2, 3, 4} ∩ {0, 1, 2, 3, 4, 6} = {0, 1, 2, 3, 4}.

Both sides are equal.

Question: Review Exercise 1, Question 3: (v) (A ∪ B) ∩ C = (A ∩ C) ∪ (B ∩ C)

Answer: Left side: {1, 2, 3, 4, 6} ∩ {0, 1, 2, 3, 4} = {1, 2, 3, 4};

right side: {1, 2, 3} ∪ {2, 4} = {1, 2, 3, 4}.

Both sides are equal.

Question: Review Exercise 1, Question 3: (vi) A ∪ (B ∩ C) = (A ∪ B) ∩ (A ∪ C)

Answer: Left side: {1, 2, 3} ∪ {2, 4} = {1, 2, 3, 4};

right side: {1, 2, 3, 4, 6} ∩ {0, 1, 2, 3, 4} = {1, 2, 3, 4}.

Both sides are equal.

Question: Review Exercise 1, Question 4: Given U = {1, 2, 3, ..., 10}, A = {1, 3, 5, 7, 9} and B = {2, 4, 6, 8, 10}, verify De Morgan's law of sets. (i)

(A ∪ B)' = A' ∩ B'

Answer: A ∪ B = {1, ..., 10} = U, so (A ∪ B)' = ∅.

A' = {2, 4, 6, 8, 10}, B' = {1, 3, 5, 7, 9}, so A' ∩ B' = ∅.

Both sides are equal.

Question: Review Exercise 1, Question 4: (ii) (A ∩ B)' = A' ∪ B'

Answer: A ∩ B = ∅, so (A ∩ B)' = U.

A' ∪ B' = {2, 4, 6, 8, 10} ∪ {1, 3, 5, 7, 9} = {1, ..., 10} = U.

Both sides are equal.

Question: Review Exercise 1, Question 5: Circle the correct answer. (i) A ∩ B is: A = {2, 4, 6, 8}, B = {1, 3, 5, 7, 9}

Answer: (b) {} — because A and B have no common member.

Question: Review Exercise 1, Question 5: (ii) A ∪ B is: A = {0, 2}, B = {1, 3}

Answer: (d) {0, 1, 2, 3} — because the union contains all members of both sets.

Question: Review Exercise 1, Question 5: (iii) The improper subset of P is: P = {0, 2, 6}

Answer: (c) {0, 2, 6} — the improper subset is the set itself.

Question: Review Exercise 1, Question 6: Tell which of the following sets

can be power sets. For any that is a power set, find its original set. (i) ∅

Answer: No, it cannot be a power set: every power set has at least one member (∅), but ∅ has no member.

Question: Review Exercise 1, Question 6: (ii) {∅, x}

Answer: No, it cannot be a power set: a power set contains ∅ together with sets of members, but here x itself is not a set.

Question: Review Exercise 1, Question 6: (iii) {∅, {x}, {y}}

Answer: No, it cannot be a power set: it is missing {x, y}, which would have to be in the power set of the original set {x, y}.

Question: Review Exercise 1, Question 6: (iv) {∅, {x}, {y}, {x, y}}

Answer: Yes, it is a power set: the original set is {x, y}.

Question: Review Exercise 1, Question 6: (v) {∅, {x}, {y}, {z}}

Answer: No, it cannot be a power set: it is missing the two-member sets {x, y}, {x, z}, {y, z} and the three-member set {x, y, z}.

Question: Review Exercise 1, Question 6: (vi) {∅}

Answer: Yes, it is a power set: the original set is ∅, because P(∅) = {∅}.

2. Real Numbers

Question: Exercise 2.1, Part A: Separate the rational and irrational numbers from the following numbers. 1. 35

Answer: It is a rational number, because it can be written as 35/1.

Question: Exercise 2.1, Part A: 2. √36

Answer: It is a rational number, because √36 = 6.

Question: Exercise 2.1, Part A: 3. -2 1/40

Answer: It is a rational number, because it is a mixed fraction: -2 1/40 = -81/40.

Question: Exercise 2.1, Part A: 4. 2√4

Answer: It is a rational number, because 2√4 = 2×2 = 4.

Question: Exercise 2.1, Part A: 5. 6π

Answer: It is an irrational number, because π is irrational and the decimal expansion of 6π neither terminates nor repeats.

Question: Exercise 2.1, Part A: 6. 22/7

Answer: It is a rational number, because it is in p/q form.

Question: Exercise 2.1, Part A: 7. 2/√5

Answer: It is an irrational number, because √5 is irrational.

Question: Exercise 2.1, Part A: 8. -√1

Answer: It is a rational number, because -√1 = -1.

Question: Exercise 2.1, Part A: 9. π/5

Answer: It is an irrational number, because π is irrational.

Question: Exercise 2.1, Part A: 10. √2/π

Answer: It is an irrational number, because both √2 and π are irrational.

Question: Exercise 2.1, Part A: 11. √9/√25

Answer: It is a rational number, because √9/√25 = 3/5.

Question: Exercise 2.1, Part A: 12. √11

Answer: It is an irrational number, because 11 is not a perfect square and the decimal expansion of √11 neither terminates nor repeats.

Question: Exercise 2.1, Part B: Separate the terminating and repeating decimals from the following. 1. 1/7

Answer: It is a repeating decimal: 1/7 = 0.142857142857...

.

Question: Exercise 2.1, Part B: 2. 2/3

Answer: It is a repeating decimal: 2/3 = 0.666...

.

Question: Exercise 2.1, Part B: 3. 0

Answer: It is a terminating decimal: 0 = 0.0.

Question: Exercise 2.1, Part B: 4. 1/11

Answer: It is a repeating decimal: 1/11 = 0.090909...

.

Question: Exercise 2.1, Part B: 5. 4/3

Answer: It is a repeating decimal: 4/3 = 1.333...

.

Question: Exercise 2.1, Part B: 6. -1/5

Answer: It is a terminating decimal: -1/5 = -0.2.

Question: Exercise 2.1, Part B: 7. -5/6

Answer: It is a repeating decimal: -5/6 = -0.8333...

.

Question: Exercise 2.1, Part B: 8. 2256

Answer: It is a terminating decimal: 2256 = 2256.0.

Question: Exercise 2.1, Part B: 9. 9/10

Answer: It is a terminating decimal: 9/10 = 0.9.

Question: Exercise 2.1, Part B: 10. 25/26

Answer: It is a repeating decimal: 25/26 = 0.961538461538...

.

Question: Exercise 2.1, Part B: 11. 243/100

Answer: It is a terminating decimal: 243/100 = 2.43.

Question: Exercise 2.1, Part B: 12. 18750/1000

Answer: It is a terminating decimal: 18750/1000 = 18.75.

Question: Exercise 2.2, Part A: Find the squares of the following. 1. 5

Answer: $5^2 = 5 \times 5 = 25$.

Question: Exercise 2.2, Part A: 2. 13

Answer: $13^2 = 13 \times 13 = 169$.

Question: Exercise 2.2, Part A: 3. 19

Answer: $19^2 = 19 \times 19 = 361$.

Question: Exercise 2.2, Part A: 4. 39

Answer: $39^2 = 39 \times 39 = 1521$.

Question: Exercise 2.2, Part A: 5. 45

Answer: $45^2 = 45 \times 45 = 2025$.

Question: Exercise 2.2, Part A: 6. 58

Answer: $58^2 = 58 \times 58 = 3364$.

Question: Exercise 2.2, Part A: 7. 63

Answer: $63^2 = 63 \times 63 = 3969$.

Question: Exercise 2.2, Part A: 8. 79

Answer: $79^2 = 79 \times 79 = 6241$.

Question: Exercise 2.2, Part A: 9. 97

Answer: $97^2 = 97 \times 97 = 9409$.

Question: Exercise 2.2, Part A: 10. 108

Answer: $108^2 = 108 \times 108 = 11664$.

Question: Exercise 2.2, Part B: Find all perfect squares lying between each of the following pairs of numbers. 1. 10 and 30

Answer: 16 and 25 ($4^2 = 16$, $5^2 = 25$).

Question: Exercise 2.2, Part B: 2. 20 and 50

Answer: 25, 36 and 49 ($5^2 = 25$, $6^2 = 36$, $7^2 = 49$).

Question: Exercise 2.2, Part B: 3. 40 and 70

Answer: 49 and 64 ($7^2 = 49$, $8^2 = 64$).

Question: Exercise 2.2, Part B: 4. 70 and 90

Answer: 81 ($9^2 = 81$).

Question: Exercise 2.2, Part C: Write the following squares as the sum of odd numbers. 1. 7^2

Answer: $7^2 = 1 + 3 + 5 + 7 + 9 + 11 + 13 = 49$.

Question: Exercise 2.2, Part C: 2. 9^2

Answer: $9^2 = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 = 81$.

Question: Exercise 2.2, Part C: 3. 15^2

Answer: $15^2 = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 + 25 + 27 + 29 = 225$.

Question: Exercise 2.2, Part C: 4. 11^2

Answer: $11^2 = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 = 121$.

Question: Exercise 2.2, Part C: 5. 13^2

Answer: $13^2 = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 + 25 = 169$.

Question: Exercise 2.2, Part C: 6. 8^2

Answer: $8^2 = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 = 64$.

Question: Exercise 2.2, Part C: 7. 12^2

Answer: $12^2 = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 = 144$.

Question: Exercise 2.2, Part C: 8. 16^2

Answer: $16^2 = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 + 25 + 27 + 29 + 31 = 256$.

Question: Exercise 2.2, Part C: 9. 20^2

Answer: $20^2 = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 + 25 + 27 + 29 + 31 + 33 + 35 + 37 + 39 = 400$.

Question: Exercise 2.2, Part C: 10. 27^2

Answer: 27^2 = the sum of the first 27 odd numbers = 729.

Question: Exercise 2.3, Part A: Find the square roots of the following by prime factors. 1. 961

Answer: 961 = 31 × 31, so √961 = 31.

Question: Exercise 2.3, Part A: 2. 1296

Answer: 1296 = $2^4 \times 3^4$, so √1296 = $2^2 \times 3^2$ = 36.

Question: Exercise 2.3, Part A: 3. 344569

Answer: 344569 = 587 × 587, so √344569 = 587.

Question: Exercise 2.3, Part A: 4. 817216

Answer: 817216 = 904 × 904, so √817216 = 904.

Question: Exercise 2.3, Part A: 5. 81/121

Answer: √(81/121) = √81/√121 = 9/11.

Question: Exercise 2.3, Part A: 6. 5929/9604

Answer: √(5929/9604) = √5929/√9604 = 77/98 = 11/14.

Question: Exercise 2.3, Part A: 7. 1764/7744

Answer: √(1764/7744) = √1764/√7744 = 42/88 = 21/44.

Question: Exercise 2.3, Part A: 8. 1 984/14641

Answer: 1 984/14641 = 15625/14641, so √(15625/14641) = 125/121 = 1 4/121.

Question: Exercise 2.3, Part A: 9. 249.64

Answer: 249.64 = 24964/100, so √249.64 = 158/10 = 15.8.

Question: Exercise 2.3, Part A: 10. 0.4096

Answer: 0.4096 = 4096/10000, so √0.4096 = 64/100 = 0.64.

Question: Exercise 2.3, Part A: 11. 2981.16

Answer: 2981.16 = 298116/100, so √2981.16 = 546/10 = 54.6.

Question: Exercise 2.3, Part A: 12. 131.1025

Answer: 131.1025 = 1311025/10000, so √131.1025 = 1145/100 = 11.45.

Question: Exercise 2.3, Part B: Find the square roots of the following by the division method. 1. 1024

Answer: By the division method: √1024 = 32.

Question: Exercise 2.3, Part B: 2. 14161

Answer: By the division method: √14161 = 119.

Question: Exercise 2.3, Part B: 3. 996004

Answer: By the division method: √996004 = 998.

Question: Exercise 2.3, Part B: 4. 10329796

Answer: By the division method: √10329796 = 3214.

Question: Exercise 2.3, Part B: 5. 169/289

Answer: √(169/289) = 13/17.

Question: Exercise 2.3, Part B: 6. 1225/2809

Answer: √(1225/2809) = 35/53.

Question: Exercise 2.3, Part B: 7. 29241/55696

Answer: √(29241/55696) = 171/236.

Question: Exercise 2.3, Part B: 8. 1 1089/1936

Answer: 1 1089/1936 = 3025/1936, so the root = 55/44 = 1 1/4.

Question: Exercise 2.3, Part B: 9. 648.7209

Answer: √648.7209 = 25.47 (because $25.47^2 = 648.7209$).

Question: Exercise 2.3, Part B: 10. 180.9025

Answer: √180.9025 = 13.45 (because $13.45^2 = 180.9025$).

Question: Exercise 2.3, Part B: 11. 727.9204

Answer: √727.9204 = 26.98 (because $26.98^2 = 727.9204$).

Question: Exercise 2.3, Part B: 12. 7613.609536

Answer: √7613.609536 = 87.256 (because $87.256^2 = 7613.609536$).

Question: Exercise 2.4, Part A: Find the square roots of the following to two decimal places. 1. 2

Answer: By the division method: √2 = 1.41 (to two decimal places).

Question: Exercise 2.4, Part A: 2. 5

Answer: √5 = 2.24 (to two decimal places).

Question: Exercise 2.4, Part A: 3. 13

Answer: √13 = 3.61 (to two decimal places).

Question: Exercise 2.4, Part A: 4. 141

Answer: √141 = 11.87 (to two decimal places).

Question: Exercise 2.4, Part A: 5. 180

Answer: √180 = 13.42 (to two decimal places).

Question: Exercise 2.4, Part A: 6. 2.5

Answer: √2.5 = 1.58 (to two decimal places).

Question: Exercise 2.4, Part A: 7. 152.7696

Answer: √152.7696 = 12.36 (to two decimal places).

Question: Exercise 2.4, Part A: 8. 0.0143

Answer: √0.0143 = 0.12 (to two decimal places).

Question: Exercise 2.4, Part A: 9. 23.085

Answer: √23.085 = 4.80 (to two decimal places).

Question: Exercise 2.4, Part A: 10. 125.08

Answer: √125.08 = 11.18 (to two decimal places).

Question: Exercise 2.4, Part B: Find the square roots of the following to three decimal places. 1. 6

Answer: By the division method: √6 = 2.449 (to three decimal places).

Question: Exercise 2.4, Part B: 2. 8

Answer: √8 = 2.828 (to three decimal places).

Question: Exercise 2.4, Part B: 3. 11

Answer: √11 = 3.317 (to three decimal places).

Question: Exercise 2.4, Part B: 4. 155

Answer: √155 = 12.450 (to three decimal places).

Question: Exercise 2.4, Part B: 5. 205

Answer: √205 = 14.318 (to three decimal places).

Question: Exercise 2.4, Part B: 6. 73

Answer: √73 = 8.544 (to three decimal places).

Question: Exercise 2.4, Part B: 7. 125.79

Answer: √125.79 = 11.216 (to three decimal places).

Question: Exercise 2.4, Part B: 8. 1468.9

Answer: √1468.9 = 38.326 (to three decimal places).

Question: Exercise 2.4, Part B: 9. 129048.27

Answer: √129048.27 = 359.233 (to three decimal places).

Question: Exercise 2.4, Part B: 10. 3355.30

Answer: √3355.30 = 57.925 (to three decimal places).

Question: Exercise 2.5: Find the number of digits in the square roots of the following perfect squares. 1. 8649

Answer: 8649 has 4 digits, so its square root has (4+1)/2 = 2 digits (√8649 = 93).

Question: Exercise 2.5: 2. 250000

Answer: 250000 has 6 digits, so its square root has 6/2 = 3 digits (√250000 = 500).

Question: Exercise 2.5: 3. 322624

Answer: 322624 has 6 digits, so its square root has 6/2 = 3 digits (√322624 = 568).

Question: Exercise 2.5: 4. 614656

Answer: 614656 has 6 digits, so its square root has 6/2 = 3 digits (√614656 = 784).

Question: Exercise 2.5: 5. 432964

Answer: 432964 has 6 digits, so its square root has 6/2 = 3 digits (√432964 = 658).

Question: Exercise 2.5: 6. 81018001

Answer: 81018001 has 8 digits, so its square root has 8/2 = 4 digits (√81018001 = 9001).

Question: Exercise 2.5: 7. 43388569

Answer: 43388569 has 8 digits, so its square root has 8/2 = 4 digits (√43388569 = 6587).

Question: Exercise 2.5: 8. 3474748809

Answer: 3474748809 has 10 digits, so its square root has 10/2 = 5 digits (√3474748809 = 58947).

Question: Exercise 2.5: 9. 73547776

Answer: 73547776 has 8 digits, so its square root has 8/2 = 4 digits (√73547776 = 8576).

Question: Exercise 2.5: 10. 33942276

Answer: 33942276 has 8 digits, so its square root has 8/2 = 4 digits (√33942276 = 5826).

Question: Exercise 2.5: 11. 7913169936

Answer: 7913169936 has 10 digits, so its square root has 10/2 = 5 digits (√7913169936 = 88956).

Question: Exercise 2.5: 12. 935566751

Answer: 935566751 has 9 digits, so its square root has (9+1)/2 = 5 digits.

[note: the printed number 935566751 is not a perfect square;

the nearest square is $30587^2 = 935564569$ — this looks like a misprint in the book.]

Question: Exercise 2.6, Question 1: A collection was made for the students' party. Each student gave as many rupees as the number of students. If the total collected amount is Rs. 8,464, tell how many rupees each student gave.

Answer: Suppose the number of students is n.

Each student gave n rupees, so the total is $n^2 = 8464$.

n = √8464 = 92.

Each student gave Rs. 92.

Question: Exercise 2.6, Question 2: A square plot of land has area 10,816 square metres. Find the measure of each of its sides.

Answer: Area of a square = (side)², so side = √10816 = 104 m.

Question: Exercise 2.6, Question 3: In a school assembly 1,156 students are standing in rows. Each row has as many students as there are rows; tell how many rows there are in all.

Answer: Suppose the number of rows is n.

Each row has n students, so $n^2 = 1156$.

n = √1156 = 34.

There are 34 rows in all.

Question: Exercise 2.6, Question 4: A rectangular plot of land has area 1,800 square metres. Its width is half its length. Find its length and width.

Answer: Suppose the width is w, then the length = 2w.

Area = $2w \times w = 2w^2 = 1800$, so $w^2 = 900$ and w = 30 m.

Length = 60 m.

Question: Exercise 2.6, Question 5: A square plot of land has area 390,625

square metres. Find the length of wire needed to fence all around it.

Answer: Side of the square = √390625 = 625 m.

Perimeter = 4 × 625 = 2500 m.

So 2500 m of wire is needed.

Question: Exercise 2.7, Part A: Write the cubes of the following. 1. 10

Answer: $10^3 = 10 \times 10 \times 10 = 1000$.

Question: Exercise 2.7, Part A: 2. 20

Answer: $20^3 = 20 \times 20 \times 20 = 8000$.

Question: Exercise 2.7, Part A: 3. 30

Answer: $30^3 = 30 \times 30 \times 30 = 27000$.

Question: Exercise 2.7, Part A: 4. 40

Answer: $40^3 = 40 \times 40 \times 40 = 64000$.

Question: Exercise 2.7, Part A: 5. 50

Answer: $50^3 = 50 \times 50 \times 50 = 125000$.

Question: Exercise 2.7, Part A: 6. 60

Answer: $60^3 = 60 \times 60 \times 60 = 216000$.

Question: Exercise 2.7, Part A: 7. 8

Answer: $8^3 = 8 \times 8 \times 8 = 512$.

Question: Exercise 2.7, Part A: 8. 12

Answer: $12^3 = 12 \times 12 \times 12 = 1728$.

Question: Exercise 2.7, Part A: 9. 15

Answer: $15^3 = 15 \times 15 \times 15 = 3375$.

Question: Exercise 2.7, Part A: 10. 18

Answer: $18^3 = 18 \times 18 \times 18 = 5832$.

Question: Exercise 2.7, Part A: 11. 24

Answer: $24^3 = 24 \times 24 \times 24 = 13824$.

Question: Exercise 2.7, Part A: 12. 25

Answer: $25^3 = 25 \times 25 \times 25 = 15625$.

Question: Exercise 2.7, Part B: Write the cubes of the following common fractions. 1. 1/2

Answer: $(1/2)^3 = 1/8$.

Question: Exercise 2.7, Part B: 2. 2/3

Answer: $(2/3)^3 = 8/27$.

Question: Exercise 2.7, Part B: 3. 3/4

Answer: $(3/4)^3 = 27/64$.

Question: Exercise 2.7, Part B: 4. 4/5

Answer: $(4/5)^3 = 64/125$.

Question: Exercise 2.7, Part B: 5. 1 1/3

Answer: 1 1/3 = 4/3, so $(4/3)^3 = 64/27 = 2\ 10/27$.

Question: Exercise 2.7, Part B: 6. 2 1/2

Answer: 2 1/2 = 5/2, so $(5/2)^3 = 125/8 = 15\ 5/8$.

Question: Exercise 2.7, Part B: 7. 1 1/4

Answer: 1 1/4 = 5/4, so $(5/4)^3 = 125/64 = 1\ 61/64$.

Question: Exercise 2.7, Part B: 8. 2 1/6

Answer: 2 1/6 = 13/6, so $(13/6)^3 = 2197/216 = 10\ 37/216$.

Question: Exercise 2.7, Part B: 9. 1 1/10

Answer: 1 1/10 = 11/10, so $(11/10)^3 = 1331/1000 = 1\ 331/1000$.

Question: Exercise 2.7, Part B: 10. 2 3/20

Answer: 2 3/20 = 43/20, so $(43/20)^3 = 79507/8000 = 9\ 7507/8000$.

Question: Exercise 2.7, Part B: 11. 1 1/9

Answer: 1 1/9 = 10/9, so $(10/9)^3 = 1000/729 = 1\ 271/729$.

Question: Exercise 2.7, Part B: 12. 3 1/3

Answer: 3 1/3 = 10/3, so $(10/3)^3 = 1000/27 = 37\ 1/27$.

Question: Exercise 2.7, Part C: Write the cubes of the following decimal fractions. 1. 0.4

Answer: $(0.4)^3 = 0.4 \times 0.4 \times 0.4 = 0.064$.

Question: Exercise 2.7, Part C: 2. 0.9

Answer: $(0.9)^3 = 0.9 \times 0.9 \times 0.9 = 0.729$.

Question: Exercise 2.7, Part C: 3. 1.2

Answer: $(1.2)^3 = 1.2 \times 1.2 \times 1.2 = 1.728$.

Question: Exercise 2.7, Part C: 4. 1.6

Answer: $(1.6)^3 = 1.6 \times 1.6 \times 1.6 = 4.096$.

Question: Exercise 2.7, Part C: 5. 2.1

Answer: $(2.1)^3 = 2.1 \times 2.1 \times 2.1 = 9.261$.

Question: Exercise 2.7, Part C: 6. 2.5

Answer: $(2.5)^3 = 2.5 \times 2.5 \times 2.5 = 15.625$.

Question: Exercise 2.7, Part C: 7. 0.01

Answer: $(0.01)^3 = 0.000001$.

Question: Exercise 2.7, Part C: 8. 0.02

Answer: $(0.02)^3 = 0.000008$.

Question: Exercise 2.7, Part C: 9. 0.05

Answer: $(0.05)^3 = 0.000125$.

Question: Exercise 2.7, Part C: 10. 1.12

Answer: $(1.12)^3 = 1.12 \times 1.12 \times 1.12 = 1.404928$.

Question: Exercise 2.7, Part C: 11. 1.05

Answer: $(1.05)^3 = 1.05 \times 1.05 \times 1.05 = 1.157625$.

Question: Exercise 2.7, Part C: 12. 1.01

Answer: $(1.01)^3 = 1.01 \times 1.01 \times 1.01 = 1.030301$.

Question: Exercise 2.7, Part D: Which of the following are cubes? 1. 512

Answer: Yes, it is a cube: $512 = 8^3$.

Question: Exercise 2.7, Part D: 2. 729

Answer: Yes, it is a cube: $729 = 9^3$.

Question: Exercise 2.7, Part D: 3. 3375

Answer: Yes, it is a cube: $3375 = 15^3$.

Question: Exercise 2.7, Part D: 4. 1728

Answer: Yes, it is a cube: $1728 = 12^3$.

Question: Exercise 2.7, Part D: 5. 27/64

Answer: Yes, it is a cube: $27/64 = (3/4)^3$.

Question: Exercise 2.7, Part D: 6. 64/125

Answer: Yes, it is a cube: $64/125 = (4/5)^3$.

Question: Exercise 2.7, Part D: 7. 4096

Answer: Yes, it is a cube: $4096 = 16^3$.

Question: Exercise 2.7, Part D: 8. 5832

Answer: Yes, it is a cube: $5832 = 18^3$.

Question: Exercise 2.7, Part D: 9. 9261

Answer: Yes, it is a cube: $9261 = 21^3$.

Question: Exercise 2.7, Part D: 10. 27000

Answer: Yes, it is a cube: $27000 = 30^3$.

Question: Exercise 2.7, Part D: 11. 32768

Answer: Yes, it is a cube: $32768 = 32^3$.

Question: Exercise 2.7, Part D: 12. 39304

Answer: Yes, it is a cube: $39304 = 34^3$.

Question: Exercise 2.8, Part A: Find the cube roots of the following. 1. 64

Answer: $64 = 4 \times 4 \times 4$, so $\sqrt[3]{64} = 4$.

Question: Exercise 2.8, Part A: 2. 343

Answer: $343 = 7 \times 7 \times 7$, so $\sqrt[3]{343} = 7$.

Question: Exercise 2.8, Part A: 3. 1331

Answer: $1331 = 11 \times 11 \times 11$, so $\sqrt[3]{1331} = 11$.

Question: Exercise 2.8, Part A: 4. 512

Answer: $512 = 8 \times 8 \times 8$, so $\sqrt[3]{512} = 8$.

Question: Exercise 2.8, Part A: 5. 729

Answer: $729 = 9 \times 9 \times 9$, so $\sqrt[3]{729} = 9$.

Question: Exercise 2.8, Part A: 6. 74,088

Answer: 74088 = 42 × 42 × 42, so ∛74088 = 42.

Question: Exercise 2.8, Part A: 7. 3375

Answer: 3375 = 15 × 15 × 15, so ∛3375 = 15.

Question: Exercise 2.8, Part A: 8. 13824

Answer: 13824 = 24 × 24 × 24, so ∛13824 = 24.

Question: Exercise 2.8, Part A: 9. 15625

Answer: 15625 = 25 × 25 × 25, so ∛15625 = 25.

Question: Exercise 2.8, Part A: 10. 35937

Answer: 35937 = 33 × 33 × 33, so ∛35937 = 33.

Question: Exercise 2.8, Part A: 11. 1/8

Answer: ∛(1/8) = ∛1/∛8 = 1/2.

Question: Exercise 2.8, Part A: 12. 2 10/27

Answer: 2 10/27 = 64/27, so ∛(64/27) = 4/3 = 1 1/3.

Question: Exercise 2.8, Part A: 13. 1 61/64

Answer: 1 61/64 = 125/64, so ∛(125/64) = 5/4 = 1 1/4.

Question: Exercise 2.8, Part A: 14. 1 91/125

Answer: 1 91/125 = 216/125, so ∛(216/125) = 6/5 = 1 1/5.

Question: Exercise 2.8, Part A: 15. 343/729

Answer: ∛(343/729) = ∛343/∛729 = 7/9.

Question: Exercise 2.8, Part B: Verify with examples that: (i) The cube of a positive number is always a positive number.

Answer: Example: $2^3 = 8$.

2 is positive and its cube 8 is also positive.

So the statement is true.

Question: Exercise 2.8, Part B: (ii) The cube of a negative number is always a negative number.

Answer: Example: $(-2)^3 = -8$.

-2 is negative and its cube -8 is also negative.

So the statement is true.

Question: Exercise 2.8, Part B: (iii) The cube of an even number is always an even number.

Answer: Example: $4^3 = 64$.

4 is even and its cube 64 is also even.

So the statement is true.

Question: Exercise 2.8, Part B: (iv) The cube of an odd number is always an odd number.

Answer: Example: $3^3 = 27$.

3 is odd and its cube 27 is also odd.

So the statement is true.

Question: Exercise 2.8, Part C: Verify the following. 1. $(3 \times 4)^3 = 3^3 \times 4^3$

Answer: Left side: $(12)^3 = 1728$;

right side: $27 \times 64 = 1728$.

Both sides are equal.

Question: Exercise 2.8, Part C: 2. $(2/3)^3 = 2^3/3^3$

Answer: Left side: $(2/3)^3 = 8/27$;

right side: 8/27.

Both sides are equal.

Question: Exercise 2.8, Part C: 3. $(2 \times 5)^3 = 2^3 \times 5^3$

Answer: Left side: $(10)^3 = 1000$;

right side: $8 \times 125 = 1000$.

Both sides are equal.

Question: Exercise 2.8, Part C: 4. $(3/4)^3 = 3^3/4^3$

Answer: Left side: $(3/4)^3 = 27/64$;

right side: 27/64.

Both sides are equal.

Question: Exercise 2.8, Part C: 5. $(4 \times 5)^3 = 4^3 \times 5^3$

Answer: Left side: $(20)^3 = 8000$;

right side: $64 \times 125 = 8000$.

Both sides are equal.

Question: Exercise 2.8, Part C: 6. $(10 \times 20)^3 = 10^3 \times 20^3$

Answer: Left side: $(200)^3 = 8000000$;

right side: $1000 \times 8000 = 8000000$.

Both sides are equal.

Question: Exercise 2.8, Part C: 7. $(5/7)^3 = 5^3/7^3$

Answer: Left side: $(5/7)^3 = 125/343$;

right side: 125/343.

Both sides are equal.

Question: Exercise 2.8, Part C: 8. $(3/5)^3 = 3^3/5^3$

Answer: Left side: $(3/5)^3 = 27/125$;

right side: 27/125.

Both sides are equal.

Question: Exercise 2.8, Part C: 9. $(5 \times 8)^3 = 5^3 \times 8^3$

Answer: Left side: $(40)^3 = 64000$;

right side: $125 \times 512 = 64000$.

Both sides are equal.

Question: Review Exercise 2, Question 1: Put a (✓) mark on the correct answer for each statement. (i) The set of real numbers is the _____ of the set of rational numbers and the set of irrational numbers.

Answer: (c) union — because R = Q ∪ Q'.

Question: Review Exercise 2, Question 1: (ii) Which of the following answers about the number 100 is NOT correct?

Answer: (c) irrational number — because 100 is a rational number.

Question: Review Exercise 2, Question 1: (iii) Which of the following numbers is not a perfect square?

Answer: (c) 12.1 — because 12.1 is not the square of any number.

Question: Review Exercise 2, Question 1: (iv) The square of 4.9 is:

Answer: (b) 24.01 — because $4.9^2 = 24.01$.

Question: Review Exercise 2, Question 1: (v) $(3/5)^2$ = ?

Answer: (c) 9/25 — because $(3/5)^2 = 9/25$.

Question: Review Exercise 2, Question 1: (vi) 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = ?

Answer: (a) 8^2 — because this sum is the sum of the first 8 odd numbers and equals $8^2 = 64$.

Question: Review Exercise 2, Question 1: (vii) A square has side 0.6 m; its area is:

Answer: (a) 0.36 m^2 — because $0.6 \times 0.6 = 0.36$.

Question: Review Exercise 2, Question 1: (viii) $\sqrt{0.04}$ = ?

Answer: (c) 0.2 — because $0.2^2 = 0.04$.

Question: Review Exercise 2, Question 1: (ix) $\sqrt{(1^2 \times 4^2)}$ = ?

Answer: (a) 4 — because $\sqrt{(1 \times 16)} = \sqrt{16} = 4$.

Question: Review Exercise 2, Question 1: (x) $\sqrt[3]{125}$ = ?

Answer: (c) 5 — because $5^3 = 125$.

Question: Review Exercise 2, Question 1: (xi) $\sqrt{(4/9)}$ = ?

Answer: (a) 2/3 — because $\sqrt{(4/9)} = 2/3$.

Question: Review Exercise 2, Question 1: (xii) $\sqrt{(a/b)}$ = ?

Answer: (c) $\sqrt{a}/\sqrt{b}$ — the division law of square roots.

Question: Review Exercise 2, Question 2: Find the number of digits in the square roots of the following numbers. (i) 12544

Answer: 12544 has 5 digits, so its square root has (5+1)/2 = 3 digits (√12544 = 112).

Question: Review Exercise 2, Question 2: (ii) 418609

Answer: 418609 has 6 digits, so its square root has 6/2 = 3 digits (√418609 = 647).

Question: Review Exercise 2, Question 2: (iii) 30349081

Answer: 30349081 has 8 digits, so its square root has 8/2 = 4 digits (√30349081 = 5509).

Question: Review Exercise 2, Question 3: Find the square roots of the following. (i) 10 26/289

Answer: 10 26/289 = 2916/289, so the root = 54/17 = 3 3/17.

Question: Review Exercise 2, Question 3: (ii) 28 4/9

Answer: 28 4/9 = 256/9, so the root = 16/3 = 5 1/3.

Question: Review Exercise 2, Question 3: (iii) 101 92/169

Answer: 101 92/169 = 17161/169, so the root = 131/13 = 10 1/13.

Question: Review Exercise 2, Question 3: (iv) 0.053361

Answer: √0.053361 = 0.231 (because $0.231^2 = 0.053361$).

Question: Review Exercise 2, Question 3: (v) 0.204304

Answer: √0.204304 = 0.452 (because $0.452^2 = 0.204304$).

Question: Review Exercise 2, Question 3: (vi) 65.61

Answer: √65.61 = 8.1 (because $8.1^2 = 65.61$).

Question: Review Exercise 2, Question 4: A square plot of land has area 161,604 square metres. Find the measure of each of its sides.

Answer: Side of the square = √161604 = 402 m.

Question: Review Exercise 2, Question 5: Find the cube roots of the following. (i) 3375

Answer: $\sqrt[3]{3375} = 15$ (because $15^3 = 3375$).

Question: Review Exercise 2, Question 5: (ii) 1728

Answer: $\sqrt[3]{1728} = 12$ (because $12^3 = 1728$).

Question: Review Exercise 2, Question 5: (iii) 343/512

Answer: $\sqrt[3]{(343/512)} = 7/8$.

Question: Review Exercise 2, Question 5: (iv) 1 602/729

Answer: 1 602/729 = 1331/729, so $\sqrt[3]{}$ = 11/9 = 1 2/9.

3. Number Systems

Question: Exercise 3.1, Question 1: (i) Give the definition of the base system of numbers.

Answer: A base system is a number system in which, for a fixed base b, only b symbols (0 to b-1) are used, and the place value of each digit is a power of b.

Question: Exercise 3.1, Question 1: (ii) Define the base-five system of numbers. Write the first twelve numbers in the base-five system.

Answer: In the base-five system only five symbols 0, 1, 2, 3, 4 are used.

The first twelve numbers: 1, 2, 3, 4, 10_5, 11_5, 12_5, 13_5, 14_5, 20_5, 21_5, 22_5.

Question: Exercise 3.1, Question 1: (iii) Define the base-eight system of numbers. Write the first sixteen numbers in that system.

Answer: In the base-eight system only eight symbols 0, 1, 2, 3, 4, 5, 6, 7 are used.

The first sixteen numbers: 1, 2, 3, 4, 5, 6, 7, 10_8, 11_8, 12_8, 13_8, 14_8, 15_8, 16_8, 17_8, 20_8.

Question: Exercise 3.1, Question 1: (iv) Define the decimal system and give some examples.

Answer: The decimal system is the base-10 system, in which ten symbols 0 to 9 are used and the place value of each digit is a power of 10.

Examples: $201 = 2\times10^2 + 0\times10^1 + 1\times10^0$, $5683 = 5\times10^3 + 6\times10^2 + 8\times10^1 + 3\times10^0$.

Question: Exercise 3.1, Question 2: Write the following decimal numbers as the sum of powers of 2. (i) 11

Answer: $11 = 8 + 2 + 1 = 2^3 + 2^1 + 2^0$.

Question: Exercise 3.1, Question 2: (ii) 15

Answer: $15 = 8 + 4 + 2 + 1 = 2^3 + 2^2 + 2^1 + 2^0$.

Question: Exercise 3.1, Question 2: (iii) 21

Answer: $21 = 16 + 4 + 1 = 2^4 + 2^2 + 2^0$.

Question: Exercise 3.1, Question 2: (iv) 29

Answer: $29 = 16 + 8 + 4 + 1 = 2^4 + 2^3 + 2^2 + 2^0$.

Question: Exercise 3.1, Question 2: (v) 37

Answer: $37 = 32 + 4 + 1 = 2^5 + 2^2 + 2^0$.

Question: Exercise 3.1, Question 2: (vi) 51

Answer: $51 = 32 + 16 + 2 + 1 = 2^5 + 2^4 + 2^1 + 2^0$.

Question: Exercise 3.1, Question 2: (vii) 63

Answer: $63 = 32 + 16 + 8 + 4 + 2 + 1 = 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$.

Question: Exercise 3.1, Question 3: Write the following decimal numbers as the sum of powers of 5. (i) 13

Answer: $13 = 2\times5 + 3 = 2\times5^1 + 3\times5^0$.

Question: Exercise 3.1, Question 3: (ii) 24

Answer: $24 = 4\times5 + 4 = 4\times5^1 + 4\times5^0$.

Question: Exercise 3.1, Question 3: (iii) 39

Answer: $39 = 1\times25 + 2\times5 + 4 = 1\times5^2 + 2\times5^1 + 4\times5^0$.

Question: Exercise 3.1, Question 3: (iv) 47

Answer: $47 = 1\times25 + 4\times5 + 2 = 1\times5^2 + 4\times5^1 + 2\times5^0$.

Question: Exercise 3.1, Question 3: (v) 59

Answer: $59 = 2\times25 + 1\times5 + 4 = 2\times5^2 + 1\times5^1 + 4\times5^0$.

Question: Exercise 3.1, Question 3: (vi) 78

Answer: $78 = 3\times25 + 0\times5 + 3 = 3\times5^2 + 0\times5^1 + 3\times5^0$.

Question: Exercise 3.1, Question 3: (vii) 99

Answer: $99 = 3\times25 + 4\times5 + 4 = 3\times5^2 + 4\times5^1 + 4\times5^0$.

Question: Exercise 3.1, Question 3: (viii) 104

Answer: $104 = 4\times25 + 0\times5 + 4 = 4\times5^2 + 0\times5^1 + 4\times5^0$.

Question: Exercise 3.1, Question 4: Write the following decimal numbers as the sum of powers of 8. (i) 30

Answer: $30 = 3\times8 + 6 = 3\times8^1 + 6\times8^0$.

Question: Exercise 3.1, Question 4: (ii) 57

Answer: $57 = 7\times8 + 1 = 7\times8^1 + 1\times8^0$.

Question: Exercise 3.1, Question 4: (iii) 78

Answer: $78 = 1\times64 + 1\times8 + 6 = 1\times8^2 + 1\times8^1 + 6\times8^0$.

Question: Exercise 3.1, Question 4: (iv) 98

Answer: $98 = 1\times64 + 4\times8 + 2 = 1\times8^2 + 4\times8^1 + 2\times8^0$.

Question: Exercise 3.1, Question 4: (v) 114

Answer: $114 = 1\times64 + 6\times8 + 2 = 1\times8^2 + 6\times8^1 + 2\times8^0$.

Question: Exercise 3.1, Question 4: (vi) 225

Answer: $225 = 3\times64 + 4\times8 + 1 = 3\times8^2 + 4\times8^1 + 1\times8^0$.

Question: Exercise 3.1, Question 4: (vii) 340

Answer: $340 = 5\times64 + 2\times8 + 4 = 5\times8^2 + 2\times8^1 + 4\times8^0$.

Question: Exercise 3.1, Question 5: Write the following decimal numbers as the sum of powers of 10. (i) 201

Answer: $201 = 2\times100 + 0\times10 + 1 = 2\times10^2 + 0\times10^1 + 1\times10^0$.

Question: Exercise 3.1, Question 5: (ii) 319

Answer: $319 = 3\times100 + 1\times10 + 9 = 3\times10^2 + 1\times10^1 + 9\times10^0$.

Question: Exercise 3.1, Question 5: (iii) 4075

Answer: $4075 = 4\times1000 + 0\times100 + 7\times10 + 5 = 4\times10^3 + 0\times10^2 + 7\times10^1 + 5\times10^0$.

Question: Exercise 3.1, Question 5: (iv) 56970

Answer: $56970 = 5\times10000 + 6\times1000 + 9\times100 + 7\times10 + 0 = 5\times10^4 +$

$6\times10^3 + 9\times10^2 + 7\times10^1 + 0\times10^0$.

Question: Exercise 3.1, Question 5: (v) 980762

Answer: $980762 = 9\times100000 + 8\times10000 + 0\times1000 + 7\times100 + 6\times10 + 2$

$= 9\times10^5 + 8\times10^4 + 0\times10^3 + 7\times10^2 + 6\times10^1 + 2\times10^0$.

Question: Exercise 3.2, Part A: Convert the following decimal numbers into the base-two system. 1. 36

Answer: By repeated division: $36 = 100100_2$.

Question: Exercise 3.2, Part A: 2. 77

Answer: By repeated division: $77 = 1001101_2$.

Question: Exercise 3.2, Part A: 3. 89

Answer: By repeated division: $89 = 1011001_2$.

Question: Exercise 3.2, Part A: 4. 156

Answer: By repeated division: $156 = 10011100_2$.

Question: Exercise 3.2, Part A: 5. 280

Answer: By repeated division: $280 = 100011000_2$.

Question: Exercise 3.2, Part A: 6. 489

Answer: By repeated division: $489 = 111101001_2$.

Question: Exercise 3.2, Part A: 7. 654

Answer: By repeated division: $654 = 1010001110_2$.

Question: Exercise 3.2, Part A: 8. 786

Answer: By repeated division: 786 = 1100010010_2.

Question: Exercise 3.2, Part A: 9. 999

Answer: By repeated division: 999 = 1111100111_2.

Question: Exercise 3.2, Part A: 10. 1600

Answer: By repeated division: 1600 = 11001000000_2.

Question: Exercise 3.2, Part A: 11. 1705

Answer: By repeated division: 1705 = 11010101001_2.

Question: Exercise 3.2, Part A: 12. 1808

Answer: By repeated division: 1808 = 11100010000_2.

Question: Exercise 3.2, Part A: 13. 1096

Answer: By repeated division: 1096 = 10001001000_2.

Question: Exercise 3.2, Part A: 14. 2001

Answer: By repeated division: 2001 = 11111010001_2.

Question: Exercise 3.2, Part A: 15. 2020

Answer: By repeated division: 2020 = 11111100100_2.

Question: Exercise 3.2, Part B: Convert the following decimal numbers into the base-five system. 1. 87

Answer: By repeated division: 87 = 322_5 (3×25 + 2×5 + 2).

Question: Exercise 3.2, Part B: 2. 98

Answer: By repeated division: $98 = 343_5$ (3×25 + 4×5 + 3).

Question: Exercise 3.2, Part B: 3. 169

Answer: By repeated division: $169 = 1134_5$ (1×125 + 1×25 + 3×5 + 4).

Question: Exercise 3.2, Part B: 4. 205

Answer: By repeated division: $205 = 1310_5$ (1×125 + 3×25 + 1×5 + 0).

Question: Exercise 3.2, Part B: 5. 370

Answer: By repeated division: $370 = 2440_5$ (2×125 + 4×25 + 4×5 + 0).

Question: Exercise 3.2, Part B: 6. 444

Answer: By repeated division: $444 = 3234_5$ (3×125 + 2×25 + 3×5 + 4).

Question: Exercise 3.2, Part B: 7. 666

Answer: By repeated division: $666 = 10131_5$ (1×625 + 0×125 + 1×25 + 3×5 + 1).

Question: Exercise 3.2, Part B: 8. 1072

Answer: By repeated division: $1072 = 13242_5$ (1×625 + 3×125 + 2×25 + 4×5 + 2).

Question: Exercise 3.2, Part B: 9. 2007

Answer: By repeated division: $2007 = 31012_5$ (3×625 + 1×125 + 0×25 + 1×5 + 2).

Question: Exercise 3.2, Part B: 10. 3333

Answer: By repeated division: $3333 = 101313_5$ (1×3125 + 0×625 + 1×125 + 3×25 + 1×5 + 3).

Question: Exercise 3.2, Part B: 11. 5005

Answer: By repeated division: $5005 = 130010_5$ (1×3125 + 3×625 + 0×125 + 0×25 + 1×5 + 0).

Question: Exercise 3.2, Part B: 12. 5050

Answer: By repeated division: $5050 = 130200_5$ (1×3125 + 3×625 + 0×125 + 2×25 + 0×5 + 0).

Question: Exercise 3.2, Part B: 13. 5016

Answer: By repeated division: $5016 = 130031_5$ (1×3125 + 3×625 + 0×125 + 0×25 + 3×5 + 1).

Question: Exercise 3.2, Part B: 14. 5602

Answer: By repeated division: $5602 = 134402_5$ (1×3125 + 3×625 + 4×125 + 4×25 + 0×5 + 2).

Question: Exercise 3.2, Part B: 15. 5005

Answer: By repeated division: $5005 = 130010_5$ (1×3125 + 3×625 + 0×125 + 0×25 + 1×5 + 0).

Question: Exercise 3.2, Part C: Convert the following decimal numbers into the base-eight system. 1. 504

Answer: By repeated division: $504 = 770_8$ (7×64 + 7×8 + 0).

Question: Exercise 3.2, Part C: 2. 1765

Answer: By repeated division: $1765 = 3345_8$ ($3\times512 + 3\times64 + 4\times8 + 5$).

Question: Exercise 3.2, Part C: 3. 20093

Answer: By repeated division: $20093 = 47175_8$ ($4\times4096 + 7\times512 + 1\times64 + 7\times8 + 5$).

Question: Exercise 3.2, Part C: 4. 33661

Answer: By repeated division: $33661 = 101575_8$ ($1\times32768 + 0\times4096 + 1\times512 + 5\times64 + 7\times8 + 5$).

Question: Exercise 3.2, Part C: 5. 48360

Answer: By repeated division: $48360 = 136350_8$ ($1\times32768 + 3\times4096 + 6\times512 + 3\times64 + 5\times8 + 0$).

Question: Exercise 3.2, Part C: 6. 50607

Answer: By repeated division: $50607 = 142657_8$ ($1\times32768 + 4\times4096 + 2\times512 + 6\times64 + 5\times8 + 7$).

Question: Exercise 3.2, Part C: 7. 61278

Answer: By repeated division: $61278 = 167536_8$ ($1\times32768 + 6\times4096 + 7\times512 + 5\times64 + 3\times8 + 6$).

Question: Exercise 3.2, Part C: 8. 79118

Answer: By repeated division: $79118 = 232416_8$ ($2\times32768 + 3\times4096 + 2\times512 + 4\times64 + 1\times8 + 6$).

Question: Exercise 3.2, Part C: 9. 80877

Answer: By repeated division: $80877 = 235755_8$ ($2\times32768 + 3\times4096 +$

5×512 + 7×64 + 5×8 + 5).

Question: Exercise 3.2, Part C: 10. 99099

Answer: By repeated division: 99099 = 301433_8 (3×32768 + 0×4096 + 1×512 + 4×64 + 3×8 + 3).

Question: Exercise 3.2, Part C: 11. 90009

Answer: By repeated division: 90009 = 257631_8 (2×32768 + 5×4096 + 7×512 + 6×64 + 3×8 + 1).

Question: Exercise 3.2, Part C: 12. 99009

Answer: By repeated division: 99009 = 301301_8 (3×32768 + 0×4096 + 1×512 + 3×64 + 0×8 + 1).

Question: Exercise 3.2, Part C: 13. 98765

Answer: By repeated division: 98765 = 300715_8 (3×32768 + 0×4096 + 0×512 + 7×64 + 1×8 + 5).

Question: Exercise 3.2, Part C: 14. 98065

Answer: By repeated division: 98065 = 277421_8 (2×32768 + 7×4096 + 7×512 + 4×64 + 2×8 + 1).

Question: Exercise 3.2, Part C: 15. 99999

Answer: By repeated division: 99999 = 303237_8 (3×32768 + 0×4096 + 3×512 + 2×64 + 3×8 + 7).

Question: Exercise 3.3, Part A: Convert the following base-two numbers into decimal numbers. 1. 1101_2

Answer: $1101_2 = 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0 = 8 + 4 + 0 + 1 = 13$.

Question: Exercise 3.3, Part A: 2. 11011_2

Answer: $11011_2 = 16 + 8 + 0 + 2 + 1 = 27$.

Question: Exercise 3.3, Part A: 3. 10111_2

Answer: $10111_2 = 16 + 0 + 4 + 2 + 1 = 23$.

Question: Exercise 3.3, Part A: 4. 110101_2

Answer: $110101_2 = 32 + 16 + 0 + 4 + 0 + 1 = 53$.

Question: Exercise 3.3, Part A: 5. 1010101_2

Answer: $1010101_2 = 64 + 0 + 16 + 0 + 4 + 0 + 1 = 85$.

Question: Exercise 3.3, Part A: 6. 1110111_2

Answer: $1110111_2 = 64 + 32 + 16 + 0 + 4 + 2 + 1 = 119$.

Question: Exercise 3.3, Part A: 7. 1100110_2

Answer: $1100110_2 = 64 + 32 + 0 + 0 + 4 + 2 + 0 = 102$.

Question: Exercise 3.3, Part A: 8. 1100111_2

Answer: $1100111_2 = 64 + 32 + 0 + 0 + 4 + 2 + 1 = 103$.

Question: Exercise 3.3, Part A: 9. 11010110_2

Answer: $11010110_2 = 128 + 64 + 0 + 16 + 0 + 4 + 2 + 0 = 214$.

Question: Exercise 3.3, Part A: 10. 111111101_2

Answer: $111111101_2 = 256 + 128 + 64 + 32 + 16 + 8 + 4 + 0 + 1 = 509$.

Question: Exercise 3.3, Part A: 11. 10010010_2

Answer: $10010010_2 = 128 + 0 + 0 + 16 + 0 + 0 + 2 + 0 = 146$.

Question: Exercise 3.3, Part A: 12. 11000101_2

Answer: $11000101_2 = 128 + 64 + 0 + 0 + 0 + 4 + 0 + 1 = 197$.

Question: Exercise 3.3, Part B: Convert the following base-five numbers into decimal numbers. 1. 21_5

Answer: $21_5 = 2\times5 + 1 = 11$.

Question: Exercise 3.3, Part B: 2. 30_5

Answer: $30_5 = 3\times5 + 0 = 15$.

Question: Exercise 3.3, Part B: 3. 40_5

Answer: $40_5 = 4\times5 + 0 = 20$.

Question: Exercise 3.3, Part B: 4. 12_5

Answer: $12_5 = 1\times5 + 2 = 7$.

Question: Exercise 3.3, Part B: 5. 22_5

Answer: $22_5 = 2\times5 + 2 = 12$.

Question: Exercise 3.3, Part B: 6. 131_5

Answer: $131_5 = 1\times25 + 3\times5 + 1 = 41$.

Question: Exercise 3.3, Part B: 7. 442_5

Answer: $442_5 = 4\times25 + 4\times5 + 2 = 122$.

Question: Exercise 3.3, Part B: 8. 232_5

Answer: $232_5 = 2\times25 + 3\times5 + 2 = 67$.

Question: Exercise 3.3, Part B: 9. 401_5

Answer: $401_5 = 4\times25 + 0\times5 + 1 = 101$.

Question: Exercise 3.3, Part B: 10. 430_5

Answer: $430_5 = 4\times25 + 3\times5 + 0 = 115$.

Question: Exercise 3.3, Part B: 11. 1310_5

Answer: $1310_5 = 1\times125 + 3\times25 + 1\times5 + 0 = 205$.

Question: Exercise 3.3, Part B: 12. 4421_5

Answer: $4421_5 = 4\times125 + 4\times25 + 2\times5 + 1 = 611$.

Question: Exercise 3.3, Part B: 13. 2322_5

Answer: $2322_5 = 2\times125 + 3\times25 + 2\times5 + 2 = 337$.

Question: Exercise 3.3, Part B: 14. 4013_5

Answer: $4013_5 = 4\times125 + 0\times25 + 1\times5 + 3 = 508$.

Question: Exercise 3.3, Part B: 15. 4304_5

Answer: $4304_5 = 4\times125 + 3\times25 + 0\times5 + 4 = 579$.

Question: Exercise 3.3, Part C: Convert the following base-eight numbers into decimal numbers. 1. 47_8

Answer: $47_8 = 4\times8 + 7 = 39$.

Question: Exercise 3.3, Part C: 2. 16_8

Answer: $16_8 = 1\times8 + 6 = 14$.

Question: Exercise 3.3, Part C: 3. 40_8

Answer: $40_8 = 4\times8 + 0 = 32$.

Question: Exercise 3.3, Part C: 4. 60_8

Answer: $60_8 = 6\times8 + 0 = 48$.

Question: Exercise 3.3, Part C: 5. 70_8

Answer: $70_8 = 7\times8 + 0 = 56$.

Question: Exercise 3.3, Part C: 6. 124_8

Answer: $124_8 = 1\times64 + 2\times8 + 4 = 84$.

Question: Exercise 3.3, Part C: 7. 242_8

Answer: $242_8 = 2\times64 + 4\times8 + 2 = 162$.

Question: Exercise 3.3, Part C: 8. 30_8

Answer: $30_8 = 3\times8 + 0 = 24$.

Question: Exercise 3.3, Part C: 9. 450_8

Answer: $450_8 = 4\times64 + 5\times8 + 0 = 296$.

Question: Exercise 3.3, Part C: 10. 700_8

Answer: $700_8 = 7\times64 + 0 + 0 = 448$.

Question: Exercise 3.3, Part C: 11. 3006_8

Answer: $3006_8 = 3\times512 + 0 + 0 + 6 = 1542$.

Question: Exercise 3.3, Part C: 12. 1054_8

Answer: $1054_8 = 1\times512 + 0 + 5\times8 + 4 = 556$.

Question: Exercise 3.3, Part C: 13. 6600_8

Answer: $6600_8 = 6\times512 + 6\times64 + 0 + 0 = 3456$.

Question: Exercise 3.3, Part C: 14. 4405_8

Answer: $4405_8 = 4\times512 + 4\times64 + 0 + 5 = 2309$.

Question: Exercise 3.3, Part C: 15. 5043_8

Answer: $5043_8 = 5\times512 + 0 + 4\times8 + 3 = 2595$.

Question: Exercise 3.4, Part A: Solve. 1. $110_2 + 11_2$

Answer: Addition in base two: $110_2 + 11_2 = 1001_2$ (in decimal 6 + 3 = 9).

Question: Exercise 3.4, Part A: 2. $1010_2 + 101_2$

Answer: $1010_2 + 101_2 = 1111_2$ (in decimal 10 + 5 = 15).

Question: Exercise 3.4, Part A: 3. $1110_2 + 111_2$

Answer: $1110_2 + 111_2 = 10101_2$ (in decimal 14 + 7 = 21).

Question: Exercise 3.4, Part A: 4. $11011_2 + 11001_2$

Answer: $11011_2 + 11001_2 = 110100_2$ (in decimal 27 + 25 = 52).

Question: Exercise 3.4, Part A: 5. $10101_2 + 110_2$

Answer: $10101_2 + 110_2 = 11011_2$ (in decimal 21 + 6 = 27).

Question: Exercise 3.4, Part A: 6. $111_2 + 11_2$

Answer: $111_2 + 11_2 = 1010_2$ (in decimal 7 + 3 = 10).

Question: Exercise 3.4, Part A: 7. $1101_2 + 110_2 + 1001_2$

Answer: $1101_2 + 110_2 + 1001_2 = 11100_2$ (in decimal 13 + 6 + 9 = 28).

Question: Exercise 3.4, Part A: 8. $1101_2 + 1001_2 + 11_2$

Answer: $1101_2 + 1001_2 + 11_2 = 11001_2$ (in decimal 13 + 9 + 3 = 25).

Question: Exercise 3.4, Part A: 9. $11011_2 + 110101_2 + 110_2$

Answer: $11011_2 + 110101_2 + 110_2 = 1010110_2$ (in decimal 27 + 53 + 6 = 86).

Question: Exercise 3.4, Part A: 10. $11111_2 + 1010_2 + 111_2$

Answer: $11111_2 + 1010_2 + 111_2 = 110000_2$ (in decimal 31 + 10 + 7 = 48).

Question: Exercise 3.4, Part A: 11. $1010_2 + 1111_2 + 10111_2$

Answer: $1010_2 + 1111_2 + 10111_2 = 110000_2$ (in decimal 10 + 15 + 23 = 48).

Question: Exercise 3.4, Part A: 12. $1111_2 + 11101_2 + 11111_2$

Answer: $1111_2 + 11101_2 + 11111_2 = 1001011_2$ (in decimal 15 + 29 + 31 = 75).

Question: Exercise 3.4, Part B: Solve. 1. $1101_2 - 11_2$

Answer: Subtraction in base two: $1101_2 - 11_2 = 1010_2$ (in decimal 13 - 3 = 10).

Question: Exercise 3.4, Part B: 2. 111_2 - 101_2

Answer: 111_2 - 101_2 = 10_2 (in decimal 7 - 5 = 2).

Question: Exercise 3.4, Part B: 3. 1110_2 - 110_2

Answer: 1110_2 - 110_2 = 1000_2 (in decimal 14 - 6 = 8).

Question: Exercise 3.4, Part B: 4. 11111_2 - 1010_2

Answer: 11111_2 - 1010_2 = 10101_2 (in decimal 31 - 10 = 21).

Question: Exercise 3.4, Part B: 5. 110111_2 - 1101_2

Answer: 110111_2 - 1101_2 = 101010_2 (in decimal 55 - 13 = 42).

Question: Exercise 3.4, Part B: 6. 10011_2 - 101_2

Answer: 10011_2 - 101_2 = 1110_2 (in decimal 19 - 5 = 14).

Question: Exercise 3.4, Part B: 7. 1001_2 - 10_2

Answer: 1001_2 - 10_2 = 111_2 (in decimal 9 - 2 = 7).

Question: Exercise 3.4, Part B: 8. 1111_2 - 1000_2

Answer: 1111_2 - 1000_2 = 111_2 (in decimal 15 - 8 = 7).

Question: Exercise 3.4, Part B: 9. 11011_2 - 1010_2

Answer: 11011_2 - 1010_2 = 10001_2 (in decimal 27 - 10 = 17).

Question: Exercise 3.4, Part B: 10. 11111_2 - 1010_2

Answer: 11111_2 - 1010_2 = 10101_2 (in decimal 31 - 10 = 21).

Question: Exercise 3.4, Part C: Simplify. 1. (1111_2 + 101_2) - (1100_2 - 101_2)

Answer: (15 + 5) - (12 - 5) = 20 - 7 = 13 = 1101_2.

Question: Exercise 3.4, Part C: 2. (1101_2 + 10010_2) - (10101_2 - 1010_2)

Answer: (13 + 18) - (21 - 10) = 31 - 11 = 20 = 10100_2.

Question: Exercise 3.4, Part C: 3. 1111_2 - 1010_2 + 101_2

Answer: 15 - 10 + 5 = 10 = 1010_2.

Question: Exercise 3.4, Part C: 4. (10101_2 - 10010_2) - (10001_2 - 1110_2)

Answer: (21 - 18) - (17 - 14) = 3 - 3 = 0 = 0_2.

Question: Exercise 3.4, Part D: Multiply. 1. 101_2 × 11_2

Answer: 101_2 × 11_2 = 1111_2 (in decimal 5 × 3 = 15).

Question: Exercise 3.4, Part D: 2. 110_2 × 101_2

Answer: 110_2 × 101_2 = 11110_2 (in decimal 6 × 5 = 30).

Question: Exercise 3.4, Part D: 3. 1111_2 × 100_2

Answer: 1111_2 × 100_2 = 111100_2 (in decimal 15 × 4 = 60).

Question: Exercise 3.4, Part D: 4. 110_2 × 1101_2

Answer: 110_2 × 1101_2 = 1001110_2 (in decimal 6 × 13 = 78).

Question: Exercise 3.4, Part D: 5. 101_2 × 111_2

Answer: 101_2 × 111_2 = 100011_2 (in decimal 5 × 7 = 35).

Question: Exercise 3.4, Part D: 6. 10101_2 × 1010_2

Answer: 10101_2 × 1010_2 = 11010010_2 (in decimal 21 × 10 = 210).

Question: Exercise 3.4, Part D: 7. $11110_2 \times 101_2$

Answer: $11110_2 \times 101_2 = 10010110_2$ (in decimal $30 \times 5 = 150$).

Question: Exercise 3.4, Part D: 8. $1001_2 \times 101_2$

Answer: $1001_2 \times 101_2 = 101101_2$ (in decimal $9 \times 5 = 45$).

Question: Exercise 3.4, Part D: 9. $10100_2 \times 11_2$

Answer: $10100_2 \times 11_2 = 111100_2$ (in decimal $20 \times 3 = 60$).

Question: Exercise 3.4, Part D: 10. $110111_2 \times 111_2$

Answer: $110111_2 \times 111_2 = 110000001_2$ (in decimal $55 \times 7 = 385$).

Question: Exercise 3.4, Part D: 11. $101010_2 \times 101_2$

Answer: $101010_2 \times 101_2 = 11010010_2$ (in decimal $42 \times 5 = 210$).

Question: Exercise 3.4, Part D: 12. $1011110_2 \times 110_2$

Answer: $1011110_2 \times 110_2 = 1000110100_2$ (in decimal $94 \times 6 = 564$).

Question: Exercise 3.5, Part A: Solve. 1. $432_5 + 134_5$

Answer: Addition in base five: $432_5 + 134_5 = 1121_5$ (in decimal $117 + 44 = 161$).

Question: Exercise 3.5, Part A: 2. $3344_5 + 1304_5$

Answer: $3344_5 + 1304_5 = 10203_5$ (in decimal $474 + 204 = 678$).

Question: Exercise 3.5, Part A: 3. $4333_5 + 343_5$

Answer: $4333_5 + 343_5 = 10231_5$ (in decimal $593 + 98 = 691$).

Question: Exercise 3.5, Part A: 4. $43_5 + 14_5$

Answer: $43_5 + 14_5 = 112_5$ (in decimal 23 + 9 = 32).

Question: Exercise 3.5, Part A: 5. $34321_5 + 24113_5$

Answer: $34321_5 + 24113_5 = 113434_5$ (in decimal 2461 + 1783 = 4244).

Question: Exercise 3.5, Part A: 6. $43_5 + 143_5$

Answer: $43_5 + 143_5 = 241_5$ (in decimal 23 + 48 = 71).

Question: Exercise 3.5, Part A: 7. $1432_5 + 2341_5 + 123_5$

Answer: $1432_5 + 2341_5 + 123_5 = 10001_5$ (in decimal 242 + 346 + 38 = 626).

Question: Exercise 3.5, Part A: 8. $222_5 + 3433_5 + 142_5$

Answer: $222_5 + 3433_5 + 142_5 = 4402_5$ (in decimal 62 + 493 + 47 = 602).

Question: Exercise 3.5, Part A: 9. $4443_5 + 343_5$

Answer: $4443_5 + 343_5 = 10341_5$ (in decimal 623 + 98 = 721).

Question: Exercise 3.5, Part A: 10. $12340_5 + 3444_5 + 444_5$

Answer: $12340_5 + 3444_5 + 444_5 = 22333_5$ (in decimal 970 + 499 + 124 = 1593).

Question: Exercise 3.5, Part A: 11. $12340_5 + 12340_5$

Answer: $12340_5 + 12340_5 = 30230_5$ (in decimal 970 + 970 = 1940).

Question: Exercise 3.5, Part A: 12. $43324_5 + 1243_5$

Answer: $43324_5 + 1243_5 = 100122_5$ (in decimal 2964 + 198 = 3162).

Question: Exercise 3.5, Part B: Solve. 1. $342_5 - 12_5$

Answer: Subtraction in base five: $342_5 - 12_5 = 330_5$ (in decimal 97 - 7 = 90).

Question: Exercise 3.5, Part B: 2. $202_5 - 14_5$

Answer: $202_5 - 14_5 = 133_5$ (in decimal 52 - 9 = 43).

Question: Exercise 3.5, Part B: 3. $3214_5 - 1403_5$

Answer: $3214_5 - 1403_5 = 1311_5$ (in decimal 434 - 228 = 206).

Question: Exercise 3.5, Part B: 4. $1203_5 - 134_5$

Answer: $1203_5 - 134_5 = 1014_5$ (in decimal 178 - 44 = 134).

Question: Exercise 3.5, Part B: 5. $4321_5 - 1234_5$

Answer: $4321_5 - 1234_5 = 3032_5$ (in decimal 586 - 194 = 392).

Question: Exercise 3.5, Part B: 6. $4231_5 - 2304_5$

Answer: $4231_5 - 2304_5 = 1422_5$ (in decimal 566 - 329 = 237).

Question: Exercise 3.5, Part B: 7. $4310_5 - 3421_5$

Answer: $4310_5 - 3421_5 = 334_5$ (in decimal 560 - 466 = 94).

Question: Exercise 3.5, Part B: 8. $20102_5 - 1424_5$

Answer: $20102_5 - 1424_5 = 13123_5$ (in decimal 1277 - 239 = 1038).

Question: Exercise 3.5, Part C: Solve. 1. $342_5 + 444_5 - 341_5$

Answer: 97 + 124 - 96 = 125 = 1000_5.

Question: Exercise 3.5, Part C: 2. $1234_5 + 4321_5 - 444_5$

Answer: 194 + 586 - 124 = 656 = 10111_5.

Question: Exercise 3.5, Part C: 3. $1321_5 + 223_5 - 44_5$

Answer: 211 + 63 - 24 = 250 = 2000_5.

Question: Exercise 3.5, Part C: 4. $44444_5 - 321_5 - 444_5$

Answer: 3124 - 86 - 124 = 2914 = 43124_5.

Question: Exercise 3.5, Part C: 5. $23222_5 - 2111_5 - 232_5$

Answer: 1687 - 281 - 67 = 1339 = 20324_5.

Question: Exercise 3.5, Part C: 6. $4304_5 - 222_5 - 333_5$

Answer: 579 - 62 - 98 = 419 = 3144_5.

Question: Exercise 3.5, Part C: 7. $1234_5 + 4321_5 - 3221_5$

Answer: 194 + 586 - 436 = 344 = 2334_5.

Question: Exercise 3.5, Part C: 8. $304021_5 - 103002_5 - 200403_5$

Answer: 9886 - 3502 - 6353 = 31 = 111_5.

Question: Exercise 3.5, Part D: Multiply. 1. $423_5 \times 321_5$

Answer: $423_5 \times 321_5 = 302333_5$ (in decimal 113 × 86 = 9718).

Question: Exercise 3.5, Part D: 2. $14_5 \times 13_5$

Answer: $14_5 \times 13_5 = 242_5$ (in decimal 9 × 8 = 72).

Question: Exercise 3.5, Part D: 3. $1134_5 \times 223_5$

Answer: $1134_5 \times 223_5 = 320042_5$ (in decimal 169 × 63 = 10647).

Question: Exercise 3.5, Part D: 4. $3221_5 \times 443_5$

Answer: $3221_5 \times 443_5 = 3204003_5$ (in decimal 436 × 123 = 53628).

Question: Exercise 3.5, Part D: 5. $233_5 \times 343_5$

Answer: $233_5 \times 343_5 = 203124_5$ (in decimal 68 × 98 = 6664).

Question: Exercise 3.5, Part D: 6. $403_5 \times 243_5$

Answer: $403_5 \times 243_5 = 220034_5$ (in decimal 103 × 73 = 7519).

Question: Exercise 3.5, Part D: 7. $1004_5 \times 113_5$

Answer: $1004_5 \times 113_5 = 114012_5$ (in decimal 129 × 33 = 4257).

Question: Exercise 3.5, Part D: 8. $4004_5 \times 303_5$

Answer: $4004_5 \times 303_5 = 2224222_5$ (in decimal 504 × 78 = 39312).

Question: Exercise 3.5, Part D: 9. $2410_5 \times 432_5$

Answer: $2410_5 \times 432_5 = 2312120_5$ (in decimal 355 × 117 = 41535).

Question: Exercise 3.5, Part D: 10. $3210_5 \times 114_5$

Answer: $3210_5 \times 114_5 = 431440_5$ (in decimal 430 × 34 = 14620).

Question: Exercise 3.5, Part D: 11. $3344_5 \times 1230_5$

Answer: $3344_5 \times 1230_5 = 10340220_5$ (in decimal 474 × 190 = 90060).

Question: Exercise 3.5, Part D: 12. $4321_5 \times 234_5$

Answer: $4321_5 \times 234_5 = 2243214_5$ (in decimal 586 × 69 = 40434).

Question: Exercise 3.6, Question 1: Solve. (i) $356_8 + 67_8$

Answer: Addition in base eight: $356_8 + 67_8 = 445_8$ (in decimal 238 + 55 = 293).

Question: Exercise 3.6, Question 1: (ii) $5034_8 + 6721_8$

Answer: $5034_8 + 6721_8 = 13755_8$ (in decimal 2588 + 3537 = 6125).

Question: Exercise 3.6, Question 1: (iii) $5003_8 + 66644_8$

Answer: $5003_8 + 66644_8 = 73647_8$ (in decimal 2563 + 28068 = 30631).

Question: Exercise 3.6, Question 1: (iv) $7356_8 + 6256_8$

Answer: $7356_8 + 6256_8 = 15634_8$ (in decimal 3822 + 3246 = 7068).

Question: Exercise 3.6, Question 1: (v) $657546_8 + 46701_8$

Answer: $657546_8 + 46701_8 = 726447_8$ (in decimal 221030 + 19905 = 240935).

Question: Exercise 3.6, Question 2: Find the value. (i) $73_8 - 46_8$

Answer: Subtraction in base eight: $73_8 - 46_8 = 25_8$ (in decimal 59 - 38 = 21).

Question: Exercise 3.6, Question 2: (ii) $200_8 - 173_8$

Answer: $200_8 - 173_8 = 5_8$ (in decimal 128 - 123 = 5).

Question: Exercise 3.6, Question 2: (iii) $4326_8 - 3754_8$

Answer: $4326_8 - 3754_8 = 352_8$ (in decimal 2262 - 2028 = 234).

Question: Exercise 3.6, Question 2: (iv) $77601_8 - 67706_8$

Answer: 77601_8 - 67706_8 = 7673_8 (in decimal 32641 - 28614 = 4027).

Question: Exercise 3.6, Question 2: (v) 163732_8 - 77766_8

Answer: 163732_8 - 77766_8 = 63744_8 (in decimal 59354 - 32758 = 26596).

Question: Exercise 3.6, Question 3: Multiply. (i) 67_8 × 45_8

Answer: 67_8 × 45_8 = 3763_8 (in decimal 55 × 37 = 2035).

Question: Exercise 3.6, Question 3: Multiply. (ii) 345_8 × 171_8

Answer: 345_8 × 171_8 = 66075_8 (in decimal 229 × 121 = 27709).

Question: Exercise 3.6, Question 3: Multiply. (iii) 13470_8 × 563_8

Answer: 13470_8 × 563_8 = 10323050_8 (in decimal 5944 × 371 = 2205224).

Question: Exercise 3.6, Question 3: Multiply. (iv) 30076_8 × 324_8

Answer: 30076_8 × 324_8 = 11771530_8 (in decimal 12350 × 212 = 2618200).

Question: Exercise 3.6, Question 3: Multiply. (v) 763541_8 × 4061_8

Answer: 763541_8 × 4061_8 = 3776450621_8 (in decimal 255841 × 2097 = 536498577).

Question: Exercise 3.7, Question 1: Simplify the following and write in the binary system. (i) 31 + 13_5 + 26_8

Answer: 31 + 8 + 22 = 61 = 111101_2.

Question: Exercise 3.7, Question 1: (ii) 65 + 31_5 + 101_8

Answer: 65 + 16 + 65 = 146 = 10010010_2.

Question: Exercise 3.7, Question 1: (iii) $71 + 121_5 + 1010_2 + 20_8$

Answer: $71 + 36 + 10 + 16 = 133 = 10000101_2$.

Question: Exercise 3.7, Question 1: (iv) $34 + 34_5 + 110_2 + 15_8$

Answer: $34 + 19 + 6 + 13 = 72 = 1001000_2$.

Question: Exercise 3.7, Question 1: (v) $45 + 44_5 + 1010_2 + 30_8$

Answer: $45 + 24 + 10 + 24 = 103 = 1100111_2$.

Question: Exercise 3.7, Question 2: Simplify the following and write in the base-five system. (i) $31 - 13_5 - 101_2$

Answer: $31 - 8 - 5 = 18 = 33_5$.

Question: Exercise 3.7, Question 2: (ii) $65 - 31_5 - 10_8$

Answer: $65 - 16 - 8 = 41 = 131_5$.

Question: Exercise 3.7, Question 2: (iii) $71 - 121_5 - 111_2 - 11_8$

Answer: $71 - 36 - 7 - 9 = 19 = 34_5$.

Question: Exercise 3.7, Question 2: (iv) $100 - 100_5 - 100_2$

Answer: $100 - 25 - 4 = 71 = 241_5$.

Question: Exercise 3.7, Question 2: (v) $1000 - 1001_5 - 1010_2 - 100_8$

Answer: $1000 - 126 - 10 - 64 = 800 = 11200_5$.

Question: Exercise 3.7, Question 3: After multiplying, express the following in base five and base eight. (i) $32 \times 114_5 \times 110_2$

Answer: $32 \times 34 \times 6 = 6528 = 202103_5 = 14600_8$.

Question: Exercise 3.7, Question 3: (ii) $32 \times 101_5 \times 111_2$

Answer: $32 \times 26 \times 7 = 5824 = 141244_5 = 13300_8$.

Question: Exercise 3.7, Question 3: (iii) $95 \times 1101_5 \times 101_8$

Answer: $95 \times 151 \times 65 = 932425 = 214314200_5 = 3435111_8$.

Question: Exercise 3.7, Question 4: Add the sum of 101_5 and 11100_2 to the sum of 3145 and 39, then convert the resulting number into the base-five, base-two and base-eight systems.

Answer: $(26 + 28) + (3145 + 39) = 54 + 3184 = 3238 = 100423_5 = 110010100110_2 = 6246_8$.

Question: Exercise 3.7, Question 5: Subtract the product of 123_5 and 11011_2 from the product of 2145_8 and 29, and convert the resulting number into base five, base two and base eight.

Answer: $1125 \times 29 - 38 \times 27 = 32625 - 1026 = 31599 = 2002344_5 = 111101101101111_2 = 75557_8$.

Question: Review Exercise 3, Question 1: Express the following base-five numbers as decimal numbers and base-eight numbers. (i) 3421_5

Answer: $3421_5 = 3\times125 + 4\times25 + 2\times5 + 1 = 586 = 1112_8$.

Question: Review Exercise 3, Question 1: (ii) 44332_5

Answer: $44332_5 = 4\times625 + 4\times125 + 3\times25 + 3\times5 + 2 = 3092 = 6024_8$.

Question: Review Exercise 3, Question 1: (iii) 1020304_5

Answer: $1020304_5 = 1\times15625 + 0 + 2\times625 + 0 + 3\times25 + 0 + 4 = 16954 = 41072_8$.

Question: Review Exercise 3, Question 2: Simplify the following and write the answer in the decimal system. (i) $12431_8 + 34211_5 + 101_2 + 289$

Answer: 5401 + 2431 + 5 + 289 = 8126.

Question: Review Exercise 3, Question 2: (ii) $40044_8 - 202023_5 - 101000_2$

Answer: 16420 - 6513 - 40 = 9867.

Question: Review Exercise 3, Question 2: (iii) $4021_8 \times 1204_5$

Answer: 2065 × 179 = 369635.

Question: Review Exercise 3, Question 2: (iv) $11000_2 \times 1100_5 \times 100_8$

Answer: 24 × 150 × 64 = 230400.

Question: Review Exercise 3, Question 3: Solve the following and convert the answer into base two, base five and base eight. (i) $408 + 4310_5 + 1011_2 + 110_8$

Answer: $408 + 580 + 11 + 72 = 1071 = 10000101111_2 = 13241_5 = 2057_8$.

Question: Review Exercise 3, Question 3: (ii) $753_8 - 4401_5 - 111101_2$

Answer: $491 - 601 - 61 = -171 = -10101011_2 = -1141_5 = -253_8$.

Question: Review Exercise 3, Question 3: (iii) $3412_5 \times 25_8 \times 110_2$

Answer: $482 \times 21 \times 6 = 60732 = 1110110100111100_2 = 3420412_5 = 166474_8$.

Question: Review Exercise 3, Question 3: (iv) $25 \times 24_8 \times 23_5 \times 100_2$

Answer: $25 \times 20 \times 13 \times 4 = 26000 = 110010110010000_2 = 1313000_5 =$

62620_8.

Question: Review Exercise 3, Question 4: Fill in the blanks: convert the decimal number 40 into a base-five and a base-eight number.

Answer: $40 = 130_5 = 50_8$.

Question: Review Exercise 3, Question 4: (ii) Convert the decimal number 82 into a base-five and a base-eight number.

Answer: $82 = 312_5 = 122_8$.

Question: Review Exercise 3, Question 4: (iii) Convert the base-five number 132 into a decimal and a base-eight number.

Answer: $132_5 = 42 = 52_8$.

Question: Review Exercise 3, Question 4: (iv) Convert the base-eight number 62 into a decimal and a base-five number.

Answer: $62_8 = 50 = 200_5$.

Question: Review Exercise 3, Question 4: (v) Convert the base-eight number 705 into a decimal and a base-five number.

Answer: $705_8 = 453 = 3303_5$.

Question: Review Exercise 3, Question 5: In the number 31204 the place value of the digit 2 is 200 in the ____ base system.

Answer: From $2 \times b^2 = 200$, $b^2 = 100$, so the base is 10 (the decimal system).

Question: Review Exercise 3, Question 5: (ii) The place value of the digit 2

is 128 in the ____ base system.

Answer: From $2 \times b^2 = 128$, $b^2 = 64$, so the base is 8 (the base-eight system).

Question: Review Exercise 3, Question 5: (iii) The place value of the digit 2 is 50 in the ____ base system.

Answer: From $2 \times b^2 = 50$, $b^2 = 25$, so the base is 5 (the base-five system).

Question: Review Exercise 3, Question 5: (iv) The place value of the digit 1 is 125 in the ____ base system.

Answer: From $1 \times b^3 = 125$, $b^3 = 125$, so the base is 5 (the base-five system).

Question: Review Exercise 3, Question 5: (v) The place value of the digit 1 is 512 in the ____ base system.

Answer: From $1 \times b^3 = 512$, $b^3 = 512$, so the base is 8 (the base-eight system).

Question: Review Exercise 3, Question 6: The place value of the digit 4 in the number 12034: (i) in base five, (ii) in base eight, (iii) in base ten.

Answer: $4 \times 5^0 = 4$ (base five), $4 \times 8^0 = 4$ (base eight), $4 \times 10^0 = 4$ (base ten).

Question: Review Exercise 3, Question 6: (ii) The place value of the digit 3: in base five, in base eight, in base ten.

Answer: $3 \times 5^1 = 15$ (base five), $3 \times 8^1 = 24$ (base eight), $3 \times 10^1 = 30$ (base ten).

Question: Review Exercise 3, Question 6: (iii) The place value of the digit 0: in base five, in base eight, in base ten.

Answer: $0\times5^2 = 0$ (base five), $0\times8^2 = 0$ (base eight), $0\times10^2 = 0$ (base ten).

Question: Review Exercise 3, Question 6: (iv) The place value of the digit 2: in base five, in base eight, in base ten.

Answer: $2\times5^3 = 250$ (base five), $2\times8^3 = 1024$ (base eight), $2\times10^3 = 2000$ (base ten).

Question: Review Exercise 3, Question 6: (v) The place value of the digit 1: in base five, in base eight, in base ten.

Answer: $1\times5^4 = 625$ (base five), $1\times8^4 = 4096$ (base eight), $1\times10^4 = 10000$ (base ten).

4. Financial Arithmetic

Question: Review Exercise IV, Question 1: Find the fourth proportional of the following. (i) 3, 4, 6

Answer: For the fourth proportional x, 3 : 4 :: 6 : x, so x = 4 × 6 ÷ 3 = 8.

Question: Review Exercise IV, Question 1: (ii) 7, 11, 14

Answer: x = 11 × 14 ÷ 7 = 22.

Question: Review Exercise IV, Question 1: (iii) 38, 19, 18

Answer: x = 19 × 18 ÷ 38 = 9.

Question: Review Exercise IV, Question 1: (iv) 5.6, 6.3, 4.8

Answer: x = 6.3 × 4.8 ÷ 5.6 = 5.4.

Question: Review Exercise IV, Question 2: The extreme terms of a proportion are 11 and 27. If one of the remaining terms is 9, find the other term.

Answer: Product of extremes = product of means, so 11 × 27 = 9 × x, giving x = 297 ÷ 9 = 33.

Question: Review Exercise IV, Question 3: The price of 25 articles is Rs. 300. What will be the price of 225 such articles?

Answer: Price = 300 × 225 ÷ 25 = Rs. 2,700.

Question: Review Exercise IV, Question 4: If 40 metres of cloth cost Rs. 1,000, how many metres of cloth can be bought for Rs. 1,315?

Answer: Cloth = 40 × 1,315 ÷ 1,000 = 52.6 metres.

Question: Review Exercise IV, Question 5: A ship covers 800 nautical miles in 40 days. Tell in how many days the ship will cover 2,440 nautical miles.

Answer: Days = 40 × 2,440 ÷ 800 = 122 days.

Question: Review Exercise IV, Question 6: A clock gains 3 minutes every twelve hours. Tell how many minutes it will gain in 14 days.

Answer: In 14 days there are 28 periods of twelve hours, so the gain = 28 × 3 = 84 minutes.

Question: Review Exercise IV, Question 7: A tower is 20 metres high and its shadow is 25 metres long. Tell what will be the height of another tower whose shadow at the same time is 75 metres.

Answer: Height = 20 × 75 ÷ 25 = 60 metres.

Question: Review Exercise IV, Question 8: 120 men can complete a work in 90 days. Tell how many men will be needed for the same work to be completed in 40 days.

Answer: Men = 120 × 90 ÷ 40 = 270 men are needed, i.e. 150 more than 120.

Question: Review Exercise IV, Question 9: In a fort there is food for 600 men for 27 days. If after 10 days 80 more men arrive in the fort, tell for how many days in total the food present there will suffice.

Answer: After 10 days the remaining food = 600 × 17 = 10,200 man-days.

Now there are 680 men, so the food will last a further 10,200 ÷ 680 = 15

days.

Total = 10 + 15 = 25 days.

Question: Review Exercise IV, Question 10: At a place there is food for 200 men for 20 days. Tell how many men must leave so that the food lasts 25 days.

Answer: Men needed = 200 × 20 ÷ 25 = 160, so 200 - 160 = 40 men must leave.

Question: Exercise 4.1, Question 1: The fare for a 12 kg load over 36 km is Rs. 24. Tell what will be the fare for an 18 kg load over 12 km.

Answer: Fare is directly proportional to both weight and distance: fare = 24 × (18 ÷ 12) × (12 ÷ 36) = 24 × 1.5 × (1 ÷ 3) = Rs. 12.

Question: Exercise 4.1, Question 2: There is ration for 4,800 men for 20 days at 1.5 kg per man. Tell for how many men the same ration will last 16 days at 1.0 kg per man.

Answer: Men are inversely proportional to both days and per-man rate: men = 4,800 × (20 ÷ 16) × (1.5 ÷ 1.0) = 4,800 × 1.25 × 1.5 = 9,000 men.

Question: Exercise 4.1, Question 3: A carpet 16 m long and 6 m wide costs Rs. 6,288. Tell what will be the price of a carpet 24 m long and 12 m wide.

Answer: Price is directly proportional to area: price = 6,288 × (24 ÷ 16) × (12 ÷ 6) = 6,288 × 1.5 × 2 = Rs. 18,864.

Question: Exercise 4.1, Question 4: 10 men can assemble 200 cycles in 8

days. Tell how many cycles 5 men will assemble in 16 days.

Answer: Cycles are directly proportional to men and days: cycles = 200 × (5 ÷ 10) × (16 ÷ 8) = 200 × 0.5 × 2 = 200 cycles.

Question: Exercise 4.1, Question 5: If 60 men use 40 kg in 12 days, in how many days will 90 men use 200 kg?

Answer: Days are directly proportional to weight and inversely to men: days = 12 × (200 ÷ 40) × (60 ÷ 90) = 12 × 5 × (2 ÷ 3) = 40 days.

Question: Exercise 4.1, Question 6: 14 men working 5 hours daily complete a work in 8 days. Tell how many hours daily must 35 men work for the work to be completed in 4 days.

Answer: Hours are inversely proportional to men and days: hours = 5 × (14 ÷ 35) × (8 ÷ 4) = 5 × 0.4 × 2 = 4 hours daily.

Question: Exercise 4.1, Question 7: A contractor hired 30 workers to complete a work in 30 days. But in 10 days only one-fourth of the work was done. Tell how many more workers must be hired so that the remaining work finishes within the fixed time.

Answer: In 10 days 30 workers did 300 man-days = 1/4 of the total work, so total work = 1,200 man-days.

Remaining work = 900 man-days, remaining time = 20 days, so workers needed = 900 ÷ 20 = 45.

There are already 30, so 15 more workers must be hired.

Question: Exercise 4.1, Question 8: The fare for carrying 1,200 kg over 200

km is Rs. 300. Tell what will be the fare for carrying 500 kg over 300 km.
Answer: Fare = 300 × (500 ÷ 1,200) × (300 ÷ 200) = 300 × (5 ÷ 12) × 1.5 = Rs. 187.50.

Question: Exercise 4.1, Question 9: 3 masons complete the flooring of a 100 m by 40 m floor in 16 days. Tell in how many days 4 masons will complete the flooring of a 160 m by 50 m floor.
Answer: Days are directly proportional to area and inversely to masons: days = 16 × (160 × 50) ÷ (100 × 40) × (3 ÷ 4) = 16 × 2 × 0.75 = 24 days.

Question: Exercise 4.1, Question 10: 14 men working 5 hours daily complete one-third of a work in 8 days. Tell how many hours daily must 35 men work for the remaining work to be completed in 4 days.
Answer: Work done = 14 × 5 × 8 = 560 man-hours = 1/3 of the total work, so total work = 1,680 man-hours and remaining work = 1,120 man-hours.

35 men × 4 days × h = 1,120, so h = 1,120 ÷ 140 = 8 hours daily.

Question: Exercise 4.2, Question 1: Maryam and Arif invested Rs. 60,000 and Rs. 80,000 respectively in a joint business. After one year the total profit was Rs. 21,000. Find each partner's profit.
Answer: Investment ratio = 60,000 : 80,000 = 3 : 4, sum of parts = 7.

Maryam's share = (3 ÷ 7) × 21,000 = Rs. 9,000.

Arif's share = (4 ÷ 7) × 21,000 = Rs. 12,000.

Question: Exercise 4.2, Question 2: Akram started a business with Rs. 70,000. After three months Aslam put in Rs. 40,000 and after four months Asghar also put in Rs. 40,000. After one year the total profit was Rs. 28,800. What was each partner's share of the profit?

Answer: Capital × time: Akram 70,000 × 12 = 840,000;

Aslam 40,000 × 9 = 360,000;

Asghar 40,000 × 8 = 320,000.

Ratio = 21 : 9 : 8, sum = 38.

Akram = (21 ÷ 38) × 28,800 = Rs. 15,915.79;

Aslam = (9 ÷ 38) × 28,800 = Rs. 6,821.05;

Asghar = (8 ÷ 38) × 28,800 = Rs. 6,063.16.

[note: the book's numbers do not give round shares, so answers are given to two decimal places.]

Question: Exercise 4.2, Question 3: Ali, Zain and Asad invested Rs. 30,000, Rs. 30,000 and Rs. 24,000 respectively in a shop. After 5 months Zain withdrew Rs. 18,000 and after 10 months the business ended. How much will each partner get out of the total profit of Rs. 37,500?

Answer: Capital × time: Ali 30,000 × 10 = 300,000;

Zain 30,000 × 5 + 12,000 × 5 = 210,000;

Asad 24,000 × 10 = 240,000.

Ratio = 10 : 7 : 8, sum = 25.

Ali = (10 ÷ 25) × 37,500 = Rs. 15,000;

Zain = (7 ÷ 25) × 37,500 = Rs. 10,500;

Asad = (8 ÷ 25) × 37,500 = Rs. 12,000.

Question: Exercise 4.2, Question 4: Ali and Usman invested different amounts for different periods in a shop. If the total profit of both is Rs. 60,000, fill in the blanks in the table below. (i) Ali Rs. 50,000, 12 months; Usman Rs. 25,000, 12 months.

Answer: This row is filled in as the example: capital × time ratio = 600,000 : 300,000 = 2 : 1 (Ali : Usman), so Ali gets Rs. 40,000 and Usman gets Rs. 20,000.

Question: Exercise 4.2, Question 4: (ii) Ali Rs. 50,000, 12 months; Usman Rs. 75,000, 12 months.

Answer: Capital × time ratio = 600,000 : 900,000 = 2 : 3 (Ali : Usman), so Ali gets (2 ÷ 5) × 60,000 = Rs. 24,000 and Usman gets (3 ÷ 5) × 60,000 = Rs. 36,000.

Question: Exercise 4.2, Question 4: (iii) Ali Rs. 25,000, 12 months; Usman Rs. 25,000, 6 months.

Answer: Capital × time ratio = 300,000 : 150,000 = 2 : 1 (Ali : Usman), so Ali gets (2 ÷ 3) × 60,000 = Rs. 40,000 and Usman gets (1 ÷ 3) × 60,000

= Rs. 20,000.

Question: Exercise 4.2, Question 4: (iv) Ali Rs. 50,000, 6 months; Usman's capital is not given, 6 months.

Answer: [note: the book does not give Usman's capital for this row, so the ratio and profit shares cannot be worked out.

This row is printed incomplete in the book.]

Question: Exercise 4.2, Question 5: A man died leaving property worth Rs. 1,800,000. His heirs are 6 daughters and 2 sons. How much will each get?

Answer: A son gets double a daughter's share: total shares = 2 × 2 + 6 × 1 = 10, each share = 1,800,000 ÷ 10 = Rs. 180,000.

Each son gets Rs. 360,000 and each daughter gets Rs. 180,000.

Question: Exercise 4.2, Question 6: Aslam died leaving property worth Rs. 1,660,000. His heirs are 2 wives, 2 sons and 1 daughter. Tell how much each heir will get, given that Rs. 60,000 was spent on burial.

Answer: Remaining property = 1,660,000 - 60,000 = Rs. 1,600,000.

Wives' share = 1/8 × 1,600,000 = Rs. 200,000, i.e. Rs. 100,000 per wife.

In the remaining Rs. 1,400,000 the children's shares: sons 2 shares each and daughter 1 share, total 5 shares, each share = Rs. 280,000.

Each son gets Rs. 560,000 and the daughter gets Rs. 280,000.

Question: Exercise 4.2, Question 7: Akram left property worth Rs. 1,520,000. The heirs are one mother, one wife (widow), 3 sons and 4 daughters. Rs. 30,000 was spent on his burial and he also owed Rs. 50,000 debt. Tell how much each heir will get.

Answer: Remaining property = 1,520,000 - 30,000 - 50,000 = Rs. 1,440,000.

Mother = 1/6 × 1,440,000 = Rs. 240,000.

Wife = 1/8 × 1,440,000 = Rs. 180,000.

In the remaining Rs. 1,020,000 the children's shares: 3 sons (6 shares) + 4 daughters (4 shares) = 10 shares, each share = Rs. 102,000.

Each son gets Rs. 204,000 and each daughter gets Rs. 102,000.

Question: Exercise 4.2, Question 8: Mrs. Abdur Rehman died, leaving her father, mother, son and husband as heirs. She left an inheritance of Rs. 480,000. Tell how much each will get.

Answer: Husband = 1/4 × 480,000 = Rs. 120,000.

Mother = 1/6 × 480,000 = Rs. 80,000.

Father = 1/6 × 480,000 = Rs. 80,000.

The son gets the remainder = 480,000 - 120,000 - 80,000 - 80,000 = Rs. 200,000.

Question: Exercise 4.2, Question 9: At the time of a man's death there are only children (parents are not alive). Fill in the blanks below for the inheritance of Rs. 800,000 in different combinations of children. (i) 1 son, 2 daughters.

Answer: This row is filled in as the example: total shares = 2 + 2 = 4, each share = Rs. 200,000.

Each son gets Rs. 400,000 and each daughter gets Rs. 200,000.

Question: Exercise 4.2, Question 9: (ii) 2 sons, 1 daughter.

Answer: Total shares = 4 + 1 = 5, each share = 800,000 ÷ 5 = Rs. 160,000.

Each son gets Rs. 320,000 and the daughter gets Rs. 160,000.

Question: Exercise 4.2, Question 9: (iii) 2 sons, 4 daughters.

Answer: Total shares = 4 + 4 = 8, each share = Rs. 100,000.

Each son gets Rs. 200,000 and each daughter gets Rs. 100,000.

Question: Exercise 4.2, Question 9: (iv) 1 son, 6 daughters.

Answer: The sons' total shares are 2 and the number of daughters is 6: total shares = 2 + 6 = 8, each share = Rs. 100,000.

The son gets Rs. 200,000 and each daughter gets Rs. 100,000.

Question: Exercise 4.2, Question 9: (v) 3 sons, 4 daughters.

Answer: The daughters' total shares are 4 and the number of sons is 3:

sons' total shares = 6, total shares = 6 + 4 = 10, each share = Rs. 80,000.

Each son gets Rs. 160,000 and each daughter gets Rs. 80,000.

Question: Exercise 4.2, Question 9: (vi) Each daughter gets Rs. 100,000; the sons' total shares are 2.

Answer: Each share = Rs. 100,000, so total shares = 800,000 ÷ 100,000 = 8.

The sons' shares are 2, so there is 1 son, and the daughters' shares are 6, so there are 6 daughters.

Each son gets Rs. 200,000.

Question: Exercise 4.3, Question 1: Convert. (i) 90,000 Chinese Yuan into rupees (1 Chinese Yuan = Rs. 16.25 Pakistani).

Answer: 90,000 × 16.25 = Rs. 1,462,500 Pakistani.

Question: Exercise 4.3, Question 1: (ii) 50,000 Pakistani rupees into Indian rupees (1 Indian rupee = Rs. 1.60 Pakistani).

Answer: 50,000 ÷ 1.60 = 31,250 Indian rupees.

Question: Exercise 4.3, Question 1: (iii) 110 British pounds into Pakistani rupees (1 British pound = Rs. 139 Pakistani).

Answer: 110 × 139 = Rs. 15,290 Pakistani.

Question: Exercise 4.3, Question 1: (iv) 250 Euros into Pakistani rupees (1 Euro = Rs. 126.0 Pakistani).

Answer: 250 × 126 = Rs. 31,500 Pakistani.

Question: Exercise 4.3, Question 1: (v) 64,200 Pakistani rupees into Turkish Lira (1 Turkish Lira = Rs. 42.80 Pakistani).

Answer: 64,200 ÷ 42.80 = 1,500 Turkish Lira.

Question: Exercise 4.3, Question 2: A Pakistani earns 9,000 Riyals per month in Saudi Arabia, of which he spends 2,500 Riyals every month. Tell his savings in Pakistani rupees. (1 Saudi Riyal = Rs. 27.15 Pakistani)

Answer: Monthly savings = 9,000 - 2,500 = 6,500 Riyals.

Savings in Pakistani rupees = 6,500 × 27.15 = Rs. 176,475.

Question: Exercise 4.3, Question 3: Kamal has Rs. 38,400 Pakistani and 4,000 Chinese Yuan, in exchange for which he wants to take American dollars. If 1 US dollar = Rs. 103.4 Pakistani and 1 Chinese Yuan = Rs. 16.25 Pakistani, tell how many US dollars he will get.

Answer: Total Pakistani rupees = 38,400 + 4,000 × 16.25 = 38,400 + 65,000 = Rs. 103,400.

Dollars = 103,400 ÷ 103.4 = 1,000 US dollars.

Question: Exercise 4.3, Question 4: Convert 48,000 Pakistani rupees into Indian rupees.

Answer: 48,000 ÷ 1.60 = 30,000 Indian rupees.

Question: Exercise 4.3, Question 5: Convert 48,000 Indian rupees into Pakistani rupees.

Answer: 48,000 × 1.60 = Rs. 76,800 Pakistani.

Question: Exercise 4.3, Question 6: A fridge costs Rs. 77,500 in the Pakistani market. The same fridge is available in Japan for 70,000 Yen. Importing it from Japan to Pakistan costs Rs. 5,000 freight and Rs. 3,000 customs duty. Tell what the total cost of importing that fridge from Japan will be, if the exchange rate is 1 Yen = Rs. 0.946 Pakistani.

Answer: Fridge price = 70,000 × 0.946 = Rs. 66,220.

Total cost = 66,220 + 5,000 + 3,000 = Rs. 74,220, which is Rs. 3,280 less than the local price of Rs. 77,500.

Question: Exercise 4.4, Question 1: How much profit will be earned on Rs. 50,000 in 6 years at the rate of 5%?

Answer: Mark-up I = P × R × T = 50,000 × (5 ÷ 100) × 6 = Rs. 15,000.

Question: Exercise 4.4, Question 2: Asad took a loan of Rs. 25,000 from a bank for four years. Find the bank's mark-up at the annual rate of 8%.

Answer: Mark-up = 25,000 × (8 ÷ 100) × 4 = Rs. 8,000.

Question: Exercise 4.4, Question 3: Naseem got Rs. 27,000 profit in 3 years at the annual mark-up rate of 12%. Tell how much money he had invested.

Answer: P = I ÷ (R × T) = 27,000 × 100 ÷ (12 × 3) = Rs. 75,000.

So Naseem had invested Rs. 75,000.

Question: Exercise 4.4, Question 4: At what annual mark-up rate will Rs.

68,000 become Rs. 90,440 after 11 years?

Answer: Mark-up = 90,440 - 68,000 = Rs. 22,440.

R = I × 100 ÷ (P × T) = 22,440 × 100 ÷ (68,000 × 11) = 3%.

Question: Exercise 4.4, Question 5: In how much time will Rs. 31,000 at a 6% mark-up rate yield Rs. 5,332 profit?

Answer: T = I × 100 ÷ (P × R) = 5,332 × 100 ÷ (31,000 × 6) = 533,200 ÷ 186,000 = 2.87 years, i.e. about 2 years and 10.4 months.

Question: Exercise 4.4, Question 6: At what mark-up rate will an amount double in 20 years?

Answer: For doubling, I = P, so R = P × 100 ÷ (P × 20) = 5%.

Question: Exercise 4.4, Question 7: In how much time will an amount double at an 8% mark-up rate?

Answer: T = P × 100 ÷ (P × 8) = 100 ÷ 8 = 12.5 years.

Question: Exercise 4.4, Question 8: Complete the table below. (i) Principal Rs. 20,000, time 5 years, mark-up rate 4%.

Answer: Mark-up = 20,000 × (4 ÷ 100) × 5 = Rs. 4,000.

Question: Exercise 4.4, Question 8: (ii) Principal Rs. 98,000, time 6 years, mark-up rate 6.5%.

Answer: Mark-up = 98,000 × (6.5 ÷ 100) × 6 = 98,000 × 0.39 = Rs. 38,220.

Question: Exercise 4.4, Question 8: (iii) Mark-up Rs. 10,500, principal Rs.

21,000, mark-up rate 5%.

Answer: Time = 10,500 × 100 ÷ (21,000 × 5) = 10 years.

Question: Exercise 4.4, Question 8: (iv) Mark-up Rs. 2,100, time 6 years, mark-up rate 7%.

Answer: Principal = 2,100 × 100 ÷ (7 × 6) = Rs. 5,000.

Question: Exercise 4.4, Question 8: (v) Mark-up Rs. 1,740, principal Rs. 5,800, time 3 years.

Answer: Rate = 1,740 × 100 ÷ (5,800 × 3) = 10%.

Question: Exercise 4.4, Question 9: Mahir bought an air conditioner with a cash price of Rs. 50,000 on credit. To get it on credit he must pay 20% in advance and pay 10% mark-up on the balance. If he must repay the loan in 2½ years, find (i) the monthly instalment (ii) how much total amount he will pay.

Answer: Advance = 20% × 50,000 = Rs. 10,000.

Balance principal = Rs. 40,000.

Mark-up = 40,000 × (10 ÷ 100) × 2.5 = Rs. 10,000.

Amount to repay = 40,000 + 10,000 = Rs. 50,000, over 30 months.

(i) Monthly instalment = 50,000 ÷ 30 = Rs. 1,666.67.

(ii) Total amount paid = 10,000 + 50,000 = Rs. 60,000.

Question: Exercise 4.4, Question 10: A company took a house on lease for 4 years. It paid Rs. 80,000 in advance. Thereafter it must pay Rs. 20,000 monthly rent. After two years the company agreed to a 5% rent increase. Tell how much total amount the owner received.

Answer: First 2 years' rent = 24 × 20,000 = Rs. 480,000.

Increased rent = 20,000 × 1.05 = Rs. 21,000 monthly, so next 2 years' rent = 24 × 21,000 = Rs. 504,000.

The owner received a total of 80,000 + 480,000 + 504,000 = Rs. 1,064,000.

Question: Exercise 4.4, Question 11: Arshad bought a flat worth Rs. 200,000 on credit. Complete the table below for different amounts, mark-up rates and periods. (i) Advance 20%, mark-up rate 10%, period 2 years.

Answer: This row is filled in as the example: advance = Rs. 40,000, balance = Rs. 160,000, mark-up = Rs. 32,000, amount to repay = Rs. 192,000.

Monthly instalment = 192,000 ÷ 24 = Rs. 8,000.

Total amount = 40,000 + 192,000 = Rs. 232,000.

Question: Exercise 4.4, Question 11: (ii) Advance 20%, mark-up rate 5%, period 2 years.

Answer: Advance = Rs. 40,000, balance = Rs. 160,000, mark-up = 160,000 × (5 ÷ 100) × 2 = Rs. 16,000, amount to repay = Rs. 176,000.

Monthly instalment = 176,000 ÷ 24 = Rs. 7,333.33.

Total amount = 40,000 + 176,000 = Rs. 216,000.

Question: Exercise 4.4, Question 11: (iii) Advance 20%, mark-up rate 10%, period 4 years.

Answer: Advance = Rs. 40,000, balance = Rs. 160,000, mark-up = 160,000 × (10 ÷ 100) × 4 = Rs. 64,000, amount to repay = Rs. 224,000.

Monthly instalment = 224,000 ÷ 48 = Rs. 4,666.67.

Total amount = 40,000 + 224,000 = Rs. 264,000.

Question: Exercise 4.4, Question 11: (iv) Advance 40%, mark-up rate 10%, period 2 years.

Answer: Advance = Rs. 80,000, balance = Rs. 120,000, mark-up = 120,000 × (10 ÷ 100) × 2 = Rs. 24,000, amount to repay = Rs. 144,000.

Monthly instalment = 144,000 ÷ 24 = Rs. 6,000.

Total amount = 80,000 + 144,000 = Rs. 224,000.

Question: Exercise 4.5, Question 1: Complete the table below. (i) Cost price Rs. 1,050, selling price Rs. 1,155.

Answer: Profit = 1,155 - 1,050 = Rs. 105.

Profit percent = 105 ÷ 1,050 × 100 = 10%.

Question: Exercise 4.5, Question 1: (ii) Cost price Rs. 1,665, profit Rs. 333.

Answer: Selling price = 1,665 + 333 = Rs. 1,998.

Profit percent = 333 ÷ 1,665 × 100 = 20%.

Question: Exercise 4.5, Question 1: (iii) Cost price Rs. 6,000, profit percent 22%.

Answer: Profit = 6,000 × 22 ÷ 100 = Rs. 1,320.

Selling price = 6,000 + 1,320 = Rs. 7,320.

Question: Exercise 4.5, Question 1: (iv) Selling price Rs. 1,250, profit percent 25%.

Answer: Cost price = 1,250 × 100 ÷ 125 = Rs. 1,000.

Profit = 1,250 - 1,000 = Rs. 250.

Question: Exercise 4.5, Question 2: A shopkeeper bought 160 chairs at Rs. 900 each. He sold 60 of them at Rs. 1,000 each and the remaining chairs at Rs. 960 each. Find his profit or loss percent.

Answer: Total cost price = 160 × 900 = Rs. 144,000.

Total selling price = 60 × 1,000 + 100 × 960 = 60,000 + 96,000 = Rs. 156,000.

Profit = Rs. 12,000.

Profit percent = 12,000 ÷ 144,000 × 100 = 8.33%.

Question: Exercise 4.5, Question 3: Jalil bought 4 plots for Rs. 900,000 and sold them for Rs. 225,000, Rs. 250,000, Rs. 300,000 and Rs. 215,000 respectively. Find his profit or loss percent.

Answer: Total selling price = 225,000 + 250,000 + 300,000 + 215,000 = Rs. 990,000.

Profit = 990,000 - 900,000 = Rs. 90,000.

Profit percent = 90,000 ÷ 900,000 × 100 = 10%.

Question: Exercise 4.5, Question 4: Khalil bought a carton of 250 eggs for Rs. 2,200. Of these 20 turned out broken and 10 rotten. The remaining eggs were sold at Rs. 11 per egg. Find his profit or loss percent.

Answer: Good eggs = 250 - 20 - 10 = 220.

Selling price = 220 × 11 = Rs. 2,420.

Profit = 2,420 - 2,200 = Rs. 220.

Profit percent = 220 ÷ 2,200 × 100 = 10%.

Question: Exercise 4.5, Question 5: An article has a marked price of Rs. 10,000. It was sold at successive discount rates. Complete the table below. (i) Successive discount rates 15% and 5%.

Answer: This row is filled in as the example: first 5% discount = Rs. 500, then 15% discount = 15% × 9,500 = Rs. 1,425.

Total discount = Rs. 1,925.

Selling price = 10,000 - 1,925 = Rs. 8,075.

Net discount rate = 1,925 ÷ 10,000 × 100 = 19.25%.

Question: Exercise 4.5, Question 5: (ii) Successive discount rates 5% and 5%.

Answer: First 5% discount = Rs. 500, then 5% discount = 5% × 9,500 = Rs. 475.

Total discount = Rs. 975.

Selling price = Rs. 9,025.

Net discount rate = 9.75%.

Question: Exercise 4.5, Question 5: (iii) Successive discount rates 10% and 15%.

Answer: First 10% discount = Rs. 1,000, then 15% discount = 15% × 9,000 = Rs. 1,350.

Total discount = Rs. 2,350.

Selling price = Rs. 7,650.

Net discount rate = 23.5%.

Question: Exercise 4.5, Question 5: (iv) Successive discount rates 12% and 18%.

Answer: First 12% discount = Rs. 1,200, then 18% discount = 18% × 8,800 = Rs. 1,584.

Total discount = Rs. 2,784.

Selling price = Rs. 7,216.

Net discount rate = 27.84%.

Question: Exercise 4.5, Question 6: A bicycle has a marked price of Rs. 8,500. Of two shopkeepers, one gives 10% discount and the other gives 5% discount. Which of the two shopkeepers has the lower selling price?

Answer: First shopkeeper's selling price = 8,500 × 0.90 = Rs. 7,650.

Second shopkeeper's selling price = 8,500 × 0.95 = Rs. 8,075.

So the shopkeeper giving 10% discount has the lower selling price (Rs. 425 less).

Question: Exercise 4.5, Question 7: Find a single discount percent equal to the three successive discounts 20%, 10% and 5%.

Answer: The fraction remaining after the three discounts = 0.80 × 0.90 × 0.95 = 0.684, so the equal single discount = (1 - 0.684) × 100 = 31.6%.

Question: Exercise 4.5, Question 8: A craftsman spends Rs. 5,000 in making an article. He sells it at 20% profit. The buyer resells the same article at 25% profit. Find its final selling price.

Answer: First selling price = 5,000 × 1.20 = Rs. 6,000.

Final selling price = 6,000 × 1.25 = Rs. 7,500.

Question: Exercise 4.6, Question 1: If the premium amount for life insurance is charged as follows: annual premium = 4.75% of the policy amount + 0.2% policy fee on the policy amount, half-yearly = 52% of the annual premium, quarterly = 27% of the annual premium, monthly = 9% of the annual premium, complete the table below. (i) Policy amount Rs. 100,000.

Answer: Annual premium = 4.95% × 100,000 = Rs. 4,950.

Half-yearly = 52% × 4,950 = Rs. 2,574.

Quarterly = 27% × 4,950 = Rs. 1,336.50.

Monthly = 9% × 4,950 = Rs. 445.50.

Question: Exercise 4.6, Question 1: (ii) Policy amount Rs. 150,000.

Answer: Annual premium = 4.95% × 150,000 = Rs. 7,425.

Half-yearly = 52% × 7,425 = Rs. 3,861.

Quarterly = 27% × 7,425 = Rs. 2,004.75.

Monthly = 9% × 7,425 = Rs. 668.25.

Question: Exercise 4.6, Question 1: (iii) Policy amount Rs. 300,000.

Answer: Annual premium = 4.95% × 300,000 = Rs. 14,850.

Half-yearly = 52% × 14,850 = Rs. 7,722.

Quarterly = 27% × 14,850 = Rs. 4,009.50.

Monthly = 9% × 14,850 = Rs. 1,336.50.

Question: Exercise 4.6, Question 2: Hamid took life insurance for Rs. 300,000. If the annual premium rate is 5.2% and the policy fee is 0.25%, find his first premium amount.

Answer: First premium = (5.2 + 0.25)% × 300,000 = 5.45% × 300,000 = Rs. 16,350.

Question: Exercise 4.6, Question 3: Bashir insured his car for 3 years at the annual rate of 2% for Rs. 500,000. If the car's value decreases by 5% each year, tell how much total premium he paid for this insurance.

Answer: First year's premium = 2% × 500,000 = Rs. 10,000.

Second year's value = Rs. 475,000, premium = Rs. 9,500.

Third year's value = Rs. 451,250, premium = Rs. 9,025.

Total premium = 10,000 + 9,500 + 9,025 = Rs. 28,525.

Question: Exercise 4.6, Question 4: A vehicle was insured at 4.0% of its value for 3 years. If the rate of decrease in value is 10% per year, complete the table below. (i) Vehicle value 100 thousand rupees.

Answer: This row is filled in as the example: first annual premium = 4% × 100 = 4.00 thousand rupees;

second = 4% × 90 = 3.60 thousand rupees;

third = 4% × 81 = 3.24 thousand rupees.

Total premium = 10.84 thousand rupees.

Question: Exercise 4.6, Question 4: (ii) Vehicle value 250 thousand rupees, first annual premium 10.00 thousand rupees.

Answer: Second year's value = 225 thousand rupees, premium = 9.00 thousand rupees.

Third year's value = 202.5 thousand rupees, premium = 8.10 thousand rupees.

Total premium = 10.00 + 9.00 + 8.10 = 27.10 thousand rupees.

Question: Exercise 4.6, Question 4: (iii) Vehicle value 300 thousand rupees.

Answer: First premium = 4% × 300 = 12.00 thousand rupees.

Second = 4% × 270 = 10.80 thousand rupees.

Third = 4% × 243 = 9.72 thousand rupees.

Total = 32.52 thousand rupees.

Question: Exercise 4.6, Question 4: (iv) Third annual premium 16.20 thousand rupees.

Answer: Third premium = 4% × 0.81 × value = 16.20, so value = 16.20 ÷ 0.0324 = 500 thousand rupees.

First premium = 20.00 thousand rupees, second = 18.00 thousand rupees.

Total = 54.20 thousand rupees.

Question: Exercise 4.6, Question 4: (v) Second annual premium 36.00 thousand rupees.

Answer: Second premium = 4% × 0.90 × value = 36.00, so value = 36.00 ÷ 0.036 = 1,000 thousand rupees.

First premium = 40.00 thousand rupees, third = 4% × 810 = 32.40 thousand rupees.

Total = 108.40 thousand rupees.

Question: Exercise 4.6, Question 5: A woman insured herself for one year at the rate of 5%. If she paid Rs. 25,000 premium, what is her insured amount (no policy fee applies)?

Answer: Insured amount = 25,000 × 100 ÷ 5 = Rs. 500,000.

Question: Exercise 4.7, Question 1: Amjad's monthly salary is Rs. 75,000. Find his annual income tax.

Answer: Annual income = 75,000 × 12 = Rs. 900,000.

By the fourth slab of the tax table: tax = 14,500 + 10% × (900,000 - 750,000) = 14,500 + 15,000 = Rs. 29,500.

Question: Exercise 4.7, Question 2: Shazia earns Rs. 495,000 in a year. How much annual income tax will be levied on her, given that she has paid Rs. 40,000 as Zakat?

Answer: Taxable income = 495,000 - 40,000 = Rs. 455,000.

By the second slab: tax = 2% × (455,000 - 400,000) = 2% × 55,000 = Rs. 1,100.

Question: Exercise 4.7, Question 3: A man earns Rs. 7,000,000 in a year. Rs. 120,000 tax has already been deducted and he has also paid Rs. 130,000 as Zakat. Tell how much remaining income tax will be levied on him at the end of the financial year.

Answer: Taxable income = 7,000,000 - 130,000 = Rs. 6,870,000.

By the eleventh slab: total tax = 597,000 + 27.5% × (6,870,000 - 4,000,000) = 597,000 + 789,250 = Rs. 1,386,250.

Remaining tax = 1,386,250 - 120,000 = Rs. 1,266,250.

Question: Exercise 4.7, Question 4: Ismail's monthly salary is Rs. 200,000. Tell how much income tax was levied at the end of the year, given that Rs. 80,000 Zakat and Rs. 20,000 wealth tax have also been paid.

Answer: Annual income = 200,000 × 12 = Rs. 2,400,000.

Taxable income = 2,400,000 - 80,000 = Rs. 2,320,000.

By the seventh slab: income tax = 137,000 + 17.5% × (2,320,000 - 1,800,000) = 137,000 + 91,000 = Rs. 228,000.

[note: wealth tax is a tax separate from income tax, so it was not deducted from the income tax.]

Question: Exercise 4.7, Question 5: Sufyan paid Rs. 497,000 income tax in a year. Tell his monthly income.

Answer: By the tenth slab: 472,000 + 25% × (annual income - 3,500,000) = 497,000, so annual income - 3,500,000 = 25,000 ÷ 0.25 = 100,000, giving annual income = Rs. 3,600,000.

Monthly income = 3,600,000 ÷ 12 = Rs. 300,000.

Question: Exercise 4.7, Question 6: Bushra paid Rs. 289,500 income tax at the end of the year. She had also paid Rs. 130,000 wealth tax and Rs. 120,000 Zakat. Find her annual income.

Answer: By the eighth slab: 259,500 + 20% × (taxable income - 2,500,000) = 289,500, so taxable income - 2,500,000 = 30,000 ÷ 0.20 = 150,000, giving taxable income = Rs. 2,650,000.

Annual income = 2,650,000 + 120,000 (Zakat) = Rs. 2,770,000.

[note: wealth tax is a tax separate from income tax, so it was not included in the income tax.]

Question: Exercise 4.7, Question 7: Complete the table below for different income limits. (i) Annual income 578 thousand rupees, tax slab number 3, minimum tax Rs. 2,000, lower limit 500 thousand rupees, income above lower limit 78 thousand rupees, tax rate 5%.

Answer: This row is filled in as the example: total income tax = 2,000 +

5% × 78,000 = 2,000 + 3,900 = Rs. 5,900.

Question: Exercise 4.7, Question 7: (ii) Annual income 480 thousand rupees.

Answer: Tax slab number 2, minimum tax Rs. 0, lower limit 400 thousand rupees, income above lower limit 80 thousand rupees, tax rate 2%, total income tax = 2% × 80,000 = Rs. 1,600.

Question: Exercise 4.7, Question 7: (iii) Annual income 900 thousand rupees, minimum tax Rs. 14,500.

Answer: Tax slab number 4, lower limit 750 thousand rupees, income above lower limit 150 thousand rupees, tax rate 10%, total income tax = 14,500 + 10% × 150,000 = 14,500 + 15,000 = Rs. 29,500.

Question: Exercise 4.7, Question 7: (iv) Annual income 6,000 thousand rupees, tax slab number 11, tax rate 27.5%, income above lower limit 2,000 thousand rupees.

Answer: Minimum tax Rs. 597,000, lower limit 4,000 thousand rupees, total income tax = 597,000 + 27.5% × 2,000,000 = 597,000 + 550,000 = Rs. 1,147,000.

Review Exercise 4, Question 1: Circle the correct answer. (i) Giving a concession on the marked amount is called:

(a) tax

(b) profit

(c) discount ✓

(d) loss.

Review Exercise 4, Question 1: (ii) The relation between two or more proportions is called:

(a) direct proportion

(b) auto proportion

(c) inverse proportion

✓ (d) compound proportion.

Review Exercise 4, Question 1: (iii) ATM stands for:

(a) Account Tele Machine

(b) Account Transfer Machine

✓ (c) Automated Teller Machine

(d) Automatic Transfer Machine.

Review Exercise 4, Question 1: (iv) The rate of Zakat is:

(a) 5%

✓ (b) 2.5%

(c) 10%

(d) 1.0%.

Review Exercise 4, Question 1: (v) The cost price of a book is Rs. 300. If a 20% discount is given on it, what will be the selling price?

✓ (a) Rs. 240

(b) Rs. 260

(c) Rs. 230

(d) Rs. 280.

Review Exercise 4, Question 1: (vi) If the exchange rate is 1 Saudi Riyal = Rs. 27.15 Pakistani, then Saudi Riyals _____ = Rs. 54,300 Pakistani:

(a) 1,900

(b) 2,100

✓ (c) 2,000

(d) 2,010.

Review Exercise 4, Question 1: (vii) The tax levied on income is called:

(a) Ushr

(b) Jazoo

✓ (c) income tax

(d) property tax.

Review Exercise 4, Question 1: (viii) A man died. His property is divided among one widow, three daughters and two sons. Each son's share in the property will be:

(a) 1/8

✓ (b) 1/4

(c) 1/6

(d) 1/3.

Review Exercise 4, Question 1: (ix) An amount will double in _____ years at an 8% mark-up rate: (a) 40 (b) 30 (c) 20 (d) 25.

Answer: the book.]

Review Exercise 4, Question 1: (x) The single discount equal to the two successive discounts 10% and 5%:

(a) 15%

✓ (b) 14.5%

(c) 15.5%

(d) 14%.

Question: Review Exercise 4, Question 2: A company's profit is divided among four partners A, B, C, D in the ratio 3:4:5:6. If D gets Rs. 5,400 as his share, tell how much A and B will get separately. What was the total profit?

Answer: D's 6 parts = Rs. 5,400, so one part = Rs. 900.

A = 3 × 900 = Rs. 2,700, B = 4 × 900 = Rs. 3,600, C = 5 × 900 = Rs. 4,500.

Total profit = 18 × 900 = Rs. 16,200.

Question: Review Exercise 4, Question 3: If 30 men use 20 kg of wheat in 5 days, tell in how many days 15 men will use 320 kg of wheat.

Answer: Days = 5 × (320 ÷ 20) × (30 ÷ 15) = 5 × 16 × 2 = 160 days.

Question: Review Exercise 4, Question 4: Which discount is greater: a one-time 25% or the successive discounts 15% and 10%?

Answer: Successive discount = (1 - 0.85 × 0.90) × 100 = (1 - 0.765) × 100 = 23.5%, which is less than 25%.

So the one-time 25% discount is greater.

Question: Review Exercise 4, Question 5: Akbar bought a car for Rs. 1,000,000. He sold it to Asghar at 20% profit. Asghar sold the same car to Anwar at 20% loss. Tell at what price Anwar got the car.

Answer: Selling price to Asghar = 1,000,000 × 1.20 = Rs. 1,200,000.

Selling price to Anwar = 1,200,000 × 0.80 = Rs. 960,000.

Question: Review Exercise 4, Question 6: Aslam got his car insured by paying Rs. 40,500 as the first premium. If the annual premium rate is 3.85% and the service fee is Rs. 2,000, find the insured amount of the car.

Answer: Subtracting the service fee from the premium: 40,500 - 2,000 = Rs. 38,500.

Insured amount = 38,500 × 100 ÷ 3.85 = Rs. 1,000,000.

Question: Review Exercise 4, Question 7: A shopkeeper marks his prepared goods 20% above the cost price and sells at some discount. If an item's cost price is Rs. 2,500 and its selling price is Rs. 2,700, what percent discount was given to the customer?

Answer: Marked price = 2,500 × 1.20 = Rs. 3,000.

Discount = 3,000 - 2,700 = Rs. 300.

Discount percent = 300 ÷ 3,000 × 100 = 10%.

Question: Review Exercise 4, Question 8: What is Ahmad's annual income if he has paid Rs. 8,900 income tax and Rs. 20,000 Zakat?

Answer: By the third slab: 2,000 + 5% × (taxable income - 500,000) = 8,900, so taxable income - 500,000 = 6,900 ÷ 0.05 = 138,000, giving taxable income = Rs. 638,000.

Annual income = 638,000 + 20,000 (Zakat) = Rs. 658,000.

5. Polynomials

Question: Exercise 5.1, Question 1: In the following algebraic expressions, identify the constant, the variable and the literal coefficient. (i) 3x + 1

Answer: Constant = 1;

variable = x;

no literal coefficient (3 is the numerical coefficient).

Question: Exercise 5.1, Question 1: (ii) 0

Answer: Constant = 0;

no variable;

no literal coefficient.

Question: Exercise 5.1, Question 1: (iii) $2a^2 + 1/3\ b - 3$

Answer: Constant = −3;

variables = a and b;

no literal coefficient.

Question: Exercise 5.1, Question 1: (iv) 3l − 2m

Answer: No constant (0);

variables = l and m;

no literal coefficient.

Question: Exercise 5.1, Question 1: (v) $a + b^2 - 2c + 4$

Answer: Constant = 4;

variables = a, b and c;

no literal coefficient.

Question: Exercise 5.1, Question 1: (vi) $px + qy + r$

Answer: Constant = r (a literal constant);

variables = x and y;

literal coefficients = p and q.

Question: Exercise 5.1, Question 1: (vii) $-5x + 9y - 4$

Answer: Constant = −4;

variables = x and y;

no literal coefficient.

Question: Exercise 5.1, Question 1: (viii) $3^2p - 2^2q$

Answer: $3^2p - 2^2q = 9p - 4q$;

no constant;

variables = p and q;

no literal coefficient.

Question: Exercise 5.1, Question 1: (ix) $y^2 - y - 1$

Answer: Constant = −1;

variable = y;

no literal coefficient.

Question: Exercise 5.1, Question 1: (x) $\sqrt{3}\,a - 9ab + \sqrt{5}$

Answer: Constant = $\sqrt{5}$;

variables = a and b;

no literal coefficient ($\sqrt{3}$ is a numerical coefficient).

Question: Exercise 5.1, Question 1: (xi) $7x^2 - 2x + 3$

Answer: Constant = 3;

variable = x;

no literal coefficient.

Question: Exercise 5.1, Question 1: (xii) −8

Answer: Constant = −8;

no variable;

no literal coefficient.

Question: Exercise 5.1, Question 2: Which of the following algebraic expressions are polynomials? (i) −1

Answer: Yes, it is a polynomial — a constant polynomial.

Question: Exercise 5.1, Question 2: (ii) 3/y

Answer: No, not a polynomial — $3/y = 3y^{-1}$ and the power of y is negative.

Question: Exercise 5.1, Question 2: (iii) 1/x − x

Answer: No, not a polynomial — because of the term $1/x = x^{-1}$.

Question: Exercise 5.1, Question 2: (iv) √5 x + y

Answer: Yes, it is a polynomial — the coefficient √5 is a real number and the powers of the variables are whole numbers.

Question: Exercise 5.1, Question 2: (v) $5xy^3$

Answer: Yes, it is a polynomial.

Question: Exercise 5.1, Question 2: (vi) 2 − x

Answer: Yes, it is a polynomial.

Question: Exercise 5.1, Question 2: (vii) √(1/3)

Answer: Yes, it is a polynomial — a constant polynomial.

Question: Exercise 5.1, Question 2: (viii) $1/(x^2 + 3)$

Answer: No, not a polynomial — the expression with the variable is in the denominator.

Question: Exercise 5.1, Question 2: (ix) $x^2 + 3x - 1$

Answer: Yes, it is a polynomial.

Question: Exercise 5.1, Question 2: (x) $3^5 + 4/x$

Answer: No, not a polynomial — because of the term $4/x = 4x^{-1}$.

Question: Exercise 5.1, Question 2: (xi) $ax^2 + bx + c$

Answer: Yes, it is a polynomial — a, b and c are literal coefficients and the powers of the variables are whole numbers.

Question: Exercise 5.1, Question 2: (xii) $2a^2 + 3a + 1$

Answer: Yes, it is a polynomial.

Question: Exercise 5.1, Question 3: State the degree of the following polynomials. (i) 5^3

Answer: $5^3 = 125$ is a constant, so the degree = 0.

Question: Exercise 5.1, Question 3: (ii) x

Answer: Degree = 1.

Question: Exercise 5.1, Question 3: (iii) $x + y^2$

Answer: The terms have degrees 1 and 2, so the degree of the polynomial = 2.

Question: Exercise 5.1, Question 3: (iv) $x^2y^2 + y^3$

Answer: The terms have degrees 4 and 3, so the degree = 4.

Question: Exercise 5.1, Question 3: (v) $x^3y^3z^3 + 1$

Answer: The degree of $x^3y^3z^3$ is 3 + 3 + 3 = 9, so the degree = 9.

Question: Exercise 5.1, Question 3: (vi) $-1/4 + 3x + 5$

Answer: $-1/4 + 3x + 5 = 3x + 19/4$, so the degree = 1.

Question: Exercise 5.1, Question 3: (vii) $x^2y + y^3 + y^7$

Answer: The terms have degrees 3, 3 and 7, so the degree = 7.

Question: Exercise 5.1, Question 3: (viii) 3^3xyz

Answer: $3^3xyz = 27xyz$, whose degree is 1 + 1 + 1 = 3.

Question: Exercise 5.1, Question 3: (ix) $x^4 + x^3y^3 + xy^4$

Answer: The terms have degrees 4, 6 and 5, so the degree = 6.

Question: Exercise 5.1, Question 3: (x) $-x^3 + 8xyz^2$

Answer: The terms have degrees 3 and 1 + 1 + 2 = 4, so the degree = 4.

Question: Exercise 5.1, Question 3: (xi) 2/13

Answer: It is a constant, so the degree = 0.

Question: Exercise 5.1, Question 3: (xii) $4^2x^5z^4y^4$

Answer: $4^2x^5z^4y^4 = 16x^5z^4y^4$, whose degree is 5 + 4 + 4 = 13.

Question: Exercise 5.1, Question 4: Write the numerical coefficients of the

following polynomials. (i) 3

Answer: Numerical coefficient = 3.

Question: Exercise 5.1, Question 4: (ii) $\sqrt{4}\,xy$

Answer: $\sqrt{4} = 2$, so the numerical coefficient = 2.

Question: Exercise 5.1, Question 4: (iii) $-x$

Answer: Numerical coefficient = -1.

Question: Exercise 5.1, Question 4: (iv) $x^2yz + 1$

Answer: The numerical coefficient of $x^2yz = 1$ (and the constant term 1 has coefficient 1 itself).

Question: Exercise 5.1, Question 4: (v) $5x^2 + 9y + z$

Answer: Numerical coefficients = 5, 9 and 1 (the coefficient of z is 1).

Question: Exercise 5.1, Question 4: (vi) $\sqrt{2}\,x^3 - \sqrt{3}\,y^4$

Answer: Numerical coefficients = $\sqrt{2}$ and $-\sqrt{3}$.

Question: Exercise 5.1, Question 4: (vii) $-2xyz$

Answer: Numerical coefficient = -2.

Question: Exercise 5.1, Question 4: (viii) $-3/4\,x^2$

Answer: Numerical coefficient = $-3/4$.

Question: Exercise 5.1, Question 5: Find the number of variables in the following polynomials. (i) $3x + 2y$

Answer: There are 2 variables: x and y.

Question: Exercise 5.1, Question 5: (ii) $5x^2 - 4y - 3z$

Answer: There are 3 variables: x, y and z.

Question: Exercise 5.1, Question 5: (iii) $x^2 - y^3 + 1$

Answer: There are 2 variables: x and y.

Question: Exercise 5.1, Question 5: (iv) $2x^2y$

Answer: There are 2 variables: x and y.

Question: Exercise 5.1, Question 5: (v) $xy + 3$

Answer: There are 2 variables: x and y.

Question: Exercise 5.1, Question 5: (vi) $3x^7$

Answer: There is 1 variable: x.

Question: Exercise 5.1, Question 5: (vii) 25

Answer: There are 0 variables — it is a constant.

Question: Exercise 5.1, Question 5: (viii) $6x^3y^3z^2$

Answer: There are 3 variables: x, y and z.

Question: Exercise 5.1, Question 6: Identify whether the following polynomials are monomial, binomial, trinomial or quadrinomial. (i) $x + 2y$

Answer: It has 2 terms, so it is a binomial polynomial.

Question: Exercise 5.1, Question 6: (ii) $3x - 5$

Answer: It has 2 terms, so it is a binomial polynomial.

Question: Exercise 5.1, Question 6: (iii) $2xy + y/5$

Answer: It has 2 terms, so it is a binomial polynomial.

Question: Exercise 5.1, Question 6: (iv) $y^3 + y - 3$

Answer: It has 3 terms, so it is a trinomial polynomial.

Question: Exercise 5.1, Question 6: (v) $x^3 + xy + 5$

Answer: It has 3 terms, so it is a trinomial polynomial.

Question: Exercise 5.1, Question 6: (vi) $x^4 - 2x^3 + 1/2$

Answer: It has 3 terms, so it is a trinomial polynomial.

Question: Exercise 5.1, Question 6: (vii) $x^3 + x^2 + \sqrt{4}$

Answer: $\sqrt{4} = 2$, so it has 3 terms and is a trinomial polynomial.

Question: Exercise 5.1, Question 6: (viii) $5x^4$

Answer: It has 1 term, so it is a monomial polynomial.

Question: Exercise 5.1, Question 6: (ix) $-9xy^2$

Answer: It has 1 term, so it is a monomial polynomial.

Question: Exercise 5.1, Question 6: (x) $x^2 + 2x^2y^2 + y^2$

Answer: It has 3 terms, so it is a trinomial polynomial.

Question: Exercise 5.1, Question 6: (xi) $x^2 - x$

Answer: It has 2 terms, so it is a binomial polynomial.

Question: Exercise 5.1, Question 6: (xii) $3xy^2z$

Answer: It has 1 term, so it is a monomial polynomial.

Question: Exercise 5.2, Question 1: Add. (i) 4x + 6y + 5z, −3x − 9y and x + 3y − 4z

Answer: Sum = (4x − 3x + x) + (6y − 9y + 3y) + (5z − 4z) = 2x + 0y + z = 2x + z.

Question: Exercise 5.2, Question 1: (ii) $7x + 2y^3 - 4xy$, $3x - 2xy^3 + 7xy$ and $2xy - 5x + 6y^3$

Answer: Sum = $(7x + 3x - 5x) + (2y^3 + 6y^3) + (-4xy + 7xy + 2xy) + (-2xy^3) = 5x + 8y^3 + 5xy - 2xy^3$.

Question: Exercise 5.2, Question 1: (iii) $4x^2 + 3y^2 - 6x + 4y - 3$, $2x - y^2 + 3x^2 - 4y + 3$ and $-6x^2 - 2y^2 - 1$

Answer: Sum = $(4x^2 + 3x^2 - 6x^2) + (3y^2 - y^2 - 2y^2) + (-6x + 2x) + (4y - 4y) + (-3 + 3 - 1) = x^2 - 4x - 1$.

Question: Exercise 5.2, Question 1: (iv) $6ab - a^2 - b^2 - 7$, $5a^2 - 7ab + 3b^2 + 9$ and $-4b^2 - 3ab - 2a^2 - 3$

Answer: Sum = $(-a^2 + 5a^2 - 2a^2) + (-b^2 + 3b^2 - 4b^2) + (6ab - 7ab - 3ab) + (-7 + 9 - 3) = 2a^2 - 2b^2 - 4ab - 1$.

Question: Exercise 5.2, Question 2: Subtract the second expression from the first (starting from the left side). (i) $z - 4x - 6y$ from $2x - 3y - z$

Answer: $(2x - 3y - z) - (z - 4x - 6y) = 2x - 3y - z - z + 4x + 6y = 6x + 3y - 2z$.

Question: Exercise 5.2, Question 2: (ii) $6y^2 - 3y^5 + 2y^4 + 3y^3 + 8$ from y^4

− 6y² − 3y³ + 4y⁵

Answer: (y⁴ − 6y² − 3y³ + 4y⁵) − (6y² − 3y⁵ + 2y⁴ + 3y³ + 8) = 4y⁵ + 3y⁵ + y⁴ − 2y⁴ − 3y³ − 3y³ − 6y² − 6y² − 8 = 7y⁵ − y⁴ − 6y³ − 12y² − 8.

Question: Exercise 5.2, Question 2: (iii) x³ + y³ + z³ − 8x + 7y − 5z − 5w from 7x − 8y + 4z − 5w

Answer: (7x − 8y + 4z − 5w) − (x³ + y³ + z³ − 8x + 7y − 5z − 5w) = −x³ − y³ − z³ + 15x − 15y + 9z.

Question: Exercise 5.2, Question 2: (iv) 10y² − 2yz − 3z² from 5z² − 3yz + 2y²

Answer: (5z² − 3yz + 2y²) − (10y² − 2yz − 3z²) = 8z² − yz − 8y².

Question: Exercise 5.2, Question 3: Subtract the sum of p² + 3pq + q² and 6p² − 7pq + 4q² from 7p² − 2pq + 3q².

Answer: The sum of the two = 7p² − 4pq + 5q².

The difference = (7p² − 2pq + 3q²) − (7p² − 4pq + 5q²) = 2pq − 2q².

Question: Exercise 5.2, Question 4: The sum of two polynomials is 3a³ + 3a + 7b + 4ab. If one of them is 4ab − 3a³, find the other polynomial.

Answer: The other polynomial = (3a³ + 3a + 7b + 4ab) − (4ab − 3a³) = 6a³ + 3a + 7b.

Question: Exercise 5.2, Question 5: Simplify. (i) 6(2x + y − 7xy) − 3(5x − 2y + 5xy)

Answer: 12x + 6y − 42xy − 15x + 6y − 15xy = −3x + 12y − 57xy.

Question: Exercise 5.2, Question 5: (ii) $4(2x - 3y + xy) - 5(3x - 2y - xy)$

Answer: $8x - 12y + 4xy - 15x + 10y + 5xy = -7x - 2y + 9xy$.

Question: Exercise 5.2, Question 5: (iii) $x(x^2 + 2xy + y^2) + 4y(x^2 + 3xy + 9y^2)$

Answer: $x^3 + 2x^2y + xy^2 + 4x^2y + 12xy^2 + 36y^3 = x^3 + 6x^2y + 13xy^2 + 36y^3$.

Question: Exercise 5.2, Question 5: (iv) $x^2(x^2 + xy + y^2) - y^2(x^2 - xy - y^2)$

Answer: $x^4 + x^3y + x^2y^2 - x^2y^2 + xy^3 + y^4 = x^4 + x^3y + xy^3 + y^4$.

Question: Exercise 5.2, Question 6: Multiply. (i) $(\sqrt{x} + \sqrt{y})(x - \sqrt{xy} + y)$

Answer: Setting $\sqrt{x} = u$ and $\sqrt{y} = v$, $(u + v)(u^2 - uv + v^2) = u^3 + v^3 = (\sqrt{x})^3 + (\sqrt{y})^3 = x\sqrt{x} + y\sqrt{y}$.

Question: Exercise 5.2, Question 6: (ii) $(x^3 - xy + y^3)(x^3 + xy + y^3)$

Answer: $(x^3 + y^3 - xy)(x^3 + y^3 + xy) = (x^3 + y^3)^2 - (xy)^2 = x^6 + 2x^3y^3 + y^6 - x^2y^2$.

Question: Exercise 5.2, Question 6: (iii) $(a^2 - b^2)(a^2 - 2ab + b^2)$

Answer: $a^4 - 2a^3b + a^2b^2 - a^2b^2 + 2ab^3 - b^4 = a^4 - 2a^3b + 2ab^3 - b^4$.

Question: Exercise 5.2, Question 6: (iv) $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

Answer: This is the well-known identity: the product $= a^3 + b^3 + c^3 - 3abc$.

Question: Exercise 5.2, Question 7: Divide. (i) $(3x^3 - 6x^2 + 12x) \div 3x$

Answer: Dividing each term by $3x$: $x^2 - 2x + 4$.

Question: Exercise 5.2, Question 7: (ii) $(4x^4 - 16x^3 + 8x^2) \div 2x^2$

Answer: Dividing each term by $2x^2$: $2x^2 - 8x + 4$.

Question: Exercise 5.2, Question 7: (iii) $(x^2 + x - 6) \div (x + 3)$

Answer: $x^2 + x - 6 = (x + 3)(x - 2)$, so the quotient = $x - 2$.

Question: Exercise 5.2, Question 7: (iv) $(9x^2 - 6xy - 8y^2) \div (3x + 2y)$

Answer: $9x^2 - 6xy - 8y^2 = (3x + 2y)(3x - 4y)$, so the quotient = $3x - 4y$.

Question: Exercise 5.2, Question 7: (v) $(a^3 + b^3) \div (a + b)$

Answer: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$, so the quotient = $a^2 - ab + b^2$.

Question: Exercise 5.2, Question 7: (vi) $(a^6 - b^6) \div (a^2 - b^2)$

Answer: $a^6 - b^6 = (a^2)^3 - (b^2)^3 = (a^2 - b^2)(a^4 + a^2b^2 + b^4)$, so the quotient = $a^4 + a^2b^2 + b^4$.

Question: Exercise 5.2, Question 8: The product of two polynomials is $x^4 - 3x^2y^2 + y^4$; one of them is $x^2 - xy - y^2$. Find the other polynomial.

Answer: The other polynomial = $(x^4 - 3x^2y^2 + y^4) \div (x^2 - xy - y^2)$.

Dividing: $x^4 \div x^2 = x^2$;

$x^2(x^2 - xy - y^2) = x^4 - x^3y - x^2y^2$;

remainder $x^3y - 2x^2y^2 + y^4$.

Then $xy(x^2 - xy - y^2) = x^3y - x^2y^2 - xy^3$;

remainder $-x^2y^2 + xy^3 + y^4$.

Then $-y^2(x^2 - xy - y^2) = -x^2y^2 + xy^3 + y^4$;

remainder 0.

So the other polynomial = $x^2 + xy - y^2$.

Question: Exercise 5.2, Question 9: What should be subtracted from $2a^4 + 3a^3 - a^2 - 5$ so that the remaining polynomial is exactly divisible by $a - 2$?

Answer: By the remainder theorem, the value at $a = 2$ is $2(16) + 3(8) - 4 - 5 = 32 + 24 - 9 = 47$.

So 47 should be subtracted;

the remainder $2a^4 + 3a^3 - a^2 - 52$ will then be exactly divisible by $a - 2$.

Question: Exercise 5.2, Question 10: Find the value of t for which $2x^4 + 3x^3 - 4x^2 + 14x + 2t$ is exactly divisible by $x^2 - 2x + 3$.

Answer: Dividing: $2x^4 + 3x^3 - 4x^2 + 14x + 2t = (x^2 - 2x + 3)(2x^2 + 7x + 4) + (x + 2t - 12)$.

The remainder $x + 2t - 12$ cannot be zero for every x, so there is no value of t that makes the division exact.

[note: as printed the question appears misprinted — if the x-term of the remainder is overlooked, setting $2t - 12 = 0$ gives $t = 6$.]

Question: Review Exercise 5, Question 1: Answer the following questions.

(i) What is meant by a trinomial? Write an example of a trinomial with one variable.

Answer: A trinomial is a polynomial with three terms.

An example of a trinomial with one variable: $x^2 + 3x + 2$.

Question: Review Exercise 5, Question 1: (ii) Define a variable and give some examples.

Answer: A variable is a symbol or letter whose value can change and which can stand for any number.

Examples: x, y, a, b.

Question: Review Exercise 5, Question 1: (iii) Which literal quantities are used in the polynomial a + 2b + 3c?

Answer: The letters a, b and c are used, but they are variables, not literal quantities (letters used for constants) — so no literal quantity is used in this expression.

Question: Review Exercise 5, Question 1: (iv) How many terms are there in $x^3 + 2x^2 + 3x^2$?

Answer: As written there are 3 terms: x^3, $2x^2$ and $3x^2$.

Combining like terms gives $x^3 + 5x^2$, which has 2 terms.

Question: Review Exercise 5, Question 1: (v) What is meant by the degree of a polynomial? Give an example.

Answer: The degree of a polynomial is the greatest power (sum of the powers of the variables) of any of its terms.

Example: $3x^2 + 2x + 1$ has degree 2.

Review Exercise 5, Question 2: Tick () the correct answer or fill in the blank. (i) The number of variables in $ax^2 + bx + c$ is: _____

(a) two

(b) three

(c) four

✓ (d) one

Review Exercise 5, Question 2: (ii) $x - 1$ is a polynomial of degree _____.

(a) two

(b) four

✓ (c) one

(d) three

Review Exercise 5, Question 2: (iii) A polynomial of degree four is _____.

(a) xy

✓ (b) $2a^4$

(c) $x + y$

(d) $a + x + 3$

Review Exercise 5, Question 2: (iv) The degree of $3x^2 + 4xy^4 + y^6 + 9$ is _____.

(a) 9

(b) 4

(c) 5

(d) 6

Review Exercise 5, Question 2: (v) The number multiplying the variable or variables in a polynomial is called _____. (a) variable (b) numerical value (c) exponent (d) none of these

Answer: the numerical coefficient. [note: printed option

Review Exercise 5, Question 2: (vi) Which of the following is a polynomial of degree three?

(a) $xbc + a^2$

(b) $a^2 + b^2 + c^2$

(c) $x^3 + 1/x^3$

(d) 3

Question: Review Exercise 5, Question 2: (vii) Which of the following algebraic expressions is NOT a polynomial? (a) $x + 1/x^{-2}$ (b) -7 (c) $x^3 - y^3$ (d) $x^{-3} + 1$

Answer: (d) $x^{-3} + 1$ — x^{-3} has a negative power.

(a) is a polynomial because $1/x^{-2} = x^2$.

Review Exercise 5, Question 2: (viii) The degree of every Quadratic polynomial is _____. (a) 1 (b) 3 (c) 4 (d) 0

Answer: 2, which is not among the given options. [note: correct answer 2 is not among the printed options (1, 3, 4, 0) — the question appears misprinted.]

Review Exercise 5, Question 2: (ix) The literal quantities in $px^2 + qx + r$ are:

(a) p, q, r

(b) x

(c) x^2, x

(d) p, q, r, x

Review Exercise 5, Question 2: (x) If one polynomial is exactly divisible by another polynomial, the remainder will be _____.

(a) 0

(b) 1

(c) −1

(d) none of these

Question: Review Exercise 5, Question 3: Find the value of the following. (i) $x - 5y$, when $x = 2a + 3b$, $y = a - 5b$

Answer: $x - 5y = (2a + 3b) - 5(a - 5b) = 2a + 3b - 5a + 25b = -3a + 28b$.

Question: Review Exercise 5, Question 3: (ii) $(x + y)(x^3 - 2x^2y + xy^2)$, when $x = -2$, $y = -3$

Answer: $x + y = -5$.

$x^3 - 2x^2y + xy^2 = -8 - 2(4)(-3) + (-2)(9) = -8 + 24 - 18 = -2$.

Product $= (-5)(-2) = 10$.

Question: Review Exercise 5, Question 3: (iii) $(a^2 + b^2)/(a^2 + 2ab + b^2)$,

when a = 0, b = 5

Answer: (0 + 25)/(0 + 0 + 25) = 25/25 = 1.

Question: Review Exercise 5, Question 3: (iv) (a + 1/a)(a² + 1 + 1/a²), when a = 3

Answer: a + 1/a = 3 + 1/3 = 10/3.

a² + 1 + 1/a² = 9 + 1 + 1/9 = 91/9.

Product = (10/3)(91/9) = 910/27.

Question: Review Exercise 5, Question 3: (v) (a + b + c)(a² + b² + c² − ab − bc − ca), when a = −2, b = 0, c = 2

Answer: By the identity, the product = a³ + b³ + c³ − 3abc = (−8) + 0 + 8 − 3(−2)(0)(2) = 0.

Question: Review Exercise 5, Question 4: Simplify. (i) (x + y)/3 − (y + z)/2 + (4x − z)/5

Answer: LCM 30: [10(x + y) − 15(y + z) + 6(4x − z)]/30 = (10x + 10y − 15y − 15z + 24x − 6z)/30 = (34x − 5y − 21z)/30.

Question: Review Exercise 5, Question 4: (ii) (√x + y)(√x − y)

Answer: (√x)² − y² = x − y².

Question: Review Exercise 5, Question 4: (iii) (x + y + 2)(x − y − 2)

Answer: (x + (y + 2))(x − (y + 2)) = x² − (y + 2)² = x² − y² − 4y − 4.

Question: Review Exercise 5, Question 4: (iv) (a³ − 2a + 4) ÷ (a + 2)

Answer: $a^3 \div a = a^2$;

$a^2(a + 2) = a^3 + 2a^2$;

remainder $-2a^2 - 2a + 4$.

$-2a^2 \div a = -2a$;

$-2a(a + 2) = -2a^2 - 4a$;

remainder $2a + 4$.

$2a \div a = 2$;

$2(a + 2) = 2a + 4$;

remainder 0.

Quotient = $a^2 - 2a + 2$.

Question: Review Exercise 5, Question 5: For what value of k will $k - 2$ exactly divide $5k^2 - 14 - 3k$?

Answer: $5k^2 - 3k - 14 = (k - 2)(5k + 7)$, so $k - 2$ already divides it exactly — no special value is needed;

the quotient is $5k + 7$ and the remainder is 0.

6. Factorization and Simultaneous Equations

Question: Exercise 6.1, Question 1: Find the value of the following using a suitable formula. (i) $(105)^2$

Answer: $(105)^2 = (100 + 5)^2 = 10000 + 1000 + 25 = 11025$.

Question: Exercise 6.1, Question 1: (ii) $(96)^2$

Answer: $(96)^2 = (100 - 4)^2 = 10000 - 800 + 16 = 9216$.

Question: Exercise 6.1, Question 1: (iii) $(57)^2$

Answer: $(57)^2 = (50 + 7)^2 = 2500 + 700 + 49 = 3249$.

Question: Exercise 6.1, Question 1: (iv) $(52)^2$

Answer: $(52)^2 = (50 + 2)^2 = 2500 + 200 + 4 = 2704$.

Question: Exercise 6.1, Question 1: (v) 104×96

Answer: $104 \times 96 = (100 + 4)(100 - 4) = 10000 - 16 = 9984$.

Question: Exercise 6.1, Question 1: (vi) 47×53

Answer: $47 \times 53 = (50 - 3)(50 + 3) = 2500 - 9 = 2491$.

Question: Exercise 6.1, Question 1: (vii) 107×93

Answer: $107 \times 93 = (100 + 7)(100 - 7) = 10000 - 49 = 9951$.

Question: Exercise 6.1, Question 1: (viii) 89×111

Answer: $89 \times 111 = (100 - 11)(100 + 11) = 10000 - 121 = 9879$.

Question: Exercise 6.1, Question 2: Find the value of the following using a suitable formula. (i) $(1.03)^2$

Answer: $(1.03)^2 = (1 + 0.03)^2 = 1 + 0.06 + 0.0009 = 1.0609$.

Question: Exercise 6.1, Question 2: (ii) $(0.99)^2$

Answer: $(0.99)^2 = (1 - 0.01)^2 = 1 - 0.02 + 0.0001 = 0.9801$.

Question: Exercise 6.1, Question 2: (iii) $(1.05)^2$

Answer: $(1.05)^2 = (1 + 0.05)^2 = 1 + 0.10 + 0.0025 = 1.1025$.

Question: Exercise 6.1, Question 2: (iv) $(0.91)^2$

Answer: $(0.91)^2 = (1 - 0.09)^2 = 1 - 0.18 + 0.0081 = 0.8281$.

Question: Exercise 6.1, Question 2: (v) 1.03×0.97

Answer: $1.03 \times 0.97 = (1 + 0.03)(1 - 0.03) = 1 - 0.0009 = 0.9991$.

Question: Exercise 6.1, Question 2: (vi) 5.02×4.98

Answer: $5.02 \times 4.98 = (5 + 0.02)(5 - 0.02) = 25 - 0.0004 = 24.9996$.

Question: Exercise 6.1, Question 3: Given the value of $x + 1/x$, find $x^2 + 1/x^2$ and $x^4 + 1/x^4$. (i) $x + 1/x = 5$

Answer: $x^2 + 1/x^2 = (x + 1/x)^2 - 2 = 25 - 2 = 23$.

$x^4 + 1/x^4 = (x^2 + 1/x^2)^2 - 2 = 529 - 2 = 527$.

Question: Exercise 6.1, Question 3: (ii) $x + 1/x = -4$

Answer: $x^2 + 1/x^2 = 16 - 2 = 14$.

$x^4 + 1/x^4 = 196 - 2 = 194$.

Question: Exercise 6.1, Question 3: (iii) $x + 1/x = 7$

Answer: $x^2 + 1/x^2 = 49 - 2 = 47$.

$x^4 + 1/x^4 = 2209 - 2 = 2207$.

Question: Exercise 6.1, Question 4: Given the value of $x - 1/x$, find $x^2 + 1/x^2$ and $x^4 + 1/x^4$. (i) $x - 1/x = 3$

Answer: $x^2 + 1/x^2 = (x - 1/x)^2 + 2 = 9 + 2 = 11$.

$x^4 + 1/x^4 = 121 - 2 = 119$.

Question: Exercise 6.1, Question 4: (ii) $x - 1/x = 0.2$

Answer: $x^2 + 1/x^2 = 0.04 + 2 = 2.04$.

$x^4 + 1/x^4 = 2.04^2 - 2 = 4.1616 - 2 = 2.1616$.

Question: Exercise 6.1, Question 4: (iii) $x - 1/x = -6$

Answer: $x^2 + 1/x^2 = 36 + 2 = 38$.

$x^4 + 1/x^4 = 1444 - 2 = 1442$.

Question: Exercise 6.1, Question 5: Verify. (i) $x^2 + 1/x^2 = (x + 1/x)^2 - 2$

Answer: $(x + 1/x)^2 = x^2 + 2(x)(1/x) + 1/x^2 = x^2 + 2 + 1/x^2$, so $x^2 + 1/x^2 = (x + 1/x)^2 - 2$.

Verified.

Question: Exercise 6.1, Question 5: Verify. (ii) $x^2 + 1/x^2 = (x - 1/x)^2 + 2$

Answer: $(x - 1/x)^2 = x^2 - 2(x)(1/x) + 1/x^2 = x^2 - 2 + 1/x^2$, so $x^2 + 1/x^2 = (x - 1/x)^2 + 2$.

Verified.

Question: Exercise 6.2, A: Factorize the following expressions. (1) $4x + 8z$

Answer: $4x + 8z = 4(x + 2z)$.

Question: Exercise 6.2, A: (2) $5x + 10y + 30z$

Answer: $5x + 10y + 30z = 5(x + 2y + 6z)$.

Question: Exercise 6.2, A: (3) $2x - 4xy + 8xz$

Answer: $2x - 4xy + 8xz = 2x(1 - 2y + 4z)$.

Question: Exercise 6.2, A: (4) $2a^3 + 10a^2 - 20a^4$

Answer: $2a^3 + 10a^2 - 20a^4 = 2a^2(1 + 5a - 10a^2)$.

Question: Exercise 6.2, A: (5) $3a^2b + 7ab^2 - 8a^2b^2$

Answer: $3a^2b + 7ab^2 - 8a^2b^2 = ab(3a + 7b - 8ab)$.

Question: Exercise 6.2, A: (6) $6a^2bc + 12ab^2c - 36abc^2$

Answer: $6a^2bc + 12ab^2c - 36abc^2 = 6abc(a + 2b - 6c)$.

Question: Exercise 6.2, A: (7) $5x + 10y + 3xz + 6yz$

Answer: $5x + 10y + 3xz + 6yz = 5(x + 2y) + 3z(x + 2y) = (x + 2y)(5 + 3z)$.

Question: Exercise 6.2, A: (8) $abc - abd + cx - xd$

Answer: $abc - abd + cx - xd = ab(c - d) + x(c - d) = (c - d)(ab + x)$.

Question: Exercise 6.2, A: (9) $x^2 + 5x + 6xy + 30y$

Answer: $x^2 + 5x + 6xy + 30y = x(x + 5) + 6y(x + 5) = (x + 5)(x + 6y)$.

Question: Exercise 6.2, A: (10) $7xy + 14yz - 5ax - 10az$

Answer: $7xy + 14yz - 5ax - 10az = 7y(x + 2z) - 5a(x + 2z) = (x + 2z)(7y - 5a)$.

Question: Exercise 6.2, A: (11) $3x^2 + 6y^2 + 6x^2z + 12y^2z$

Answer: $3x^2 + 6y^2 + 6x^2z + 12y^2z = 3(x^2 + 2y^2) + 6z(x^2 + 2y^2) = (x^2 + 2y^2)(3 + 6z) = 3(x^2 + 2y^2)(1 + 2z)$.

Question: Exercise 6.2, A: (12) $x^2 - 7xy - xz + 7yz$

Answer: $x^2 - 7xy - xz + 7yz = x(x - 7y) - z(x - 7y) = (x - 7y)(x - z)$.

Question: Exercise 6.2, B: Factorize the following expressions. (1) $a^2 + 10a + 25$

Answer: $a^2 + 10a + 25 = (a + 5)^2$.

Question: Exercise 6.2, B: (2) $x^2 + 12xy + 36y^2$

Answer: $x^2 + 12xy + 36y^2 = (x + 6y)^2$.

Question: Exercise 6.2, B: (3) $4x^2 + 12xy + 9y^2$

Answer: $4x^2 + 12xy + 9y^2 = (2x + 3y)^2$.

Question: Exercise 6.2, B: (4) $16a^2 + 40ab + 25b^2$

Answer: $16a^2 + 40ab + 25b^2 = (4a + 5b)^2$.

Question: Exercise 6.2, B: (5) $b^2 + c^2 - 2bc$

Answer: $b^2 + c^2 - 2bc = (b - c)^2$.

Question: Exercise 6.2, B: (6) $49p^2 - 14p + 1$

Answer: $49p^2 - 14p + 1 = (7p - 1)^2$.

Question: Exercise 6.2, B: (7) $81c^2 - 36cd + 4d^2$

Answer: $81c^2 - 36cd + 4d^2 = (9c - 2d)^2$.

Question: Exercise 6.2, B: (8) $144x^4 - 72x^2y^2 + 9y^4$

Answer: $144x^4 - 72x^2y^2 + 9y^4 = (12x^2 - 3y^2)^2$.

Question: Exercise 6.2, B: (9) $3a^2 - 6ab + 3b^2$

Answer: $3a^2 - 6ab + 3b^2 = 3(a^2 - 2ab + b^2) = 3(a - b)^2$.

Question: Exercise 6.2, B: (10) $1/4\ x^2 - 1/3\ xy + 1/9\ y^2$

Answer: $1/4\ x^2 - 1/3\ xy + 1/9\ y^2 = (x/2 - y/3)^2$, since $(x/2)^2 = x^2/4$, $2(x/2)(y/3) = xy/3$ and $(y/3)^2 = y^2/9$.

Question: Exercise 6.2, C: Factorize the following expressions. (1) $81x^2 - 4y^2$

Answer: $81x^2 - 4y^2 = (9x + 2y)(9x - 2y)$.

Question: Exercise 6.2, C: (2) $169a^2 - 100b^2$

Answer: $169a^2 - 100b^2 = (13a + 10b)(13a - 10b)$.

Question: Exercise 6.2, C: (3) $3a^2 - 27b^4$

Answer: $3a^2 - 27b^4 = 3(a^2 - 9b^4) = 3(a + 3b^2)(a - 3b^2)$.

Question: Exercise 6.2, C: (4) $2p^2 - 18q^2$

Answer: $2p^2 - 18q^2 = 2(p^2 - 9q^2) = 2(p + 3q)(p - 3q)$.

Question: Exercise 6.2, C: (5) $5x^2 - 125y^2$

Answer: $5x^2 - 125y^2 = 5(x^2 - 25y^2) = 5(x + 5y)(x - 5y)$.

Question: Exercise 6.2, C: (6) $25/x^2 - y^2/16$

Answer: $25/x^2 - y^2/16 = (5/x + y/4)(5/x - y/4)$.

Question: Exercise 6.2, C: (7) $x^2/144 - y^2$

Answer: $x^2/144 - y^2 = (x/12 + y)(x/12 - y)$.

Question: Exercise 6.2, C: (8) $36/25\ l^2 - 49/4\ d^2$

Answer: $36/25\ l^2 - 49/4\ d^2 = (6l/5 + 7d/2)(6l/5 - 7d/2)$.

Question: Exercise 6.2, C: (9) $c^2 - (a - b)^2$

Answer: $c^2 - (a - b)^2 = (c + a - b)(c - a + b)$.

Question: Exercise 6.2, C: (10) $(x + y)^2 - z^2$

Answer: $(x + y)^2 - z^2 = (x + y + z)(x + y - z)$.

Question: Exercise 6.2, C: (11) $(a + b)^2 - (p + q)^2$

Answer: $(a + b)^2 - (p + q)^2 = (a + b + p + q)(a + b - p - q)$.

Question: Exercise 6.2, C: (12) $144 - y^2$

Answer: $144 - y^2 = (12 + y)(12 - y)$.

Question: Exercise 6.2, C: (13) $32a^2 - 50b^2$

Answer: $32a^2 - 50b^2 = 2(16a^2 - 25b^2) = 2(4a + 5b)(4a - 5b)$.

Question: Exercise 6.2, C: (14) $x^4 - y^4$

Answer: $x^4 - y^4 = (x^2 + y^2)(x^2 - y^2) = (x^2 + y^2)(x + y)(x - y)$.

Question: Exercise 6.2, C: (15) $(2a + b)^2 - 9c^2$

Answer: $(2a + b)^2 - 9c^2 = (2a + b + 3c)(2a + b - 3c)$.

Question: Exercise 6.2, D: Factorize the following expressions. (1) $x^2 + 2xy + y^2 - z^2$

Answer: $x^2 + 2xy + y^2 - z^2 = (x + y)^2 - z^2 = (x + y + z)(x + y - z)$.

Question: Exercise 6.2, D: (2) $a^2 + b^2 - 2ab - c^2$

Answer: $a^2 + b^2 - 2ab - c^2 = (a - b)^2 - c^2 = (a - b + c)(a - b - c)$.

Question: Exercise 6.2, D: (3) $64a^2 + 48ab + 9b^2 - c^2$

Answer: $64a^2 + 48ab + 9b^2 - c^2 = (8a + 3b)^2 - c^2 = (8a + 3b + c)(8a + 3b - c)$.

Question: Exercise 6.2, D: (4) $4p^2 - 12pq + 9q^2 - 49r^2$

Answer: $4p^2 - 12pq + 9q^2 - 49r^2 = (2p - 3q)^2 - (7r)^2 = (2p - 3q + 7r)(2p - 3q - 7r)$.

Question: Exercise 6.2, D: (5) $x^4 + 2x^2y^2 + y^4 - 25z^2$

Answer: $x^4 + 2x^2y^2 + y^4 - 25z^2 = (x^2 + y^2)^2 - (5z)^2 = (x^2 + y^2 + 5z)(x^2 + y^2$

− 5z).

Question: Exercise 6.2, D: (6) $x^2/4 - xy + y^2 - c^2/36$

Answer: $x^2/4 - xy + y^2 - c^2/36 = (x/2 - y)^2 - (c/6)^2 = (x/2 - y + c/6)(x/2 - y - c/6)$.

Question: Exercise 6.2, D: (7) $2a^2 + 4ab + 2b^2 - 2c^2$

Answer: $2a^2 + 4ab + 2b^2 - 2c^2 = 2[(a + b)^2 - c^2] = 2(a + b + c)(a + b - c)$.

Question: Exercise 6.2, D: (8) $81c^2 - 16p^2 + 18cd + d^2$

Answer: $81c^2 - 16p^2 + 18cd + d^2 = (9c + d)^2 - (4p)^2 = (9c + d + 4p)(9c + d - 4p)$.

Question: Exercise 6.2, D: (9) $4x^2 + 12x + 9 - 16y^2$

Answer: $4x^2 + 12x + 9 - 16y^2 = (2x + 3)^2 - (4y)^2 = (2x + 3 + 4y)(2x + 3 - 4y)$.

Question: Exercise 6.2, D: (10) $36 - 25a^2 + 70ab - 49b^2$

Answer: $36 - 25a^2 + 70ab - 49b^2 = 36 - (5a - 7b)^2 = (6 + 5a - 7b)(6 - 5a + 7b)$.

Question: Exercise 6.3, Question 1: Find the cube of the following expressions. (i) $(2x + 3)^3$

Answer: $(2x + 3)^3 = 8x^3 + 3(4x^2)(3) + 3(2x)(9) + 27 = 8x^3 + 36x^2 + 54x + 27$.

Question: Exercise 6.3, Question 1: (ii) $(5x - 1)^3$

Answer: $(5x - 1)^3 = 125x^3 - 3(25x^2)(1) + 3(5x)(1) - 1 = 125x^3 - 75x^2 + 15x - 1$.

Question: Exercise 6.3, Question 1: (iii) $(2x - 3y)^3$

Answer: $(2x - 3y)^3 = 8x^3 - 3(4x^2)(3y) + 3(2x)(9y^2) - 27y^3 = 8x^3 - 36x^2y + 54xy^2 - 27y^3$.

Question: Exercise 6.3, Question 1: (iv) $(2a + 5b)^3$

Answer: $(2a + 5b)^3 = 8a^3 + 3(4a^2)(5b) + 3(2a)(25b^2) + 125b^3 = 8a^3 + 60a^2b + 150ab^2 + 125b^3$.

Question: Exercise 6.3, Question 1: (v) $(x + 7z)^3$

Answer: $(x + 7z)^3 = x^3 + 3x^2(7z) + 3x(49z^2) + 343z^3 = x^3 + 21x^2z + 147xz^2 + 343z^3$.

Question: Exercise 6.3, Question 1: (vi) $(x/2 - 1)^3$

Answer: $(x/2 - 1)^3 = x^3/8 - 3(x^2/4)(1) + 3(x/2)(1) - 1 = x^3/8 - 3x^2/4 + 3x/2 - 1$.

Question: Exercise 6.3, Question 1: (vii) $(5 + 3a)^3$

Answer: $(5 + 3a)^3 = 125 + 3(25)(3a) + 3(5)(9a^2) + 27a^3 = 125 + 225a + 135a^2 + 27a^3$.

Question: Exercise 6.3, Question 1: (viii) $(5 - a/4)^3$

Answer: $(5 - a/4)^3 = 125 - 3(25)(a/4) + 3(5)(a^2/16) - a^3/64 = 125 - 75a/4 + 15a^2/16 - a^3/64$.

Question: Exercise 6.3, Question 1: (ix) $(5a - 3b)^3$

Answer: $(5a - 3b)^3 = 125a^3 - 3(25a^2)(3b) + 3(5a)(9b^2) - 27b^3 = 125a^3 - 225a^2b + 135ab^2 - 27b^3$.

Question: Exercise 6.3, Question 1: (x) $(x/3 + y/5)^3$

Answer: $(x/3 + y/5)^3 = x^3/27 + 3(x^2/9)(y/5) + 3(x/3)(y^2/25) + y^3/125 = x^3/27 + x^2y/15 + xy^2/25 + y^3/125$.

Question: Exercise 6.3, Question 2: Find the cube of each of the following using the formula. (i) 18

Answer: $18^3 = (20 - 2)^3 = 8000 - 3(400)(2) + 3(20)(4) - 8 = 8000 - 2400 + 240 - 8 = 5832$.

Question: Exercise 6.3, Question 2: (ii) 13

Answer: $13^3 = (10 + 3)^3 = 1000 + 3(100)(3) + 3(10)(9) + 27 = 1000 + 900 + 270 + 27 = 2197$.

Question: Exercise 6.3, Question 2: (iii) 105

Answer: $105^3 = (100 + 5)^3 = 1000000 + 3(10000)(5) + 3(100)(25) + 125 = 1000000 + 150000 + 7500 + 125 = 1157625$.

Question: Exercise 6.3, Question 2: (iv) 10.1

Answer: $10.1^3 = (10 + 0.1)^3 = 1000 + 3(100)(0.1) + 3(10)(0.01) + 0.001 = 1000 + 30 + 0.3 + 0.001 = 1030.301$.

Question: Exercise 6.3, Question 2: (v) 0.98

Answer: $0.98^3 = (1 - 0.02)^3 = 1 - 3(1)(0.02) + 3(1)(0.0004) - 0.000008 = 1 - 0.06 + 0.0012 - 0.000008 = 0.941192$.

Question: Exercise 6.3, Question 2: (vi) 2.25

Answer: $2.25^3 = (2 + 0.25)^3 = 8 + 3(4)(0.25) + 3(2)(0.0625) + 0.015625$ $= 8 + 3 + 0.375 + 0.015625 = 11.390625$.

Question: Exercise 6.3, Question 3: Find the value of $x^3 + 1/x^3$, when. (i) $x + 1/x = 6$

Answer: $x^3 + 1/x^3 = (x + 1/x)^3 - 3(x + 1/x) = 216 - 18 = 198$.

Question: Exercise 6.3, Question 3: (ii) $x + 1/x = -5$

Answer: $x^3 + 1/x^3 = (-125) - 3(-5) = -125 + 15 = -110$.

Question: Exercise 6.3, Question 3: (iii) $x + 1/x = 1/2$

Answer: $x^3 + 1/x^3 = 1/8 - 3/2 = 1/8 - 12/8 = -11/8$.

Question: Exercise 6.3, Question 3: (iv) $x + 1/x = 10.1$

Answer: $x^3 + 1/x^3 = 1030.301 - 30.3 = 1000.001$.

Question: Exercise 6.3, Question 4: Find the value of $x^3 - 1/x^3$, when. (i) $x - 1/x = 5$

Answer: $x^3 - 1/x^3 = (x - 1/x)^3 + 3(x - 1/x) = 125 + 15 = 140$.

Question: Exercise 6.3, Question 4: (ii) $x - 1/x = -2$

Answer: $x^3 - 1/x^3 = (-8) + 3(-2) = -8 - 6 = -14$.

Question: Exercise 6.3, Question 4: (iii) $x - 1/x = 1/3$

Answer: $x^3 - 1/x^3 = 1/27 + 3(1/3) = 1/27 + 1 = 28/27$.

Question: Exercise 6.3, Question 4: (iv) $x - 1/x = 9.9$

Answer: $x^3 - 1/x^3 = 970.299 + 29.7 = 999.999$.

Question: Exercise 6.4, Question 1: Which of the following equations are first-degree equations in one variable, and which are simultaneous linear equations in two variables? (i) 2x + 3y = 6

Answer: A single linear equation in two variables (x, y).

Question: Exercise 6.4, Question 1: (ii) 2x − 7 = 0

Answer: A first-degree equation in one variable (x).

Question: Exercise 6.4, Question 1: (iii) 2x = 6, x + y = 8

Answer: Simultaneous linear equations in two variables.

Question: Exercise 6.4, Question 1: (iv) x − y = 7, x + y = 10

Answer: Simultaneous linear equations in two variables.

Question: Exercise 6.4, Question 1: (v) 5y = 8

Answer: A first-degree equation in one variable (y).

Question: Exercise 6.4, Question 1: (vi) 3y − 7 = 8x

Answer: A single linear equation in two variables (x, y).

Question: Exercise 6.4, Question 2: Form a first-degree linear equation for each of the following statements. (i) The sum of the weights of two boxes is 50 kg.

Answer: If the weights of the boxes are x and y, then x + y = 50.

Question: Exercise 6.4, Question 2: (ii) The price of 6 pens is three times

the price of one marble.

Answer: If the price of a pen is x and of a marble is y, then 6x = 3y, i.e. 2x = y.

Question: Exercise 6.4, Question 2: (iii) One number is 5 greater than the other, and the sum of the two is 25.

Answer: If the numbers are x and y, then x − y = 5 and x + y = 25.

Question: Exercise 6.4, Question 2: (iv) The price of 3 kg of apples is Rs 45 less than the price of 6 kg of mangoes.

Answer: If the price of apples is x per kg and of mangoes is y per kg, then 6y − 3x = 45.

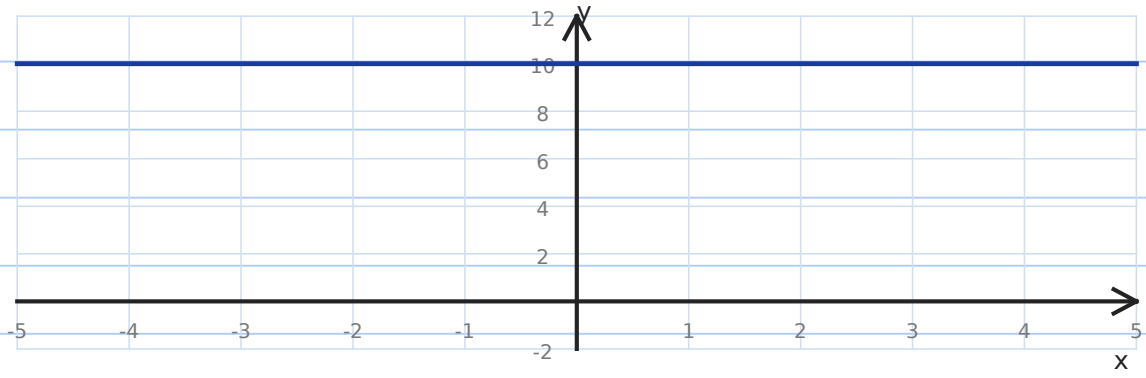
Question: Exercise 6.4, Question 2: (v) In a school, the number of girl students is equal to 2/3 of the number of boy students.

Answer: If the number of girls is g and of boys is b, then g = (2/3)b, i.e. 3g = 2b.

Question: Exercise 6.4, Question 3: Solve by estimation and verify with a graph. (i) x + y = 10, (ii) 2x + y = 10

Answer: Subtracting the first equation from the second gives x = 0, then y = 10.

The solution is {(0, 10)} — the intersection point of the two lines is (0, 10).

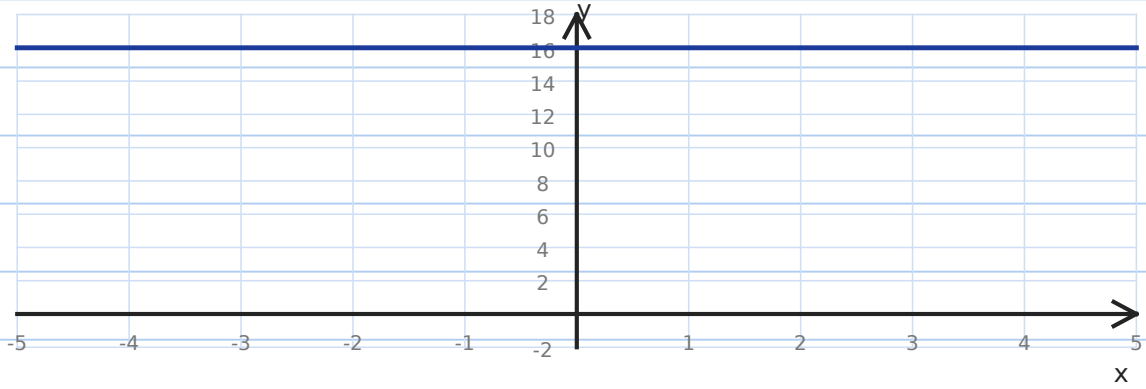


Question: Exercise 6.4, Question 4: Solve by estimation and verify with a graph. (i) x − y = 2, (ii) 2x + y = 16

Answer: Adding the two equations gives 3x = 18, i.e. x = 6;

then y = 4.

The solution is {(6, 4)} — the lines intersect at (6, 4) on the graph.



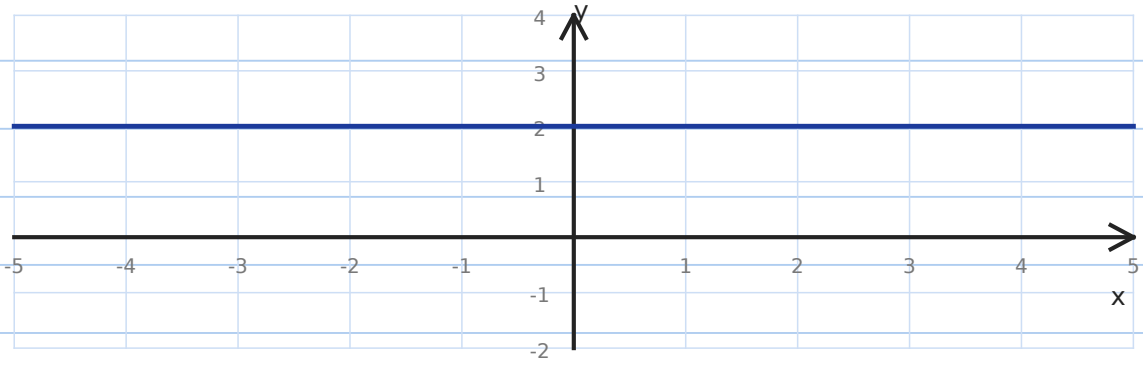
Question: Exercise 6.4, Question 5: Solve by estimation and verify with a graph. 4x − y = 2, 3x − 2y = 2

Answer: From the first equation y = 4x − 2.

Substituting in the second: 3x − 2(4x − 2) = 2, i.e. −5x = −2, x = 0.4;

then y = −0.4.

The solution is {(0.4, −0.4)}.



Question: Exercise 6.4, Question 6: Check which of the pairs (6, 0) and (1, 5) is the solution of the simultaneous equations 6x − y = 1 and 2x + y = 7.

Answer: (6, 0): 6(6) − 0 = 36 ≠ 1, so it is not a solution.

(1, 5): 6 − 5 = 1 and 2 + 5 = 7;

both equations hold, so (1, 5) is the solution.

Question: Exercise 6.4, Question 7: Form a pair of simultaneous equations from the following statements. (i) A book costs Rs 50 more than a copy, and their combined price is Rs 115.

Answer: If the price of the book is x and of the copy is y, then x = y + 50 and x + y = 115.

Question: Exercise 6.4, Question 7: (ii) The sum of two numbers is 27 and the difference is 17.

Answer: If the numbers are x and y, then x + y = 27 and x − y = 17.

Question: Exercise 6.4, Question 7: (iii) One of two complementary angles is double the other.

Answer: If the angles are x and y, then x + y = 90° (complementary angles sum to 90°) and x = 2y.

Question: Exercise 6.4, Question 7: (iv) One pen costs Rs 50; two pens and five pencils cost Rs 200.

Answer: If the price of a pen is x and of a pencil is y, then x = 50 and 2x + 5y = 200.

Question: Exercise 6.5, Question 1: Solve the following equations by the method of equalizing coefficients (elimination). (i) 2x + y = 10, 5x − 2y = 7

Answer: Multiplying the first equation by 2 gives 4x + 2y = 20.

Adding the second gives 9x = 27, x = 3.

Then y = 10 − 6 = 4.

The solution set is {(3, 4)}.

Question: Exercise 6.5, Question 1: (ii) 5x − y = 3, 3x + y = 4

Answer: Adding the two gives 8x = 7, x = 7/8.

Then y = 4 − 21/8 = 11/8.

The solution set is {(7/8, 11/8)}.

Question: Exercise 6.5, Question 1: (iii) 3x + 7y = 27, 5x + 2y = 16

Answer: First × 2: 6x + 14y = 54.

Second × 7: 35x + 14y = 112.

Subtracting gives −29x = −58, x = 2.

Then 7y = 21, y = 3.

The solution set is {(2, 3)}.

Question: Exercise 6.5, Question 1: (iv) 2x + 3y = 2, 4x − 9y = −1

Answer: First × 3: 6x + 9y = 6.

Adding the second gives 10x = 5, x = 1/2.

Then 3y = 1, y = 1/3.

The solution set is {(1/2, 1/3)}.

Question: Exercise 6.5, Question 1: (v) 2x + y = 8, 6x − 2y = −1

Answer: First × 2: 4x + 2y = 16.

Adding the second gives 10x = 15, x = 3/2.

Then y = 5.

The solution set is {(3/2, 5)}.

Question: Exercise 6.5, Question 1: (vi) 2x + 3y = 14, 3x + 2y = 5

Answer: First × 3: 6x + 9y = 42.

Second × 2: 6x + 4y = 10.

Subtracting gives 5y = 32, y = 32/5.

Then 2x = 14 − 96/5 = −26/5, x = −13/5.

The solution set is {(−13/5, 32/5)}.

Question: Exercise 6.5, Question 2: Solve the following simultaneous equations by the substitution method. (i) 3x − 5y = 5, 2x + y = 12

Answer: From the second equation y = 12 − 2x.

Substituting in the first: 3x − 5(12 − 2x) = 5, i.e. 13x = 65, x = 5.

Then y = 2.

The solution set is {(5, 2)}.

Question: Exercise 6.5, Question 2: (ii) 4x + 3y = 2, 5x + 2y = −1

Answer: From the first, x = (2 − 3y)/4.

Substituting in the second: 5(2 − 3y)/4 + 2y = −1, i.e. 10 − 15y + 8y = −4, y = 2.

Then x = −1.

The solution set is {(−1, 2)}.

Question: Exercise 6.5, Question 2: (iii) 13x − 9y = 1, 11x − 12y = 14

Answer: From the first, x = (1 + 9y)/13.

Substituting in the second: 11(1 + 9y)/13 − 12y = 14, i.e. 11 + 99y − 156y = 182, y = −3.

Then x = −2.

The solution set is {(−2, −3)}.

Question: Exercise 6.5, Question 2: (iv) 7x − 4y = 1, 2x − 5y = 8

Answer: From the first, x = (1 + 4y)/7.

Substituting in the second: 2(1 + 4y)/7 − 5y = 8, i.e. 2 + 8y − 35y = 56, y = −2.

Then x = −1.

The solution set is {(−1, −2)}.

Question: Exercise 6.5, Question 2: (v) x + y = 1, 3x + 6y = 5

Answer: From the first, x = 1 − y.

Substituting in the second: 3(1 − y) + 6y = 5, i.e. 3y = 2, y = 2/3.

Then x = 1/3.

The solution set is {(1/3, 2/3)}.

Question: Exercise 6.5, Question 2: (vi) 3x − 7y = 1, 4x + y = 22

Answer: From the second, y = 22 − 4x.

Substituting in the first: 3x − 7(22 − 4x) = 1, i.e. 31x = 155, x = 5.

Then y = 2.

The solution set is {(5, 2)}.

Question: Exercise 6.5, Question 3: Solve the following equations by the cross-multiplication method. (i) 2x + 5y = 7, 5x − y = 4

Answer: Standard form: 2x + 5y − 7 = 0, 5x − y − 4 = 0.

x/(5×(−4) − (−7)(−1)) = y/((−7)(5) − 2(−4)) = 1/(2(−1) − 5×5), i.e. x/(−27) = y/(−27) = 1/(−27), so x = 1, y = 1.

The solution set is {(1, 1)}.

Question: Exercise 6.5, Question 3: (ii) −2x + 5y = 5, 3x − 2y = −13

Answer: Standard form: −2x + 5y − 5 = 0, 3x − 2y + 13 = 0.

x/(5×13 − (−5)(−2)) = y/((−5)(3) − (−2)(13)) = 1/((−2)(−2) − 5×3), i.e. x/55 = y/11 = 1/(−11), so x = −5, y = −1.

The solution set is {(−5, −1)}.

Question: Exercise 6.5, Question 3: (iii) 2x + y = 10, x + 3y = 8

Answer: Standard form: 2x + y − 10 = 0, x + 3y − 8 = 0.

x/(1×(−8) − (−10)(3)) = y/((−10)(1) − 2(−8)) = 1/(2×3 − 1×1), i.e. x/22 = y/6 = 1/5, so x = 22/5, y = 6/5.

The solution set is {(22/5, 6/5)}.

Question: Exercise 6.5, Question 3: (iv) x − 4y = 2, 2x − 7y = 5

Answer: Standard form: x − 4y − 2 = 0, 2x − 7y − 5 = 0.

x/((−4)(−5) − (−2)(−7)) = y/((−2)(2) − 1(−5)) = 1/(1(−7) − (−4)(2)), i.e. x/6 = y/1 = 1/1, so x = 6, y = 1.

The solution set is {(6, 1)}.

Question: Exercise 6.5, Question 3: (v) 10x + 2y = 8, 5x + y = 2

Answer: Standard form: 10x + 2y − 8 = 0, 5x + y − 2 = 0.

In cross-multiplication the denominator of the 1-term becomes 10×1 − 2×5 = 0.

Doubling the second equation gives 10x + 2y = 4, which contradicts the first (10x + 2y = 8) — the two lines are parallel, so there is no solution.

Question: Exercise 6.5, Question 3: (vi) 13x − y = 14, 2x + 7y = 9

Answer: Standard form: 13x − y − 14 = 0, 2x + 7y − 9 = 0.

x/((−1)(−9) − (−14)(7)) = y/((−14)(2) − 13(−9)) = 1/(13×7 − (−1)(2)), i.e. x/107 = y/89 = 1/93, so x = 107/93, y = 89/93.

The solution set is {(107/93, 89/93)}.

Question: Exercise 6.6, Question 1: Ayesha bought 6 kg of potatoes for Rs 300. If the price of tomatoes is three times the price of potatoes, find the price of tomatoes.

Answer: If the price of potatoes is x per kg and of tomatoes is y, then 6x = 300, i.e. x = 50.

Then y = 3x = 150.

The price of tomatoes is Rs 150 per kg.

Question: Exercise 6.6, Question 2: Students bought pens at Rs 100 each and balls at Rs 50 each. If they bought 8 items in total for Rs 600, tell how many pens and how many balls they bought?

Answer: If pens are x and balls are y, then x + y = 8 and 100x + 50y = 600.

Substituting x = 8 − y gives 100(8 − y) + 50y = 600, i.e. 800 − 50y = 600, y = 4.

Then x = 4.

They bought 4 pens and 4 balls.

Question: Exercise 6.6, Question 3: The length of a rectangular playground is double its width. Fencing the field costs Rs 9,000. If the cost is Rs 30 per metre, find the length and width of the field.

Answer: The perimeter = 9000 ÷ 30 = 300 m.

If length is l and width is w, then l = 2w and 2(l + w) = 300, i.e. l + w = 150.

Then 3w = 150, w = 50, l = 100.

The length is 100 m and the width is 50 m.

Question: Exercise 6.6, Question 4: In a school, the number of boys is double the number of girls. If the total number of students studying in the school is 450, find the number of boys.

Answer: If boys are b and girls are g, then b = 2g and b + g = 450.

Then 3g = 450, g = 150, b = 300.

The number of boys is 300.

Question: Exercise 6.6, Question 5: A worker's income-to-expense ratio is 15:14. If his saving is Rs 2000, find his income.

Answer: If income is 15x and expense is 14x, then saving = 15x − 14x = x = 2000.

Income = 15 × 2000 = Rs 30,000.

Question: Exercise 6.7, Question 1: Eliminate x from the following simultaneous equations by substitution. (i) x − y + 1 = 0, 3x + y − 3 = 0

Answer: From the first, x = y − 1.

Substituting in the second: 3(y − 1) + y − 3 = 0, i.e. 4y − 6 = 0, so 2y − 3 = 0.

Question: Exercise 6.7, Question 1: (ii) x + 5y = 2, 6x + 3y = 7

Answer: From the first, x = 2 − 5y.

Substituting in the second: 6(2 − 5y) + 3y = 7, i.e. 12 − 27y = 7, so 27y − 5 = 0.

Question: Exercise 6.7, Question 1: (iii) 2x + y = 7, x + 3y = 4

Answer: From the second, x = 4 − 3y.

Substituting in the first: 2(4 − 3y) + y = 7, i.e. 8 − 5y = 7, so 5y − 1 = 0.

Question: Exercise 6.7, Question 1: (iv) $x^2 + y = 0$, $ax^2 + b = 0$

Answer: From the first, $x^2 = -y$.

Substituting in the second: a(−y) + b = 0, i.e. ay − b = 0.

Question: Exercise 6.7, Question 1: (v) x − 2y + 1 = 0, 3x + y − 3 = 0

Answer: From the first, x = 2y − 1.

Substituting in the second: $3(2y - 1) + y - 3 = 0$, i.e. $7y - 6 = 0$.

Question: Exercise 6.7, Question 1: (vi) $4x + 3y + 8 = 0$, $x + 5y - 2 = 0$

Answer: From the second, $x = 2 - 5y$.

Substituting in the first: $4(2 - 5y) + 3y + 8 = 0$, i.e. $16 - 17y = 0$, so $17y - 16 = 0$.

Question: Exercise 6.7, Question 2: Eliminate V_i from the following simultaneous equations by substitution. (i) $V_f = V_i + at$, $S = V_i t + 1/2 at^2$

Answer: From the first, $V_i = V_f - at$.

Substituting in the second: $S = (V_f - at)t + (1/2)at^2 = V_f t - at^2 + (1/2)at^2 = V_f t - (1/2)at^2$.

Question: Exercise 6.7, Question 2: (ii) $V_f = V_i + gt$, $S = V_i t - 1/2\ gt^2$

Answer: From the first, $V_i = V_f - gt$.

Substituting in the second: $S = (V_f - gt)t - (1/2)gt^2 = V_f t - (3/2)gt^2$.

Question: Exercise 6.7, Question 2: (iii) $V_f^2 - V_i^2 = 2gS$, $V_f = V_i + gt$

Answer: From the second, $V_i = V_f - gt$.

Substituting in the first: $V_f^2 - (V_f - gt)^2 = 2gS$, i.e. $2V_f gt - g^2t^2 = 2gS$.

Dividing by $2g$ gives $S = V_f t - (1/2)gt^2$.

Question: Exercise 6.7, Question 3: Eliminate x from the following simultaneous equations using formulas. (i) $x + 1/x = 2a$, $x - 1/x = 2b + 1$

Answer: Squaring both gives $x^2 + 2 + 1/x^2 = 4a^2$ and $x^2 - 2 + 1/x^2 = (2b + 1)^2$.

Subtracting gives $4 = 4a^2 - (2b + 1)^2$.

Question: Exercise 6.7, Question 3: (ii) $x^2 - 1/x^2 = 2a$, $x^2 + 1/x^2 = 3b$

Answer: Squaring both gives $x^4 - 2 + 1/x^4 = 4a^2$ and $x^4 + 2 + 1/x^4 = 9b^2$.

Subtracting gives $-4 = 4a^2 - 9b^2$, i.e. $9b^2 - 4a^2 = 4$.

Question: Exercise 6.7, Question 3: (iii) $\sqrt{x} + 1/\sqrt{x} = p$, $x - 1/x = q^2$

Answer: Squaring the first gives $x + 2 + 1/x = p^2$.

$(\sqrt{x} - 1/\sqrt{x})^2 = x - 2 + 1/x = p^2 - 4$.

And $x - 1/x = (\sqrt{x} + 1/\sqrt{x})(\sqrt{x} - 1/\sqrt{x}) = q^2$, so $q^2 = p\sqrt{(p^2 - 4)}$, i.e. $q^4 = p^2(p^2 - 4)$.

Question: Exercise 6.7, Question 3: (iv) $x^2 + 1/x^2 = p^2$, $x^4 + 1/x^4 = q^4$

Answer: Squaring the first gives $x^4 + 2 + 1/x^4 = p^4$.

So $q^4 = p^4 - 2$.

Question: Exercise 6.7, Question 3: (v) $x + 1/x = 2y + 1$, $x - 1/x = y - 2$

Answer: Squaring gives $x^2 + 2 + 1/x^2 = (2y + 1)^2 = 4y^2 + 4y + 1$ and $x^2 - 2 + 1/x^2 = (y - 2)^2 = y^2 - 4y + 4$.

Subtracting gives $4 = 3y^2 + 8y - 3$, i.e. $3y^2 + 8y - 7 = 0$.

Question: Exercise 6.7, Question 3: (vi) $x^3/y^3 + y^3/x^3 = m$, $x^3/y^3 - y^3/x^3 = n$

Answer: Setting $A = x^3/y^3$ and $B = y^3/x^3$ gives $A + B = m$ and $A - B = n$, with $AB = 1$.

Squaring both and subtracting gives $4AB = m^2 - n^2$, i.e. $m^2 - n^2 = 4$.

Review Exercise 6, Question 1: Tick the correct answer. (i) $(x + y)^2 =$ _____

(a) $x^2 + y^2$

✓ (b) $x^2 + 2xy + y^2$

(c) $x^2 - y^2$

(d) both a and b

Review Exercise 6, Question 1: (ii) $x^2 - y^2 =$ _____

(a) $(x + y)^2$

(b) $(x - y)^2$

(c) $x^2 + y^2$

✓ (d) $(x - y)(x + y)$

Review Exercise 6, Question 1: (iii) $y^3 + 3y^2x + 3yx^2 + x^3 =$ _____

(a) $(y - x)^3$

(b) $(x - y)^3$

✓ (c) $(x + y)^3$

(d) none

Review Exercise 6, Question 1: (iv) _____ is a solution of the equation $x + y + 6 = 0$.

(a) (1, 2)

(b) (−2, 4)

(c) (−4, −2)

(d) (4, 2)

Review Exercise 6, Question 1: (v) _____ is the solution of the equations x + y = 5 and x − y = 1.

(a) (2, 3)

(b) (3, 2)

(c) (1, 4)

(d) (4, 1)

Question: Review Exercise 6, Question 2: If $x + 1/x = 3$ and $x - 1/x = 5$, find the value of the following. (i) $x^2 + 1/x^2$

Answer: From $x + 1/x = 3$, $x^2 + 1/x^2 = 9 - 2 = 7$.

[note: the two given conditions cannot hold together — they would give $x^2 + 1/x^2 = 7$ and 27 at the same time;

the values below use the "+" condition for "+" parts and the "−" condition for "−" parts.]

Question: Review Exercise 6, Question 2: (ii) $x^4 + 1/x^4$

Answer: $(x^2 + 1/x^2)^2 - 2 = 49 - 2 = 47$.

Question: Review Exercise 6, Question 2: (iii) $x^4 - 1/x^4$

Answer: $x^4 - 1/x^4 = (x^2 + 1/x^2)(x^2 - 1/x^2)$.

$x^2 - 1/x^2 = (x + 1/x)(x - 1/x) = 3 \times 5 = 15$, so the answer = $7 \times 15 = 105$.

Question: Review Exercise 6, Question 2: (iv) $x^3 + 1/x^3$

Answer: $(x + 1/x)^3 - 3(x + 1/x) = 27 - 9 = 18$.

Question: Review Exercise 6, Question 2: (v) $x^3 - 1/x^3$

Answer: $(x - 1/x)^3 + 3(x - 1/x) = 125 + 15 = 140$.

Question: Review Exercise 6, Question 2: (vi) $x^6 - 1/x^6$

Answer: $x^6 - 1/x^6 = (x^3 - 1/x^3)(x^3 + 1/x^3) = 140 \times 18 = 2520$.

Question: Review Exercise 6, Question 3: Find the value with the help of a formula. (i) $(103)^2$

Answer: $(103)^2 = (100 + 3)^2 = 10000 + 600 + 9 = 10609$.

Question: Review Exercise 6, Question 3: (ii) $(97)^2$

Answer: $(97)^2 = (100 - 3)^2 = 10000 - 600 + 9 = 9409$.

Question: Review Exercise 6, Question 3: (iii) $(103) \times (97)$

Answer: $(103)(97) = (100 + 3)(100 - 3) = 10000 - 9 = 9991$.

Question: Review Exercise 6, Question 4: Factorize. (i) $5x + 10y + 15z$

Answer: $5x + 10y + 15z = 5(x + 2y + 3z)$.

Question: Review Exercise 6, Question 4: (ii) $x^2 + 14x + 49$

Answer: $x^2 + 14x + 49 = (x + 7)^2$.

Question: Review Exercise 6, Question 4: (iii) $144x^2 - 121y^2$

Answer: $144x^2 - 121y^2 = (12x + 11y)(12x - 11y)$.

Question: Review Exercise 6, Question 4: (iv) $25x^2 - 10xy + y^2$

Answer: $25x^2 - 10xy + y^2 = (5x - y)^2$.

Question: Review Exercise 6, Question 4: (v) $a^2 + 4ab + 4b^2 - 9y^2$

Answer: $a^2 + 4ab + 4b^2 - 9y^2 = (a + 2b)^2 - (3y)^2 = (a + 2b + 3y)(a + 2b - 3y)$.

Question: Review Exercise 6, Question 4: (vi) $1 - 1/x^2$

Answer: $1 - 1/x^2 = (1 + 1/x)(1 - 1/x)$.

Question: Review Exercise 6, Question 5: Find the cube. (i) $2x + 3y$

Answer: $(2x + 3y)^3 = 8x^3 + 3(4x^2)(3y) + 3(2x)(9y^2) + 27y^3 = 8x^3 + 36x^2y + 54xy^2 + 27y^3$.

Question: Review Exercise 6, Question 5: (ii) $3x - 4y$

Answer: $(3x - 4y)^3 = 27x^3 - 3(9x^2)(4y) + 3(3x)(16y^2) - 64y^3 = 27x^3 - 108x^2y + 144xy^2 - 64y^3$.

Question: Review Exercise 6, Question 5: (iii) $x + 1/x$

Answer: $(x + 1/x)^3 = x^3 + 3x + 3/x + 1/x^3$.

Question: Review Exercise 6, Question 5: (iv) $x - 1/x$

Answer: $(x - 1/x)^3 = x^3 - 3x + 3/x - 1/x^3$.

Question: Review Exercise 6, Question 6: Find the cube using a formula. (i)

29

Answer: $29^3 = (30 - 1)^3 = 27000 - 2700 + 90 - 1 = 24389$.

Question: Review Exercise 6, Question 6: (ii) 31

Answer: $31^3 = (30 + 1)^3 = 27000 + 2700 + 90 + 1 = 29791$.

Question: Review Exercise 6, Question 7: Solve the following equations. (a) By the substitution method. (i) $2x - y = 6$, $x + 3y = 10$

Answer: From the first, $y = 2x - 6$.

Substituting in the second: $x + 3(2x - 6) = 10$, i.e. $7x = 28$, $x = 4$.

Then $y = 2$.

The solution set is $\{(4, 2)\}$.

Question: Review Exercise 6, Question 7: (a) By the substitution method. (ii) $2x + 5y = 1$, $x - 2y = 4$

Answer: From the second, $x = 4 + 2y$.

Substituting in the first: $2(4 + 2y) + 5y = 1$, i.e. $9y = -7$, $y = -7/9$.

Then $x = 22/9$.

The solution set is $\{(22/9, -7/9)\}$.

Question: Review Exercise 6, Question 7: (b) By the elimination method. (i) $3x + 2y = 7$, $2x + 3y = 8$

Answer: First × 3: 9x + 6y = 21.

Second × 2: 4x + 6y = 16.

Subtracting gives 5x = 5, x = 1.

Then y = 2.

The solution set is {(1, 2)}.

Question: Review Exercise 6, Question 7: (b) By the elimination method.

(ii) 3x − 2y = 7, 5x + y = 29

Answer: Second × 2: 10x + 2y = 58.

Adding the first gives 13x = 65, x = 5.

Then y = 4.

The solution set is {(5, 4)}.

Question: Review Exercise 6, Question 7: (c) By the cross-multiplication method. (i) 2x + y = 5, 3x − 2y = 4

Answer: Standard form: 2x + y − 5 = 0, 3x − 2y − 4 = 0.

x/(1×(−4) − (−5)(−2)) = y/((−5)(3) − 2(−4)) = 1/(2(−2) − 1×3), i.e.

x/(−14) = y/(−7) = 1/(−7), so x = 2, y = 1.

The solution set is {(2, 1)}.

Question: Review Exercise 6, Question 7: (c) By the cross-multiplication method. (ii) $x + y = 56$, $x + 2y = 29$

Answer: Standard form: $x + y - 56 = 0$, $x + 2y - 29 = 0$.

$x/(1\times(-29) - (-56)(2)) = y/((-56)(1) - 1(-29)) = 1/(1\times2 - 1\times1)$, i.e. $x/83 = y/(-27) = 1/1$, so $x = 83$, $y = -27$.

The solution set is $\{(83, -27)\}$.

Question: Review Exercise 6, Question 8: The width of a rectangle is half its length. If the perimeter of the rectangle is 24 m, find its area.

Answer: If width is w and length is l, then $w = l/2$ and $2(l + w) = 24$, i.e. $l + w = 12$.

Then $l + l/2 = 12$, $l = 8$, $w = 4$.

Area $= 8 \times 4 = 32$ square metres.

Question: Review Exercise 6, Question 9: Eliminate x using formulas. (i) $x - 1/x = 2k - 1$, $x + 1/x = 5k$

Answer: Squaring gives $x^2 - 2 + 1/x^2 = (2k - 1)^2 = 4k^2 - 4k + 1$ and $x^2 + 2 + 1/x^2 = 25k^2$.

Subtracting gives $-4 = 4k^2 - 4k + 1 - 25k^2$, i.e. $21k^2 + 4k - 5 = 0$.

Question: Review Exercise 6, Question 9: (ii) $x - 1/x = m$, $x^2 + 1/x^2 = n^2$

Answer: Squaring the first gives $x^2 - 2 + 1/x^2 = m^2$, i.e. $n^2 - 2 = m^2$, so $n^2 - m^2 = 2$.

7. Fundamentals of Geometry

Question: Exercise 7.1, Question 1: In each figure below, find the measure of ∠1, given that the measure of one angle is stated in each figure. (a) l and m are parallel; at the upper intersection the upper-right angle is 120°, and ∠1 is the lower-right angle at the lower intersection.

Answer: The vertical angle to the 120° angle at the upper intersection is 120°, and ∠1 is the same-side exterior angle to it, so ∠1 = 180° − 120° = 60°.

Question: Exercise 7.1, Question 1: (b) l and m are parallel; at the lower intersection the lower-left angle is 135°, and ∠1 is the upper-right angle at the upper intersection.

Answer: The vertical angle to the 135° angle at the lower intersection is 135° at the upper right, and ∠1 is its corresponding angle, so ∠1 = 135°.

Question: Exercise 7.1, Question 1: (c) l and m are parallel; with transversal n, at the lower intersection the lower-right angle is 60°, and ∠1 is the upper-right angle at the upper intersection.

Answer: ∠1 and the 60° angle are same-side exterior angles, so ∠1 = 180° − 60° = 120°.

Question: Exercise 7.1, Question 1: (d) Line n is cut by two parallel lines l and m; at the intersection with l the lower-left angle is 100°, and ∠1 is the lower-right angle at the intersection with m.

Answer: Since l and m are parallel, the corresponding angle to the 100° angle at the l-intersection is 100° at the lower left of the m-intersection,

and ∠1 is its linear-pair neighbour, so ∠1 = 180° − 100° = 80°.

Question: Exercise 7.1, Question 2: In the given figure, l ∥ m and m∠3 = 65° is given in the figure. Find the measures of all the angles.

Answer: ∠1 = ∠3 = 65° (vertical angles).

∠2 = ∠4 = 180° − 65° = 115° (neighbouring angles).

By corresponding angles: ∠5 = ∠1 = 65°, ∠6 = ∠2 = 115°, ∠7 = ∠3 = 65°, ∠8 = ∠4 = 115°.

Question: Exercise 7.1, Question 3: CD ∥ AB and EF cuts them at P and Q respectively, such that m∠PQD = 50°. Draw the figure according to the statement and find the measures of all the angles.

Answer: Draw two parallel lines CD (upper) and AB (lower) and cut them with transversal EF at P and Q, so that the given angle at Q is 50°.

At each intersection two angles are 50° and two are 130°: the given angle, its vertical opposite, its corresponding angle, and the vertical opposite of the corresponding angle — all four are 50°;

the other four angles (their neighbours) are 180° − 50° = 130° each.

Question: Exercise 7.1, Question 4: CD ∥ AB and EF cuts them at G and H respectively. State: (i) the angle equal to ∠GHC as a corresponding angle.

Answer: The corresponding angle to ∠GHC is ∠AGE (both are in the same corresponding upper-left position at their intersections).

Question: Exercise 7.1, Question 4: (ii) the angle equal to ∠GHC as an alternate angle.

Answer: The alternate interior angle to ∠GHC is ∠BGH (both lie between the parallel lines, on opposite sides of the transversal).

Question: Exercise 7.1, Question 4: (iii) the angle equal to ∠GHC as a vertically opposite angle.

Answer: The vertically opposite angle to ∠GHC is ∠DHF (the opposite angle at the same intersection).

Exercise 7.1, Question 4: (iv) the supplementary angle of ∠GHC when it is:

✓ (a) the same-side interior angle.

Question: Exercise 7.1, Question 4: (iv) (b) an interior angle.

Answer: (b) The neighbouring interior angle is ∠GHD, which together with ∠GHC makes 180° (a linear pair at the same intersection).

Question: Exercise 7.1, Question 4: (iv) (c) an exterior angle.

Answer: (c) The neighbouring exterior angle is ∠CHF, which together with ∠GHC makes 180° (a linear pair at the same intersection).

Question: Exercise 7.1, Question 5: In the following figures it is shown that l ∥ m and p ∥ q. Find the values of the marked angles x, y or z in them. (i) At the upper intersection the upper-right angle is 105°, and at the lower intersection the lower-right angle is (10x − 5)°.

Answer: The two angles are same-side exterior angles, so (10x − 5) + 105 = 180, i.e. 10x = 80, x = 8.

Question: Exercise 7.1, Question 5: (ii) l and m are cut by p and q; at the upper intersection with q the upper-right angle is (z − 5)°, at the upper intersection with p the lower-right angle is (5y)°, at the lower intersection with p the upper-left angle is (10x + 5)°, and at the lower intersection with q the lower-right angle is 55°.

Answer: For q: (z − 5) and 55 are same-side exterior angles, so z − 5 + 55 = 180, z = 130.

Since p ∥ q with transversal l, (5y)° equals the angle at the lower right of the q-intersection, i.e. 5y = 180 − (z − 5) = 55, y = 11.

From alternate interior angles with p, 5y = 10x + 5, i.e. 55 = 10x + 5, x = 5.

Question: Exercise 7.1, Question 5: (iii) l and m are cut by two parallel transversals; at the upper intersection with the left transversal the upper-right angle is 120°, at the lower intersection with the same transversal the lower-right angle is (y − 15)°, and at the lower intersection with the right transversal the lower-right angle is (z + 15)°.

Answer: 120° and (y − 15)° are same-side exterior angles, so y − 15 + 120 = 180, y = 75.

Since the two transversals are parallel, by corresponding angles with transversal m, (z + 15) = (y − 15) = 60, so z = 45.

Question: Exercise 7.1, Question 6: In the given parallelogram figure, AB ∥ DC and AD ∥ BC. Find the values of the marked angles x, y and z in the figure. (x at A, 100° at B, z at C, y at D.)

Answer: Consecutive angles are supplementary: x + 100° = 180°, so x = 80°.

Opposite angles are equal: z = x = 80°, y = 100°.

Question: Exercise 7.2, Question 1: Circle the wrong statements from the statements given below. (i) A rectangle is a parallelogram.

Answer: Correct — the opposite sides of a rectangle are parallel, so it is a parallelogram.

Question: Exercise 7.2, Question 1: (ii) A square is a parallelogram and at the same time also a rectangle.

Answer: Correct — the opposite sides of a square are parallel and all four of its angles are 90°.

Question: Exercise 7.2, Question 1: (iii) In a parallelogram, the pair of interior angles on the same side of a transversal is a pair of supplementary angles.

Answer: Correct — in a parallelogram, same-side interior angles together make 180°.

Question: Exercise 7.2, Question 1: (iv) If a transversal line does not meet two lines in the same plane at any point, then the alternate interior angles are always equal.

Answer: Wrong (circle it).

As printed, a transversal that never meets the lines forms no alternate interior angles at all;

and equality of alternate interior angles requires the two lines to be parallel.

Correct statement: if a transversal cuts two parallel lines in the same plane, the alternate interior angles are always equal.

[note: as printed, the statement omits the parallel condition.]

Question: Exercise 7.2, Question 1: (v) If the pair of same-side interior angles of a transversal is supplementary, then the two lines cut by the transversal do not intersect each other in the same plane.

Answer: Correct — same-side interior angles being supplementary is the condition for the lines to be parallel.

Question: Exercise 7.2, Question 1: (vi) If the diagonals of a quadrilateral bisect each other, then that quadrilateral is a parallelogram.

Answer: Correct — the diagonals bisecting each other is a property of a parallelogram.

Question: Exercise 7.2, Question 1: (vii) In a rectangle, its diagonals do not bisect each other.

Answer: Wrong (circle it).

The diagonals of a rectangle do bisect each other.

Correct statement: in a rectangle, its diagonals bisect each other.

Question: Exercise 7.2, Question 1: (viii) In a parallelogram, if a diagonal is

drawn, it forms equal pairs of alternate angles.

Answer: Correct — a diagonal of a parallelogram forms equal pairs of alternate interior angles.

Question: Exercise 7.2, Question 1: (ix) A square is a regular quadrilateral.

Answer: Correct — all four sides of a square are equal and all four angles are equal.

Question: Exercise 7.2, Question 1: (x) If two lines do not intersect each other, then those two lines may or may not be parallel lines.

Answer: Wrong (circle it).

By the book's definition, lines in the same plane that never meet are parallel — the "may or may not" does not hold in the book's plane geometry.

Correct statement: if two lines in the same plane do not intersect each other, they are parallel lines.

[note: skew lines in space are outside the book's scope.]

Question: Exercise 7.2, Question 2: The perimeter of parallelogram PQRS is 18 cm, while the length of one of its sides is double the other side. Find the measure of each side.

Answer: If one side is x and the other is 2x, the perimeter = 2(x + 2x) = 6x = 18, so x = 3.

The sides are 3 cm, 6 cm, 3 cm, 6 cm.

Question: Exercise 7.2, Question 3: In a parallelogram ABCD, the sum of a pair of opposite angles is 130°. Find the measure of each angle of the quadrilateral.

Answer: Opposite angles are equal, so each = 130° ÷ 2 = 65°.

Consecutive angles are supplementary: 180° − 65° = 115°.

The angles are 65°, 115°, 65°, 115°.

Question: Exercise 7.2, Question 4: In parallelogram WXYZ, if the measure of ∠X is double the measure of ∠W, find the measure of each angle of the quadrilateral.

Answer: ∠X and ∠W are consecutive angles, so ∠X + ∠W = 180°.

Since ∠X = 2∠W, 3∠W = 180°, ∠W = 60°, ∠X = 120°.

Opposite angles are equal: ∠Y = ∠W = 60°, ∠Z = ∠X = 120°.

Question: Exercise 7.2, Question 5: In a parallelogram ABCD, the length of the first diagonal AC is double the second diagonal BD. If the sum of the lengths of both diagonals of the quadrilateral is 9 cm, find the length of each diagonal. If the two diagonals intersect each other at P, find the lengths of PA and PB, and also find the relation mPA : mPB.

Answer: If BD = d, then AC = 2d.

d + 2d = 9, so d = 3: BD = 3 cm, AC = 6 cm.

The diagonals bisect each other, so PA = AC ÷ 2 = 3 cm, PB = BD ÷ 2 =

1.5 cm.

mPA : mPB = 3 : 1.5 = 2 : 1.

Question: Exercise 7.2, Question 6: State the definitions of the following:

(i) parallelogram.

Answer: A parallelogram is a quadrilateral whose opposite sides are parallel.

Question: Exercise 7.2, Question 6: (ii) regular pentagon.

Answer: A regular pentagon is a closed figure with all five sides equal and all five angles equal.

Question: Exercise 7.2, Question 6: (iii) regular octagon.

Answer: A regular octagon is a closed figure with all eight sides equal and all eight angles equal.

Question: Exercise 7.2, Question 6: (iv) chord.

Answer: A chord is the line segment joining two points of a circle.

Question: Exercise 7.2, Question 6: (v) secant.

Answer: A secant is the line that cuts a circle at two points.

Question: Exercise 7.2, Question 6: (vi) sector.

Answer: A sector is the part of a circle bounded by two radii and an arc.

Question: Exercise 7.2, Question 6: (vii) tangent line.

Answer: A tangent line is the line that touches the circle at exactly one

point.

Question: Exercise 7.2, Question 6: (viii) concyclic points.

Answer: Concyclic points are points that lie on the same circle.

Question: Exercise 7.2, Question 6: (ix) concentric circles.

Answer: Concentric circles are circles with the same centre but different radii.

Question: Exercise 7.2, Question 6: (x) diameter.

Answer: A diameter is the chord that passes through the centre of the circle;

it is double the radius.

Question: Exercise 7.2, Question 7: A circle has centre O, and its radius is 3 cm. P is a point at a distance of 3 cm from O. Q and R are in the same plane; the distance of Q from O is 2.5 cm and the distance of R from O is 4 cm. Tell what the position of each point is with respect to the given circle.

Answer: P's distance equals the radius (3 cm), so P is on the circle.

Q's distance is less than the radius (2.5 cm), so Q is inside the circle.

R's distance is greater than the radius (4 cm), so R is outside the circle.

Question: Exercise 7.2, Question 8: Circle the correct statements from the statements below, and if a statement is wrong, write it correctly. (i) A diameter which passes through the centre of the circle is called a chord.

Answer: Correct — a diameter is itself a chord (the longest chord, passing through the centre).

Question: Exercise 7.2, Question 8: (ii) The part of the circular region which lies between two radii and a sector is called an arc.

Answer: Wrong.

Correct statement: the part of the circular region which lies between two radii and an arc is called a sector.

[note: as printed, the words for sector and arc appear swapped.]

Question: Exercise 7.2, Question 8: (iii) A secant, i.e. a cutting line, is a line that cuts the circle at two points.

Answer: Correct — a secant cuts the circle at two points.

Question: Exercise 7.2, Question 8: (iv) Any point lying outside the circle is at a distance from the centre of the circle greater than the radius of the circle.

Answer: Correct — an exterior point is at a distance from the centre greater than the radius.

Question: Exercise 7.2, Question 8: (v) Three or four concyclic points, if a circle can pass through those points.

Answer: Correct — points through which a circle can pass are called concyclic points.

Question: Exercise 7.2, Question 8: (vi) The four angles of a parallelogram

are supplementary, i.e. a circle can pass through the four vertices of a parallelogram.

Answer: Wrong.

Correct statement: the consecutive (same-side) angles of a parallelogram are supplementary;

a circle can pass through all four vertices only when the sum of opposite angles is 180° (as in a rectangle).

Question: Exercise 7.2, Question 8: (vii) If there are any four concyclic points, then the pair of opposite angles of the quadrilateral related to those points is a pair of complementary angles.

Answer: Wrong.

Correct statement: if there are any four concyclic points, then the opposite angles of the quadrilateral related to those points are a pair of supplementary angles (their sum is 180°, not 90°).

Question: Exercise 7.2, Question 8: (viii) An arc of any circle is not a subset of the circle.

Answer: Wrong.

Correct statement: an arc of any circle is a subset of the circle (an arc is a part of the circumference).

Question: Exercise 7.2, Question 8: (ix) A circle's diameter is double its radius.

Answer: Correct — diameter = 2 × radius.

Question: Exercise 7.2, Question 8: (x) The tangent line of a circle passes through the centre of the line parallel to it.

Answer: Wrong.

Correct statement: the tangent line of a circle touches the circle at exactly one point and is perpendicular to the radius at the point of contact;

it does not pass through the centre.

[note: the printed wording of this item is garbled.]

Question: Review Exercise 7, Question 1: Fill in the blanks below with suitable words. (i) Two lines which lie in the same plane and never meet are called parallel.

Answer: Completed sentence: "Two lines which lie in the same plane and never meet are called parallel." Filled words: in the same plane, never meeting.

Question: Review Exercise 7, Question 1: (ii) Two lines which are parallel to the same line are also parallel to each other.

Answer: Completed sentence: "Two lines which are parallel to the same line are also parallel to each other." Filled words: two, parallel, parallel.

Question: Review Exercise 7, Question 1: (iii) A line which passes through the midpoint of one side of a triangle and is parallel to the second side bisects the third side.

Answer: Completed sentence: "A line which passes through the midpoint of one side of a triangle and is parallel to the second side bisects the third side." Filled words: midpoint, parallel to the second side.

Question: Review Exercise 7, Question 1: (iv) When two parallel lines are cut by a transversal, the pairs of angles formed are as follows: (a) corresponding angles (b) alternate angles (c) same-side interior angles.

Answer: Filled answer: (a) corresponding angles (b) alternate angles (c) same-side interior angles.

Question: Review Exercise 7, Question 1: (v) When a transversal cuts two parallel lines, then (a) pairs of corresponding angles are equal (b) pairs of alternate angles are equal (c) pairs of same-side interior angles are supplementary.

Answer: Filled answer: (a) corresponding (b) alternate (c) same-side interior.

Question: Review Exercise 7, Question 1: (vi) A polygon is a closed figure consisting of three or more than three sides.

Answer: Completed sentence: "A polygon is a closed figure consisting of three or more than three sides." Filled words: three, three.

Question: Review Exercise 7, Question 1: (vii) A parallelogram is a quadrilateral whose opposite sides are parallel and equal.

Answer: Completed sentence: "A parallelogram is a quadrilateral whose opposite sides are parallel and equal." Filled words: parallel, equal.

Question: Review Exercise 7, Question 1: (viii) The quadrilateral whose

diagonals bisect each other is a parallelogram.

Answer: Completed sentence: "The quadrilateral whose diagonals bisect each other is a parallelogram." Filled word: parallelogram.

Question: Review Exercise 7, Question 1: (ix) In a parallelogram, the same-side (consecutive) angles are supplementary.

Answer: Completed sentence: "In a parallelogram, the same-side (consecutive) angles are supplementary." Filled words: same-side, consecutive.

Question: Review Exercise 7, Question 1: (x) A polygon whose all sides are equal is called a regular polygon.

Answer: Completed sentence: "A polygon whose all sides are equal is called a regular polygon." Filled word: regular.

Question: Review Exercise 7, Question 1: (xi) A circle is the set of points which are at an equal distance from a fixed point, i.e. the centre.

Answer: Completed sentence: "A circle is the set of points which are at an equal distance from a fixed point, i.e. the centre." Filled word: circle.

Question: Review Exercise 7, Question 1: (xii) A line segment which joins two points of a circle is called a chord.

Answer: Completed sentence: "A line segment which joins two points of a circle is called a chord." Filled word: chord.

Question: Review Exercise 7, Question 2: Look at the two parallelograms given below. m∠P = 70°, mPS = 3 cm, mRS = 4 cm, RS ∥ CD and PS ∥ AD. Find the measures of the remaining angles and sides of parallelogram

PQRS.

Answer: Opposite angles are equal: ∠R = ∠P = 70°.

Consecutive angles are supplementary: ∠Q = ∠S = 180° − 70° = 110°.

Opposite sides are equal: PQ = RS = 4 cm, QR = PS = 3 cm.

Question: Review Exercise 7, Question 2: If the sides of parallelogram ABCD are 1 1/2 times the corresponding sides of parallelogram PQRS, find the measures of the angles of ABCD. Is (m∠A = 3/2 m∠P)?

Answer: Since the corresponding sides are parallel and in the same ratio (1 1/2), the two quadrilaterals are similar, so corresponding angles are equal: ∠A = ∠P = 70°, ∠C = 70°, ∠B = ∠D = 110°.

Making the sides bigger does not change the angles, so the claim is wrong: m∠A = 70°, not 3/2 × 70° = 105°.

Question: Review Exercise 7, Question 3: In parallelogram ABCD, side AD is extended to E and side DC is extended to F. m∠CDE = 65°. Find the measures of the remaining angles.

Answer: ∠ADC and ∠CDE are neighbouring angles on the same straight line: ∠ADC = 180° − 65° = 115°.

Opposite angles are equal: ∠ABC = ∠ADC = 115°, ∠BCD = ∠DAB.

Consecutive angles are supplementary: ∠DAB = 180° − 115° = 65°, ∠BCD = 65°.

(By corresponding angles also ∠DAB = ∠CDE = 65°.)

Question: Review Exercise 7, Question 3: Write the pairs of angles in the figure above: (i) corresponding angles.

Answer: (i) Corresponding angles: (∠DAB, ∠CDE) = 65°, (∠ABC, ∠BCF) = 115°, where ∠BCF is the exterior angle at C.

Question: Review Exercise 7, Question 3: (ii) alternate angles.

Answer: (ii) Alternate angles: with transversal AD, the alternate interior angle to ∠DAB is the 65° angle at D between AD and the leftward extension of CD;

with transversal CD, the alternate interior angle to ∠BCD is the 65° angle at B between BC and the rightward extension of AB.

Question: Review Exercise 7, Question 3: (iii) pairs of supplementary angles.

Answer: (iii) Pairs of supplementary angles: (∠DAB, ∠ABC), (∠ABC, ∠BCD), (∠BCD, ∠CDA), (∠CDA, ∠DAB) — each pair 65° + 115° = 180°;

and (∠ADC, ∠CDE), (∠BCD, ∠BCF) — the exterior-angle pairs are also 180°.

Question: Review Exercise 7, Question 4: Name the terms of the following figures and write definitions: (i) parallel lines.

Answer: Parallel lines are lines that lie in the same plane and never meet.

Question: Review Exercise 7, Question 4: (ii) parallelogram.

Answer: A parallelogram is a quadrilateral whose opposite sides are parallel.

Question: Review Exercise 7, Question 4: (iii) arc.

Answer: An arc is a part of the circumference of a circle, i.e. the part between two points on the circle.

Question: Review Exercise 7, Question 4: (iv) secant.

Answer: A secant is the line that cuts a circle at two points.

Question: Review Exercise 7, Question 4: (v) tangent line.

Answer: A tangent line is the line that touches the circle at exactly one point.

Question: Review Exercise 7, Question 4: (vi) concyclic points.

Answer: Concyclic points are points that lie on the same circle.

Question: Review Exercise 7, Question 4: (vii) concentric circles.

Answer: Concentric circles are circles with the same centre but different radii.

Question: Review Exercise 7, Question 5: Write the definitions of the following polygons and find the interior angle in each case: (i) regular pentagon.

Answer: A regular pentagon is a closed figure with all five sides equal and all five angles equal.

The sum of interior angles S = (2n − 4) right angles = (10 − 4) × 90° =

540°.

Each interior angle = 540° ÷ 5 = 108°.

Question: Review Exercise 7, Question 5: (ii) regular hexagon.

Answer: A regular hexagon is a closed figure with all six sides equal and all six angles equal.

The sum of interior angles S = (12 − 4) × 90° = 720°.

Each interior angle = 720° ÷ 6 = 120°.

Question: Review Exercise 7, Question 5: (iii) regular octagon.

Answer: A regular octagon is a closed figure with all eight sides equal and all eight angles equal.

The sum of interior angles S = (16 − 4) × 90° = 1080°.

Each interior angle = 1080° ÷ 8 = 135°.

Question: Review Exercise 7, Question 6: (i) A circle has a radius of 4 cm; what will be the length of its diameter?

Answer: Diameter = 2 × radius = 2 × 4 = 8 cm.

Question: Review Exercise 7, Question 6: (ii) Three concentric circles are drawn with centre O; their diameters are 5 cm, 7 cm, 10 cm respectively. Find their radii.

Answer: Radius = diameter ÷ 2: first circle 5 ÷ 2 = 2.5 cm, second 7 ÷ 2

= 3.5 cm, third 10 ÷ 2 = 5 cm.

Question: Review Exercise 7, Question 6: (iii) How many circles can pass through the following: (a) three non-collinear points (b) four concyclic points.

Answer: (a) Exactly one circle can pass through three non-collinear points.

(b) Exactly one circle can pass through four concyclic points too — the circle on which all four points lie.

(Four arbitrary points need not be concyclic;

then no circle passes through them.)

8. Practical Geometry

Question: Exercise 8.1, Question 1: Draw the following converging lines in your scale notebook and (i) estimate the angle associated with the converging lines without extending them, (ii) find its measure. Figure (i): lines PQ and RS.

Answer: By the book's method (draw any transversal and subtract the sum of the two same-side angles from 180°): in the printed figure, line PQ makes an angle of about 25° with the horizontal RS.

So the meeting angle is estimated at about 25°, and measuring with a protractor also gives about 25°.

[note: the value is read from the printed figure with a protractor.]

Question: Exercise 8.1, Question 1: Figure (ii): lines AB and CD.

Answer: By the book's method: in the printed figure, line CD makes an angle of about 25° with the horizontal AB.

So the meeting angle is estimated at about 25°, and measuring with a protractor also gives about 25°.

[note: the value is read from the printed figure with a protractor.]

Question: Exercise 8.1, Question 2: Construct square ABCD with the following measure: (a) (i) mAC = 6 cm (diagonal).

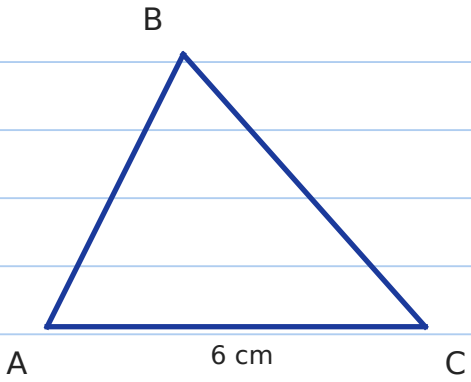
Answer: (1) Draw the diagonal segment AC = 6 cm.

(2) Draw the perpendicular bisector of AC, cutting AC at its midpoint P.

(3) With centre P and radius PA = 3 cm, draw arcs cutting the perpendicular bisector at B and D.

(4) Join AB, BC, CD, DA.

Check: the side of the square = 6 ÷ √2 ≈ 4.24 cm.



mAC = 6 cm

Question: Exercise 8.1, Question 2: (a) (ii) mBC = 7 cm (side).

Answer: (1) Draw side BC = 7 cm.

(2) Draw perpendiculars at B and C.

(3) With centres B and C and radius 7 cm, draw arcs cutting the perpendiculars at A and D.

(4) Join AB, CD, DA.

Check: all sides are 7 cm and all angles are 90°.

Question: Exercise 8.1, Question 2: (a) (iii) mAC = 8 cm (diagonal).

Answer: (1) Draw diagonal AC = 8 cm.

(2) Draw the perpendicular bisector of AC, cutting AC at its midpoint P.

(3) With centre P and radius PA = 4 cm, draw arcs cutting the perpendicular bisector at B and D.

(4) Join AB, BC, CD, DA.

Check: the side of the square = 8 ÷ √2 ≈ 5.66 cm.

Question: Exercise 8.1, Question 2: (b) Construct square PQRS in which the measure of one of its sides is: (i) 2.5 cm.

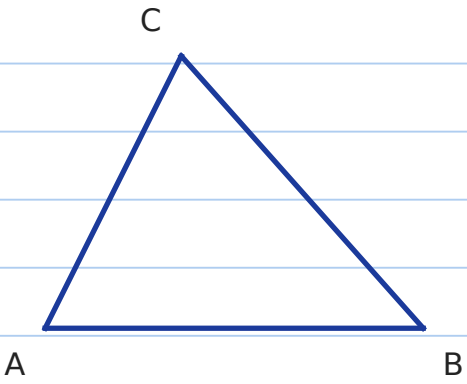
Answer: (1) Draw side PQ = 2.5 cm.

(2) Draw perpendiculars at P and Q.

(3) With centres P and Q and radius 2.5 cm, draw arcs cutting the perpendiculars at S and R.

(4) Join PS, SR, RQ.

Check: all sides are 2.5 cm and all angles are 90°.



Question: Exercise 8.1, Question 2: (b) (ii) 3 cm.

Answer: Construct the square the same way with side PQ = 3 cm: perpendiculars at P and Q, arcs of radius 3 cm, then join S and R.

Check: all sides are 3 cm.

Question: Exercise 8.1, Question 2: (b) (iii) 3.5 cm.

Answer: Construct the square the same way with side PQ = 3.5 cm: perpendiculars at P and Q, arcs of radius 3.5 cm, then join S and R.

Check: all sides are 3.5 cm.

Question: Exercise 8.1, Question 2: (b) (iv) 4.5 cm.

Answer: Construct the square the same way with side PQ = 4.5 cm: perpendiculars at P and Q, arcs of radius 4.5 cm, then join S and R.

Check: all sides are 4.5 cm.

Question: Exercise 8.1, Question 2: (c) Construct square WXYZ whose one side measures: (i) 10 cm.

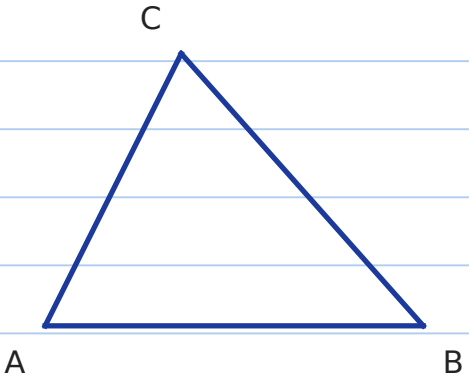
Answer: (1) Draw side WX = 10 cm.

(2) Draw perpendiculars at W and X.

(3) With centres W and X and radius 10 cm, draw arcs cutting the perpendiculars at Z and Y.

(4) Join WZ, ZY, YX.

Check: all sides are 10 cm and all angles are 90°.



Question: Exercise 8.1, Question 2: (c) (ii) 12 cm.

Answer: Construct the square the same way with side WX = 12 cm: perpendiculars at W and X, arcs of radius 12 cm, then join Z and Y.

Check: all sides are 12 cm.

Question: Exercise 8.1, Question 2: (c) (iii) 11 cm.

Answer: Construct the square the same way with side WX = 11 cm: perpendiculars at W and X, arcs of radius 11 cm, then join Z and Y.

Check: all sides are 11 cm.

Question: Exercise 8.1, Question 3: Construct rectangle PQRS whose two sides measure: (a) (i) mPQ = 6 cm, mQR = 4 cm.

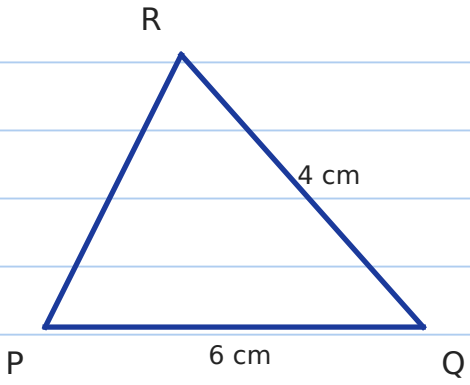
Answer: (1) Draw side PQ = 6 cm.

(2) Draw perpendiculars at P and Q.

(3) With centres P and Q and radius 4 cm, draw arcs cutting the perpendiculars at S and R.

(4) Join PS, SR, RQ.

Check: diagonal PR = √(36 + 16) = √52 ≈ 7.21 cm.



mPQ = 6 cm, mQR = 4 cm

Question: Exercise 8.1, Question 3: (a) (ii) mPQ = 4 cm, mQR = 7 cm.

Answer: The same way: draw PQ = 4 cm, perpendiculars at P and Q, arcs of radius 7 cm for S and R, then join.

Check: diagonal = √(16 + 49) = √65 ≈ 8.06 cm.

Question: Exercise 8.1, Question 3: Construct rectangle ABCD, given one side and the diagonal as follows: (b) (i) mAC = 6 cm, mAB = 4 cm.

Answer: (1) Draw side AB = 4 cm.

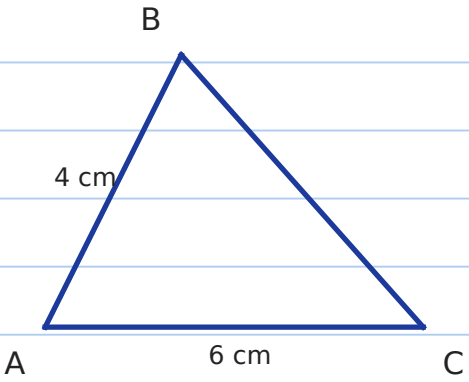
(2) Draw a perpendicular at B.

(3) With centre A and radius 6 cm, draw an arc cutting the perpendicular at C.

(4) Draw a perpendicular at C and, with centre A and radius equal to BC, draw an arc cutting it at D.

(5) Join AD, DC.

Check: BC = √(36 − 16) = √20 ≈ 4.47 cm.



mAC = 6 cm, mAB = 4 cm

Question: Exercise 8.1, Question 3: (b) (ii) mBD = 7 cm, mAB = 5 cm.

Answer: The same way: draw AB = 5 cm, a perpendicular at B, get D with an arc of radius 7 cm centred at A, then complete the rectangle.

Check: AD = √(49 − 25) = √24 ≈ 4.90 cm.

Question: Exercise 8.1, Question 4: Construct rhombus ABCD with one side and one angle measuring as follows: (a) (i) side = 4 cm, m∠A = 120°.

Answer: (1) Draw AB = 4 cm.

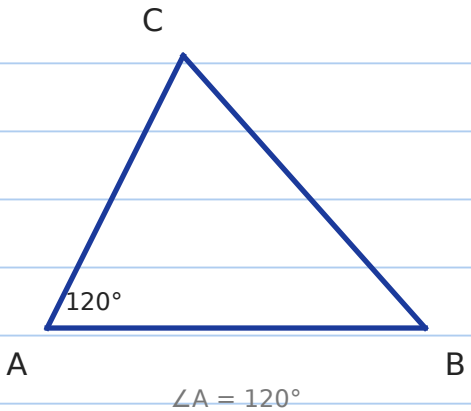
(2) Make a 120° angle at A and cut AD = 4 cm on its arm.

(3) With centres B and D and radius 4 cm, draw arcs meeting at C.

(4) Join BC, DC.

Check: ∠B = ∠D = 60°, ∠C = 120°;

diagonals AC = 4 cm, BD = 4√3 ≈ 6.93 cm.



∠A = 120°

Question: Exercise 8.1, Question 4: (a) (ii) mAB = 5 cm, m∠A = 60°.

Answer: The same way: draw AB = 5 cm, make a 60° angle at A and cut AD = 5 cm, get C from arcs of radius 5 cm centred at B and D, then join.

Check: ∠B = ∠D = 120°, ∠C = 60°;

diagonals BD = 5 cm, AC = 5√3 ≈ 8.66 cm.

Question: Exercise 8.1, Question 4: Construct rhombus PQRS with one side and one diagonal measuring as follows: (b) (i) mPQ = 5 cm, mQS = 4 cm.

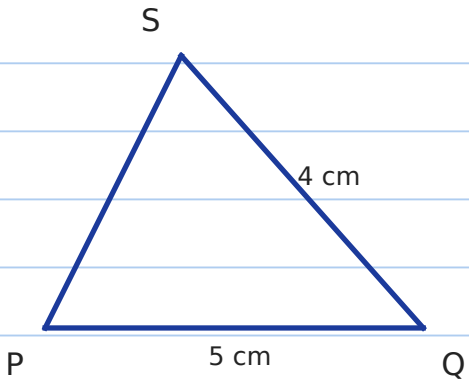
Answer: (1) Draw diagonal QS = 4 cm.

(2) Draw the perpendicular bisector of QS, cutting QS at its midpoint O.

(3) With centres Q and S and radius 5 cm, draw arcs cutting the perpendicular bisector at P and R.

(4) Join PQ, QR, RS, SP.

Check: the other diagonal PR = $2\sqrt{(25 - 4)} = 2\sqrt{21} \approx 9.17$ cm.



mPQ = 5 cm, mQS = 4 cm

Question: Exercise 8.1, Question 4: (b) (ii) mPQ = 4.5 cm, mPR = 7 cm.

Answer: The same way: draw diagonal PR = 7 cm, its perpendicular bisector, get Q and S from arcs of radius 4.5 cm centred at P and R, then join.

Check: the other diagonal QS = $2\sqrt{(20.25 - 12.25)} = 2\sqrt{8} \approx 5.66$ cm.

Question: Exercise 8.1, Question 5: Construct parallelogram PQRS given

one side, one diagonal, and the angle between the diagonals as follows:

(a) (i) mQR = 8 cm, mQS = 6 cm, m∠QOR = 60° (O is the intersection of the diagonals).

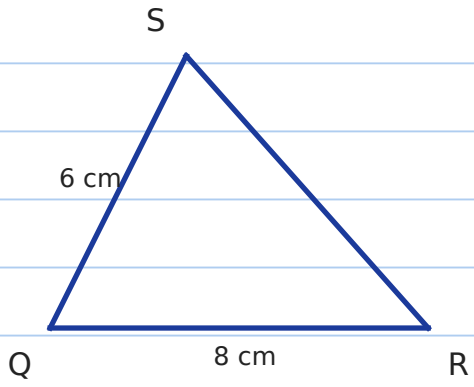
Answer: (1) Draw diagonal QS = 6 cm and mark its midpoint O.

(2) Make a 60° angle with QS at O and cut OR = 7 cm on its arm (in triangle OQR: OQ = 3, QR = 8, ∠QOR = 60° gives OR = 7).

(3) With centre O, extend OR to the other side to get P (OP = OR).

(4) Join PQ, QR, RS, SP.

Check: PR = 14 cm, PQ ≈ 8.89 cm.



mQR = 8 cm, mQS = 6 cm

Question: Exercise 8.1, Question 5: (a) (ii) mPR = 6 cm, mQS = 8 cm, m∠QOR = 120°.

Answer: (1) Cross the two diagonals at their midpoint O at a 120° angle, so that OP = OR = 3 cm and OQ = OS = 4 cm.

(2) Join P, Q, R, S in order.

Check: the sides are PQ = √(9 + 16 − 2×3×4×cos120°) = √37 ≈ 6.08 cm and QR = √(9 + 16 − 2×3×4×cos60°) = √13 ≈ 3.61 cm.

Question: Exercise 8.1, Question 5: Construct parallelogram ABCD with two adjacent sides and the included angle measuring as follows: (b) (i) mAB = 6 cm, mBC = 4 cm, m∠B = 70°.

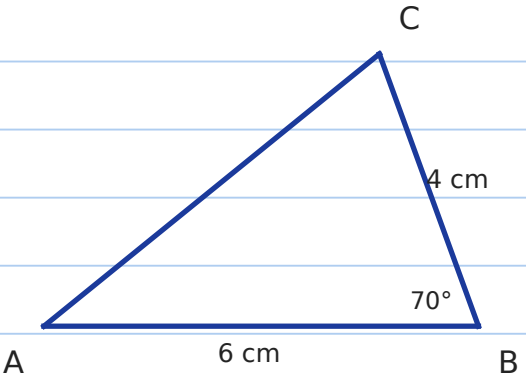
Answer: (1) Draw AB = 6 cm.

(2) Make a 70° angle at B and cut BC = 4 cm on its arm.

(3) With centres A and C and radii equal to BC and AB respectively, draw arcs meeting at D.

(4) Join AD, DC.

Check: diagonal AC = √(36 + 16 − 2×6×4×cos70°) ≈ 5.96 cm.



Question: Exercise 8.1, Question 5: (b) (ii) mAB = 4 cm, mAD = 5 cm, m∠A = 100°.

Answer: The same way: draw AB = 4 cm, make a 100° angle at A and cut AD = 5 cm, get C from arcs of radii AD and AB centred at B and D, then

join.

Check: diagonal BD = √(16 + 25 − 2×4×5×cos100°) ≈ 6.92 cm.

Question: Exercise 8.1, Question 6: Construct kite ABCD with two pairs of equal sides and one diagonal measuring as follows: (i) mAB = 4 cm, mBC = 7 cm, mAC = 9 cm.

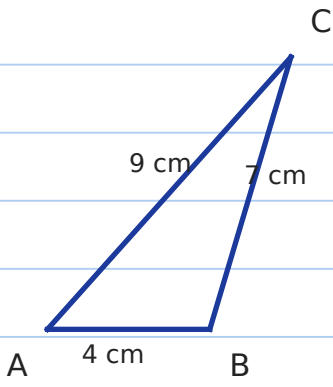
Answer: (1) Draw diagonal AC = 9 cm.

(2) With centres A and C and radius 4 cm, draw arcs meeting at B.

(3) With centres A and C and radius 7 cm, draw arcs meeting at D (on the opposite side from B).

(4) Join AB, BC, CD, DA.

Check: the other diagonal BD ≈ 5.96 cm (the diagonals cut each other perpendicularly).



Question: Exercise 8.1, Question 6: (ii) mAB = 4 cm, mBC = 6 cm, mBD = 4 cm.

Answer: (1) Draw diagonal BD = 4 cm.

(2) With centres B and D and radius 4 cm, draw arcs meeting at A.

(3) With centres B and D and radius 6 cm, draw arcs meeting at C (on the opposite side from A).

(4) Join AB, BC, CD, DA.

Check: the other diagonal AC ≈ 9.12 cm.

Question: Exercise 8.1, Question 7: Construct regular pentagon ABCDE with each side measuring: (i) 3.5 cm.

Answer: (1) Draw AB = 3.5 cm.

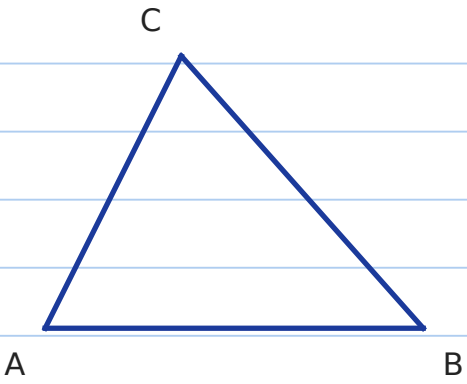
(2) Make interior angles of 108° at A and B (each interior angle of a regular pentagon = (2×5 − 4) × 90° ÷ 5 = 108°).

(3) Cut AE = BC = 3.5 cm on those arms.

(4) With centres E and C and radius 3.5 cm, draw arcs meeting at D.

(5) Join ED, DC.

Check: diagonal ≈ 1.618 × 3.5 ≈ 5.66 cm.



Question: Exercise 8.1, Question 7: (ii) 4 cm.

Answer: The same way with side 4 cm: AB = 4 cm, 108° angles at A and B, AE = BC = 4 cm, D from arcs of radius 4 cm centred at E and C, then join.

Check: diagonal ≈ 6.47 cm.

Question: Exercise 8.1, Question 7: (iii) 4.5 cm.

Answer: The same way with side 4.5 cm: AB = 4.5 cm, 108° angles at A and B, AE = BC = 4.5 cm, D from arcs of radius 4.5 cm centred at E and C, then join.

Check: diagonal ≈ 7.28 cm.

Question: Exercise 8.1, Question 7: (iv) 5 cm.

Answer: The same way with side 5 cm: AB = 5 cm, 108° angles at A and B, AE = BC = 5 cm, D from arcs of radius 5 cm centred at E and C, then join.

Check: diagonal ≈ 8.09 cm.

Question: Exercise 8.1, Question 8: Construct regular hexagon ABCDEF

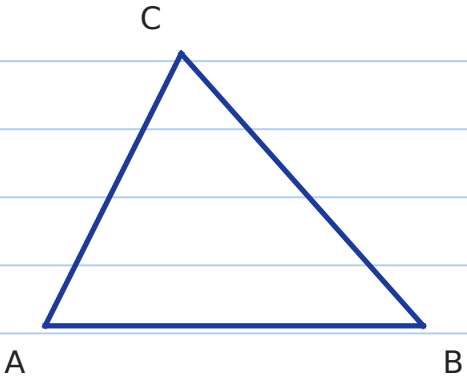
with each side measuring: (i) 3 cm.

Answer: (1) Take centre O and draw a circle of radius 3 cm (in a regular hexagon the circumradius equals the side).

(2) Take any point A on the circle and, with the same radius (3 cm), mark off arcs on the circle to get B, C, D, E, F.

(3) Join AB, BC, CD, DE, EF, FA.

Check: each interior angle is 120°.



Question: Exercise 8.1, Question 8: (ii) 3.5 cm.

Answer: The same way with a circle of radius 3.5 cm: from A mark off arcs of the same radius to get B, C, D, E, F, then join.

Check: each side is 3.5 cm and each interior angle is 120°.

Question: Exercise 8.1, Question 8: (iii) 2.5 cm.

Answer: The same way with a circle of radius 2.5 cm: from A mark off arcs of the same radius to get B, C, D, E, F, then join.

Check: each side is 2.5 cm and each interior angle is 120°.

Question: Exercise 8.1, Question 8: (iv) 4 cm.

Answer: The same way with a circle of radius 4 cm: from A mark off arcs of the same radius to get B, C, D, E, F, then join.

Check: each side is 4 cm and each interior angle is 120°.

Question: Exercise 8.2, Question 1: Construct the following right triangles when: (a) hypotenuse = 10 cm and the measure of one of its sides = 8 cm.

Answer: (1) Draw hypotenuse BC = 10 cm.

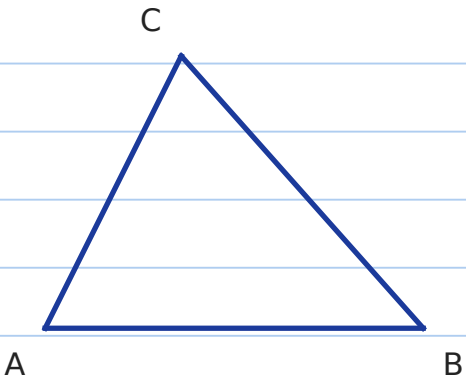
(2) Taking BC as diameter, draw a semicircle on it.

(3) With centre B and radius 8 cm, draw an arc cutting the semicircle at A.

(4) Join AB, AC.

Check: ∠A = 90° (angle in a semicircle);

the third side AC = √(100 − 64) = 6 cm.



Question: Exercise 8.2, Question 1: (b) hypotenuse = 13 cm and one of its sides = 12 cm.

Answer: The same way: a semicircle on BC = 13 cm, A from an arc of radius 12 cm centred at B, then join AB, AC.

Check: ∠A = 90°;

the third side = √(169 − 144) = 5 cm.

Question: Exercise 8.2, Question 1: (c) one side PQ = 5 cm and side SQ = 6 cm.

Answer: (1) Draw PQ = 5 cm.

(2) Draw a perpendicular at Q and cut QS = 6 cm on it.

(3) Join P and S.

Check: ∠Q = 90°;

hypotenuse PS = √(25 + 36) = √61 ≈ 7.81 cm.

Question: Exercise 8.2, Question 2: Construct the following right triangles when: (a) hypotenuse = 10 cm and the perpendicular height from the hypotenuse is 3 cm.

Answer: (1) Draw hypotenuse BC = 10 cm.

(2) Taking BC as diameter, draw a semicircle on it.

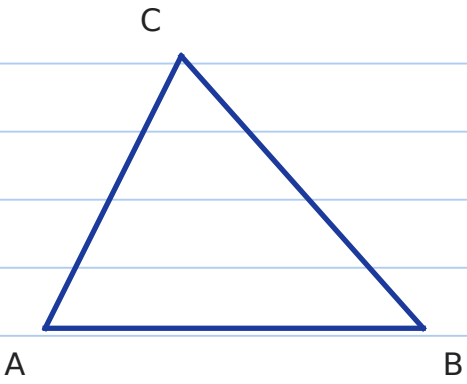
(3) Draw a line parallel to BC at a distance of 3 cm from BC, cutting the semicircle at A.

(4) Join AB, AC.

Check: ∠A = 90°;

the altitude from A to BC is 3 cm;

the sides are √10 ≈ 3.16 cm and 3√10 ≈ 9.49 cm.



Question: Exercise 8.2, Question 2: (b) hypotenuse = 12 cm and the perpendicular height from the hypotenuse is 4 cm.

Answer: The same way: a semicircle on BC = 12 cm, get A from a line

parallel to BC at a distance of 4 cm, then join AB, AC.

Check: ∠A = 90°;

the altitude is 4 cm;

the sides are about 4.28 cm and 11.21 cm.

Question: Exercise 8.2, Question 2: (c) perpendicular height PM = 3 cm and perpendicular height SN = 3.5 cm.

Answer: (1) Draw base SN = 3.5 cm.

(2) Taking SN as diameter, draw a semicircle on it.

(3) Draw a line parallel to SN at a distance of 3 cm from SN, cutting the semicircle at P (PM = 3 cm is the altitude).

(4) Join PS, PN.

Check: ∠SPN = 90° (angle in a semicircle);

the altitude from P to SN is 3 cm.

Question: Exercise 8.2, Question 3: Construct right triangles ΔABC and ΔDBC with mBC = 10 cm as the common hypotenuse, when: (a) altitude AL = 3.5 cm and mDC = 5 cm.

Answer: (1) Draw the common hypotenuse BC = 10 cm.

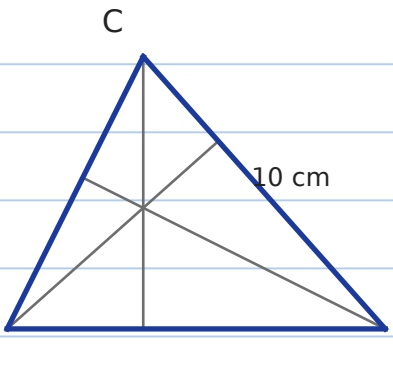
(2) Taking BC as diameter, draw a circle on it.

(3) Draw a line parallel to BC at a distance of 3.5 cm from BC, cutting the circle at A (on one side) — ΔABC is formed.

(4) With centre C and radius 5 cm, draw an arc cutting the circle at D (on the other side) — ΔDBC is formed.

Check: both triangles are right-angled on BC;

in ΔDBC, DB = √(100 − 25) = √75 ≈ 8.66 cm.



mBC = 10 cm, mDC = 5 cm, mAL = 3.5 cm

Question: Exercise 8.2, Question 3: (b) altitude AL = 3 cm and mDB = 6 cm.

Answer: The same way: a circle on BC = 10 cm, A from a line parallel to BC at a distance of 3 cm, and D on the other side from an arc of radius 6 cm centred at B, then join.

Check: both triangles are right-angled on BC;

in ΔDBC, DC = √(100 − 36) = 8 cm.

Review Exercise 8, Question 1: For each statement, answers are given; mark () any one correct answer. (i) The measure of the angle associated with converging lines (without extending them) equals:

(a) the difference of the measures of the two angles

(b) the difference of 180° and the sum of the two angles

✓ (c) 180° − (any transversal line cutting the two lines and the pair of associated angles from it)

(d) none of these.

Review Exercise 8, Question 1: (ii) The polygon whose sum of all interior angles is four right angles is _____.

(a) triangle

✓ (b) quadrilateral

(c) pentagon

(d) hexagon.

Review Exercise 8, Question 1: (iii) A polygon whose sum of all interior angles is two right angles is _____.

✓ (a) triangle

(b) quadrilateral

(c) pentagon

(d) hexagon.

Review Exercise 8, Question 1: (iv) A polygon whose sum of all interior angles is eight right angles is _____.

(a) triangle

(b) quadrilateral

(c) pentagon

(d) hexagon.

Review Exercise 8, Question 1: (v) A polygon whose sum of all interior angles is six right angles is _____.

(a) triangle

(b) quadrilateral

(c) pentagon

(d) hexagon.

Review Exercise 8, Question 1: (vi) In a rectangle, the diagonal _____.

(a) bisect each other at a right angle

(b) are perpendicular to each other

(c) cut each other at the midpoint

(d) none of these.

Review Exercise 8, Question 1: (vii) A quadrilateral is a _____.

(a) rectangle

(b) regular polygon

(c) that quadrilateral whose all sides are not equal

(d) none of these.

Review Exercise 8, Question 1: (viii) Each interior angle of a regular hexagon measures _____.

(a) 180°

(b) 120°

(c) 140°

(d) none of these.

9. Areas and Volumes

Question: Exercise 9.1, Question 1: Verify Pythagoras' theorem for the following right triangles. (i) Triangle ABC: AC = 3 cm, BC = 3 cm, AB = $3\sqrt{2}$ cm.

Answer: By Pythagoras: $(\text{hypotenuse})^2 = (\text{base})^2 + (\text{perpendicular})^2$.

Here: $(3\sqrt{2})^2 = 9 \times 2 = 18$;

$(3)^2 + (3)^2 = 9 + 9 = 18$.

Both sides are equal, so the theorem is verified.

Question: Exercise 9.1, Question 1: (ii) Triangle PQR: PQ = 12 cm, QR = 5 cm, PR = 13 cm.

Answer: $(13)^2 = 169$;

$(12)^2 + (5)^2 = 144 + 25 = 169$.

Both sides are equal, so the theorem is verified.

Question: Exercise 9.1, Question 1: (iii) Triangle ABC: AB = 2 cm, BC = 1.2 cm, AC = 1.6 cm.

Answer: $(2)^2 = 4$;

$(1.6)^2 + (1.2)^2 = 2.56 + 1.44 = 4$.

Both sides are equal, so the theorem is verified.

Question: Exercise 9.1, Question 2: Verify Pythagoras' theorem using the following data. (i) Triangle LMN, with m∠M = 90°: mLM = 9 cm, mMN = 12 cm, mLN = 15 cm.

Answer: Hypotenuse LN = 15: $(15)^2 = 225$;

$(9)^2 + (12)^2 = 81 + 144 = 225$.

Both sides are equal, so the theorem is verified.

Question: Exercise 9.1, Question 2: (ii) Triangle PQR, with m∠Q = 90°: mPR = 20 cm, mPQ = 16 cm, mQR = 12 cm.

Answer: Hypotenuse PR = 20: $(20)^2 = 400$;

$(16)^2 + (12)^2 = 256 + 144 = 400$.

Both sides are equal, so the theorem is verified.

Question: Exercise 9.1, Question 2: (iii) Triangle XYZ, with m∠Y = 90°: mXZ = 25 cm, mXY = 15 cm, mYZ = 20 cm.

Answer: Hypotenuse XZ = 25: $(25)^2 = 625$;

$(15)^2 + (20)^2 = 225 + 400 = 625$.

Both sides are equal, so the theorem is verified.

Question: Exercise 9.2, Question 1: Find the unknown sides of the

following right triangles: (i) legs 9 cm and 12 cm, hypotenuse x.

Answer: $x^2 = 9^2 + 12^2 = 81 + 144 = 225$, so $x = \sqrt{225} = 15$ cm.

Question: Exercise 9.2, Question 1: (ii) leg 15 cm, hypotenuse 25 cm, unknown leg y.

Answer: $y^2 = 25^2 - 15^2 = 625 - 225 = 400$, so $y = \sqrt{400} = 20$ cm.

Question: Exercise 9.2, Question 1: (iii) legs 18 cm and 30 cm, hypotenuse p.

Answer: $p^2 = 18^2 + 30^2 = 324 + 900 = 1224$, so $p = \sqrt{1224} = 6\sqrt{34} \approx 34.99$ cm.

Question: Exercise 9.2, Question 1: (iv) legs 4.5 cm and 6 cm, hypotenuse z.

Answer: $z^2 = 4.5^2 + 6^2 = 20.25 + 36 = 56.25$, so $z = \sqrt{56.25} = 7.5$ cm.

Question: Exercise 9.2, Question 1: (v) legs 4.8 cm and 3.6 cm, hypotenuse x.

Answer: $x^2 = 4.8^2 + 3.6^2 = 23.04 + 12.96 = 36$, so $x = \sqrt{36} = 6$ cm.

Question: Exercise 9.2, Question 1: (vi) leg 6.4 cm, hypotenuse 8 cm, unknown leg s.

Answer: $s^2 = 8^2 - 6.4^2 = 64 - 40.96 = 23.04$, so $s = \sqrt{23.04} = 4.8$ cm.

Question: Exercise 9.2, Question 1: (vii) legs 2.1 cm and 2.8 cm, hypotenuse r.

Answer: $r^2 = 2.1^2 + 2.8^2 = 4.41 + 7.84 = 12.25$, so $r = \sqrt{12.25} = 3.5$ cm.

Question: Exercise 9.2, Question 1: (viii) legs 9 cm and 5.4 cm, hypotenuse y.

Answer: $y^2 = 9^2 + 5.4^2 = 81 + 29.16 = 110.16$, so $y = \sqrt{110.16} \approx 10.50$ cm.

Question: Exercise 9.2, Question 2: l, m, n are the side measures of triangle LMN. Decide from the following data which sets form a right triangle (any one of ∠L, ∠M, ∠N may be the right angle, or none). (i) l = 9, m = 12, n = 15.

Answer: Largest side n = 15: $9^2 + 12^2 = 81 + 144 = 225 = 15^2$.

Equal, so it is a right triangle (the right angle is opposite n).

Question: Exercise 9.2, Question 2: (ii) l = 7, m = 8, n = 9.

Answer: $7^2 + 8^2 = 49 + 64 = 113 \neq 81 = 9^2$;

$7^2 + 9^2 = 130 \neq 64 = 8^2$;

$8^2 + 9^2 = 145 \neq 49 = 7^2$.

No equality holds, so it is not a right triangle.

Question: Exercise 9.2, Question 2: (iii) l = 6, m = 8, n = 10.

Answer: $6^2 + 8^2 = 36 + 64 = 100 = 10^2$.

Equal, so it is a right triangle.

Question: Exercise 9.2, Question 2: (iv) l = 12, m = 16, n = 20.

Answer: $12^2 + 16^2 = 144 + 256 = 400 = 20^2$.

Equal, so it is a right triangle.

Question: Exercise 9.2, Question 2: (v) l = 2, m = 4, n = 6.

Answer: $2^2 + 4^2 = 4 + 16 = 20 \neq 36 = 6^2$;

$2^2 + 6^2 = 40 \neq 16 = 4^2$;

$4^2 + 6^2 = 52 \neq 4 = 2^2$.

No equality holds, so it is not a right triangle.

Question: Exercise 9.2, Question 2: (vi) l = 1.5, m = 2, n = 2.5.

Answer: $1.5^2 + 2^2 = 2.25 + 4 = 6.25 = 2.5^2$.

Equal, so it is a right triangle.

Question: Exercise 9.2, Question 3: The square of the hypotenuse of an isosceles right triangle is given below. Find the measure of one of its sides.

(i) $H^2 = 50$ cm^2.

Answer: In an isosceles right triangle, $H^2 = 2 \times (\text{side})^2$, so $(\text{side})^2 = 50 \div 2 = 25$, side = 5 cm.

Question: Exercise 9.2, Question 3: (ii) $H^2 = 72$ cm^2.

Answer: $(\text{side})^2 = 72 \div 2 = 36$, side = 6 cm.

Question: Exercise 9.2, Question 3: (iii) $H^2 = 98$ cm^2.

Answer: (side)² = 98 ÷ 2 = 49, side = 7 cm.

Question: Exercise 9.2, Question 3: (iv) H² = 128 cm².

Answer: (side)² = 128 ÷ 2 = 64, side = 8 cm.

Question: Exercise 9.2, Question 3: (v) H² = 2.42 cm².

Answer: (side)² = 2.42 ÷ 2 = 1.21, side = 1.1 cm.

Question: Exercise 9.2, Question 3: (vi) H² = 9.68 cm².

Answer: (side)² = 9.68 ÷ 2 = 4.84, side = 2.2 cm.

Question: Exercise 9.2, Question 3: (vii) H² = 4.5 cm².

Answer: (side)² = 4.5 ÷ 2 = 2.25, side = 1.5 cm.

Question: Exercise 9.2, Question 3: (viii) H² = 60.5 cm².

Answer: (side)² = 60.5 ÷ 2 = 30.25, side = 5.5 cm.

Question: Exercise 9.2, Question 3: (ix) H² = 112.5 cm².

Answer: (side)² = 112.5 ÷ 2 = 56.25, side = 7.5 cm.

Question: Exercise 9.2, Question 4: A ladder 6.5 m long is placed so that it reaches a height of 5.2 m on a wall. Find the distance of the foot of the ladder from the wall.

Answer: The ladder is the hypotenuse: distance² = 6.5² − 5.2² = 42.25 − 27.04 = 15.21, so distance = √15.21 = 3.9 m.

Question: Exercise 9.2, Question 5: An electric pole is 12 m high. A 20 m wire from the top of the pole is tied to a peg on the ground. Tell what the

distance between the peg and the pole is.

Answer: The wire is the hypotenuse: distance² = 20² − 12² = 400 − 144 = 256, so distance = √256 = 16 m.

Question: Exercise 9.2, Question 6: Find the diagonal of the following rectangles, whose length and width are as follows: (i) length = 6 m, width = 8 m.

Answer: diagonal² = 6² + 8² = 36 + 64 = 100, so diagonal = 10 m.

Question: Exercise 9.2, Question 6: (ii) length = 44 m, width = 33 m.

Answer: diagonal² = 44² + 33² = 1936 + 1089 = 3025, so diagonal = 55 m.

Question: Exercise 9.2, Question 6: (iii) length = 5.6 m, width = 4.2 m.

Answer: diagonal² = 5.6² + 4.2² = 31.36 + 17.64 = 49, so diagonal = 7 m.

Question: Exercise 9.2, Question 6: (iv) length = 6.4 m, width = 4.8 m.

Answer: diagonal² = 6.4² + 4.8² = 40.96 + 23.04 = 64, so diagonal = 8 m.

Question: Exercise 9.2, Question 7: A student uses the shortcut path (HS) from home (H) to school (S). If the student first goes from home to the garden (B) and then from the garden to school, how much more distance will he have to cover? In the figure: HG = 450 m, HS = 750 m.

Answer: First find GS: GS² = 750² − 450² = 562500 − 202500 = 360000, GS = 600 m.

Via the garden: 450 + 600 = 1050 m.

Shortcut: 750 m.

Extra distance = 1050 − 750 = 300 m.

Question: Exercise 9.2, Question 8: A rectangular plot has width 48 m and its diagonal is 80 m long. Find its area.

Answer: length² = 80² − 48² = 6400 − 2304 = 4096, length = 64 m.

Area = 64 × 48 = 3072 m².

Question: Exercise 9.2, Question 9: Look carefully at the given figures; find the measures of the unknown quantities x and y. Figure (i): triangle PQR, PS perpendicular to QR, QS = 18 cm, SR = 32 cm, PQ = 30 cm, PS = x.

Answer: In ΔPQS: $x^2 = 30^2 - 18^2 = 900 - 324 = 576$, so x = 24 cm.

Question: Exercise 9.2, Question 9: Figure (i): PR = y.

Answer: In ΔPSR: $y^2 = x^2 + 32^2 = 576 + 1024 = 1600$, so y = 40 cm.

Question: Exercise 9.2, Question 9: Figure (ii): lower right triangle: legs 12 cm and 9 cm, hypotenuse x.

Answer: $x^2 = 12^2 + 9^2 = 144 + 81 = 225$, so x = 15 cm.

Question: Exercise 9.2, Question 9: Figure (ii): upper right triangle: hypotenuse 25 cm, one leg x = 15 cm, other leg y.

Answer: $y^2 = 25^2 - 15^2 = 625 - 225 = 400$, so y = 20 cm.

Question: Exercise 9.2, Question 9: Figure (iii): the diagonals of two right triangles: one triangle with legs 24 cm and 32 cm, the other with legs 18

cm and 32 cm. x is the diagonal of the triangle with legs 18 cm and 32 cm.

Answer: $x^2 = 18^2 + 32^2 = 324 + 1024 = 1348$, so $x = \sqrt{1348} = 2\sqrt{337} \approx 36.72$ cm.

Question: Exercise 9.2, Question 9: Figure (iii): y is the diagonal of the triangle with legs 24 cm and 32 cm.

Answer: $y^2 = 24^2 + 32^2 = 576 + 1024 = 1600$, so $y = 40$ cm.

Question: Exercise 9.3, Question 1: Find the areas of the following triangles by Heron's formula, given their three sides: (i) ΔABC: a = 16 cm, b = 20 cm, c = 16 cm.

Answer: $S = (16 + 20 + 16) \div 2 = 26$.

Area $= \sqrt{(26 \times 10 \times 6 \times 10)} = \sqrt{15600} = 20\sqrt{39} \approx 124.90$ cm².

Question: Exercise 9.3, Question 1: (ii) ΔDEF: d = 10 cm, e = 17 cm, f = 21 cm.

Answer: $S = (10 + 17 + 21) \div 2 = 24$.

Area $= \sqrt{(24 \times 14 \times 7 \times 3)} = \sqrt{7056} = 84$ cm².

Question: Exercise 9.3, Question 1: (iii) ΔLMN: l = 6 cm, m = 8 cm, n = 10 cm.

Answer: $S = (6 + 8 + 10) \div 2 = 12$.

Area $= \sqrt{(12 \times 6 \times 4 \times 2)} = \sqrt{576} = 24$ cm².

Question: Exercise 9.3, Question 1: (iv) ΔPQR: p = 20 cm, q = 24 cm, r =

18 cm.

Answer: S = (20 + 24 + 18) ÷ 2 = 31.

Area = √(31 × 11 × 7 × 13) = √31031 ≈ 176.16 cm².

Question: Exercise 9.3, Question 1: (v) ΔXYZ: x = 15 cm, y = 21 cm, z = 14 cm.

Answer: S = (15 + 21 + 14) ÷ 2 = 25.

Area = √(25 × 10 × 4 × 11) = √11000 = 10√110 ≈ 104.88 cm².

Question: Exercise 9.3, Question 2: Find the areas of the following isosceles triangles using Heron's formula, given their side measures: (i) ΔABC: a = 10 cm, b = 10 cm, c = 12 cm.

Answer: S = (10 + 10 + 12) ÷ 2 = 16.

Area = √(16 × 6 × 6 × 4) = √2304 = 48 cm².

Question: Exercise 9.3, Question 2: (ii) ΔDCE: d = 12 cm, e = 16 cm, c = 12 cm.

Answer: S = (12 + 16 + 12) ÷ 2 = 20.

Area = √(20 × 8 × 4 × 8) = √5120 = 32√5 ≈ 71.55 cm².

Question: Exercise 9.3, Question 2: (iii) ΔLMN: l = 9 cm, m = 9 cm, n = 12 cm.

Answer: S = (9 + 9 + 12) ÷ 2 = 15.

Area = √(15 × 6 × 6 × 3) = √1620 = 18√5 ≈ 40.25 cm².

Question: Exercise 9.3, Question 2: (iv) ΔPQR: p = 24 cm, q = 24 cm, r = 40 cm.

Answer: S = (24 + 24 + 40) ÷ 2 = 44.

Area = √(44 × 20 × 20 × 4) = √70400 = 80√11 ≈ 265.33 cm².

Question: Exercise 9.3, Question 3: Find the area of an equilateral triangle with each side of length 'a' cm by Heron's formula.

Answer: S = (a + a + a) ÷ 2 = 3a/2.

Area = √((3a/2)(a/2)(a/2)(a/2)) = √(3a⁴/16) = (√3/4)a² cm².

Question: Exercise 9.3, Question 4: Find the remaining side length and the triangle's area from the following measures: (i) a = 13 cm, b = 15 cm, S = 20 cm, c = ___, area = ___.

Answer: c = 2S − a − b = 40 − 13 − 15 = 12 cm.

Area = √(20 × 7 × 5 × 8) = √5600 = 20√14 ≈ 74.83 cm².

Question: Exercise 9.3, Question 4: (ii) d = 3.6 cm, e = 4.5 cm, S = 6.0 cm, f = ___, area = ___.

Answer: f = 2S − d − e = 12 − 3.6 − 4.5 = 3.9 cm.

Area = √(6 × 2.4 × 1.5 × 2.1) = √45.36 ≈ 6.73 cm².

Question: Exercise 9.3, Question 4: (iii) l = 7.2 cm, m = 6.4 cm, S = 10.0

cm, n = ___, area = ___.

Answer: n = 2S − l − m = 20 − 7.2 − 6.4 = 6.4 cm.

Area = √(10 × 2.8 × 3.6 × 3.6) = √362.88 ≈ 19.05 cm².

Question: Exercise 9.3, Question 4: (iv) p = 4.5 cm, q = 4.5 cm, S = 7.0 cm, r = ___, area = ___.

Answer: r = 2S − p − q = 14 − 4.5 − 4.5 = 5 cm.

Area = √(7 × 2.5 × 2.5 × 2) = √87.5 ≈ 9.35 cm².

Question: Exercise 9.4, Question 1: Find the area of quadrilateral ABCD given: mAB = 10 cm, mCD = 8 cm, mAC = 12 cm, mAD = 8 cm, and mBC = 6 cm.

Answer: Diagonal AC divides the quadrilateral into two triangles.

ΔABC: sides 10, 6, 12;

S = 14;

area = √(14 × 4 × 8 × 2) = √896 = 8√14 ≈ 29.93.

ΔADC: sides 8, 8, 12;

S = 14;

area = √(14 × 6 × 6 × 2) = √1008 = 12√7 ≈ 31.75.

Total area ≈ 29.93 + 31.75 = 61.68 cm².

Question: Exercise 9.4, Question 2: Find the area of quadrilateral PQRS given: mQR = mPS = 4 cm, mPQ = mSR = 3 cm, mPR = 5 cm.

Answer: Diagonal PR divides the quadrilateral into two triangles.

ΔPQR: sides 3, 4, 5 — a right triangle ($3^2 + 4^2 = 5^2$);

area = (1/2) × 3 × 4 = 6.

ΔPSR: sides 4, 3, 5 — a right triangle;

area = (1/2) × 4 × 3 = 6.

Total area = 6 + 6 = 12 cm².

Question: Exercise 9.4, Question 3: Find the area of quadrilateral DEFG given: mDE = 8 cm, mEF = 24 cm, mDF = 25 cm, mDG = 9 cm, and mGF = 23 cm.

Answer: Diagonal DF divides the quadrilateral into two triangles.

ΔDEF: sides 8, 24, 25;

S = 28.5;

area = √(28.5 × 20.5 × 4.5 × 3.5) ≈ 95.93.

ΔDGF: sides 9, 23, 25;

S = 28.5;

area = √(28.5 × 19.5 × 5.5 × 3.5) ≈ 103.43.

Total area ≈ 95.93 + 103.43 = 199.36 cm².

Question: Exercise 9.4, Question 4: Find the area of the following quadrilateral plot whose measures are given in the figure: UV = 120 m, VW = 50 m, WX = 90 m, XU = 80 m, diagonal UW = 130 m.

Answer: Diagonal UW divides the plot into two triangles.

ΔUVW: sides 120, 50, 130 — a right triangle (50² + 120² = 130²);

area = (1/2) × 50 × 120 = 3000.

ΔUWX: sides 80, 90, 130;

S = 150;

area = √(150 × 70 × 60 × 20) = √12600000 = 600√35 ≈ 3549.65.

Total area ≈ 3000 + 3549.65 = 6549.65 m².

Question: Exercise 9.4, Question 5: LMNO is a quadrilateral; find its area. Given: mLM = 5 cm, mMN = 12 cm, mLN = 13 cm, mLO = 14 cm, and mNO = 15 cm.

Answer: Diagonal LN divides the quadrilateral into two triangles.

ΔLMN: sides 5, 12, 13 — a right triangle ($5^2 + 12^2 = 13^2$);

area = $(1/2) \times 5 \times 12 = 30$.

ΔLNO: sides 14, 15, 13;

S = 21;

area = $\sqrt{(21 \times 7 \times 6 \times 8)} = \sqrt{7056} = 84$.

Total area = 30 + 84 = 114 cm².

Question: Exercise 9.5, Question 1: Find the surface areas of spheres with the following radii (π = 22/7). (i) r = 35 cm.

Answer: $A = 4\pi r^2 = 4 \times (22/7) \times 35^2 = 4 \times 22 \times 5 \times 35 = 15400$ cm².

Question: Exercise 9.5, Question 1: (ii) r = 56 cm.

Answer: $A = 4 \times (22/7) \times 56^2 = 4 \times 22 \times 8 \times 56 = 39424$ cm².

Question: Exercise 9.5, Question 1: (iii) r = 0.42 cm.

Answer: $A = 4 \times (22/7) \times 0.42^2 = 4 \times (22/7) \times 0.1764 = 2.2176$ cm².

Question: Exercise 9.5, Question 1: (iv) r = 0.63 cm.

Answer: $A = 4 \times (22/7) \times 0.63^2 = 4 \times (22/7) \times 0.3969 = 4.9896$ cm².

Question: Exercise 9.5, Question 1: (v) r = 0.98 cm.

Answer: $A = 4 \times (22/7) \times 0.98^2 = 4 \times (22/7) \times 0.9604 = 12.0736$ cm².

Question: Exercise 9.5, Question 2: Find the radii of the following spheres, whose surface areas are: (i) A = 616 cm².

Answer: $r^2 = A \div (4\pi) = 616 \times 7 \div 88 = 49$, so r = 7 cm.

Question: Exercise 9.5, Question 2: (ii) A = 2464 cm².

Answer: $r^2 = 2464 \times 7 \div 88 = 196$, so r = 14 cm.

Question: Exercise 9.5, Question 2: (iii) A = 88,704 cm².

Answer: $r^2 = 88704 \times 7 \div 88 = 7056$, so r = 84 cm.

Question: Exercise 9.5, Question 2: (iv) A = 985.6 m².

Answer: $r^2 = 985.6 \times 7 \div 88 = 78.4$, so $r = \sqrt{78.4} \approx 8.85$ m.

Question: Exercise 9.5, Question 2: (v) A = 154 m².

Answer: $r^2 = 154 \times 7 \div 88 = 12.25$, so r = 3.5 m.

Question: Exercise 9.5, Question 3: Find the volumes of spheres with the following radii: (i) r = 21 cm.

Answer: $V = (4/3)\pi r^3 = (4/3) \times (22/7) \times 21^3 = 38808$ cm³.

Question: Exercise 9.5, Question 3: (ii) r = 49 cm.

Answer: $V = (4/3) \times (22/7) \times 49^3 = (88/21) \times 117649 \approx 493005.33$ cm³.

Question: Exercise 9.5, Question 3: (iii) r = 2.8 m.

Answer: $V = (4/3) \times (22/7) \times 2.8^3 = (88/21) \times 21.952 \approx 91.99$ m³.

Question: Exercise 9.5, Question 3: (iv) r = 77 cm.

Answer: $V = (4/3) \times (22/7) \times 77^3 = (88/21) \times 456533 \approx 1913090.67$ cm³.

Question: Exercise 9.5, Question 3: (v) r = 3.5 m.

Answer: V = (4/3) × (22/7) × 3.5^3 = (88/21) × 42.875 ≈ 179.67 m^3.

Question: Exercise 9.5, Question 4: How many litres of water will fit in the following sphere, if its radius is: (i) r = 70 cm.

Answer: V = (4/3) × (22/7) × 70^3 = 1437333.33 cm^3.

1 litre = 1000 cm^3, so water = 1437333.33 ÷ 1000 ≈ 1437.33 litres.

Question: Exercise 9.5, Question 4: (ii) r = 105 cm.

Answer: V = (4/3) × (22/7) × 105^3 = 4851000 cm^3 = 4851 litres.

Question: Exercise 9.5, Question 4: (iii) r = 2.8 m.

Answer: V = (4/3) × (22/7) × 2.8^3 ≈ 91.99 m^3.

1 m^3 = 1000 litres, so water ≈ 91989 litres.

Question: Exercise 9.5, Question 4: (iv) r = 175 cm.

Answer: V = (4/3) × (22/7) × 175^3 = 22458333.33 cm^3 ≈ 22458.33 litres.

Question: Exercise 9.5, Question 4: (v) r = 2.45 m.

Answer: V = (4/3) × (22/7) × 2.45^3 ≈ 61.63 m^3 ≈ 61626 litres.

Question: Exercise 9.5, Question 5: Find the volume of the following sphere and tell how many litres of water it will hold, when its surface area is: (i) A = 2464 cm^2.

Answer: r^2 = 2464 × 7 ÷ 88 = 196, r = 14 cm.

V = (4/3) × (22/7) × 14^3 = 11498.67 cm^3 ≈ 11.50 litres.

Question: Exercise 9.5, Question 5: (ii) A = 9856 cm^2.

Answer: r^2 = 9856 × 7 ÷ 88 = 784, r = 28 cm.

V = (4/3) × (22/7) × 28^3 = 91989.33 cm^3 ≈ 91.99 litres.

Question: Exercise 9.5, Question 5: (iii) A = 29.8144 m^2.

Answer: r^2 = 29.8144 × 7 ÷ 88 = 2.3716, r = 1.54 m.

V = (4/3) × (22/7) × 1.54^3 ≈ 15.30 m^3 ≈ 15305 litres.

Question: Exercise 9.5, Question 5: (iv) A = 43.12 m^2.

Answer: r^2 = 43.12 × 7 ÷ 88 = 3.43, r = √3.43 ≈ 1.85 m.

V = (4/3) × (22/7) × 1.85^3 ≈ 26.62 m^3 ≈ 26620 litres.

Question: Exercise 9.5, Question 5: (v) A = 13.86 m^2.

Answer: r^2 = 13.86 × 7 ÷ 88 = 1.1025, r = 1.05 m.

V = (4/3) × (22/7) × 1.05^3 = 4.851 m^3 = 4851 litres.

Question: Exercise 9.5, Question 6: If the radius of a sphere is doubled, how much will (a) its surface area and (b) its volume change?

Answer: (a) Area A = $4\pi r^2$;

doubling r gives A′ = $4\pi(2r)^2 = 4 \times 4\pi r^2$ — the area becomes 4 times.

(b) Volume $V = (4/3)\pi r^3$;

doubling r gives $V' = (4/3)\pi(2r)^3 = 8 \times (4/3)\pi r^3$ — the volume becomes 8 times.

Question: Exercise 9.5, Question 7: The radius of sphere A is three times the radius of sphere B. Find the ratio between the two spheres' (i) surface areas. (ii) volumes.

Answer: Suppose B's radius is r, so A's radius is 3r.

(i) Ratio of areas = $4\pi(3r)^2 : 4\pi r^2 = 9 : 1$.

(ii) Ratio of volumes = $(4/3)\pi(3r)^3 : (4/3)\pi r^3 = 27 : 1$.

Question: Exercise 9.5, Question 8: An iron sphere has surface area 77,000 cm^2. If it is melted, how many spheres of radius 1 cm can be made from the material obtained?

Answer: Radius R of the big sphere: $R^2 = 77000 \times 7 \div 88 = 6125$, $R = \sqrt{6125} \approx 78.26$ cm.

Number of small spheres = volume of big sphere ÷ volume of small sphere = $R^3 \div 1^3 = 6125 \times \sqrt{6125} \approx 479357$.

So about 479357 small spheres can be made.

Question: Exercise 9.5, Question 9: A copper sphere of radius 7 cm is melted and the copper obtained is made into a wire. If the diameter of the wire is 0.7 cm, find the length of the wire.

Answer: Volume of the sphere = (4/3) × (22/7) × 7^3 = 1437.33 cm^3.

The wire is a cylinder, radius = 0.7 ÷ 2 = 0.35 cm.

Volume of the wire = π × 0.35^2 × L = (22/7) × 0.1225 × L = 0.385L.

So L = 1437.33 ÷ 0.385 = 3733.33 cm = 37.33 m.

Question: Exercise 9.5, Question 10: Find the volume of a football of radius 12 cm.

Answer: V = (4/3) × (22/7) × 12^3 = (88/21) × 1728 = 7241.14 cm^3.

Question: Exercise 9.6, Question 1: In the following chart, find the unknown quantities about a cone: (i) r = 21 cm, h = 28 cm.

Answer: s = √(28^2 + 21^2) = √(784 + 441) = √1225 = 35 cm.

Circular surface area A_1 = πr^2 = (22/7) × 441 = 1386 cm^2.

Curved surface area A_2 = πrs = (22/7) × 21 × 35 = 2310 cm^2.

Total area A = πr(r + s) = 1386 + 2310 = 3696 cm^2.

Question: Exercise 9.6, Question 1: (ii) h = 56 cm, s = 70 cm.

Answer: r = √(70^2 − 56^2) = √(4900 − 3136) = √1764 = 42 cm.

A_1 = πr^2 = (22/7) × 1764 = 5544 cm^2.

A_2 = πrs = (22/7) × 42 × 70 = 9240 cm^2.

A = 5544 + 9240 = 14784 cm².

Question: Exercise 9.6, Question 1: (iii) s = 32 cm (only).

Answer: This row gives only s = 32 cm;

no r, h, or area is given, so the cone is not determined.

[note: as printed, this row gives only one value, so the cone is not determined.]

Question: Exercise 9.6, Question 1: (iv) h = 84 cm (only).

Answer: This row gives only h = 84 cm;

no r, s, or area is given, so the cone is not determined.

[note: as printed, this row gives only one value, so the cone is not determined.]

Question: Exercise 9.6, Question 1: (v) r = 84 cm, total area A = 33264 cm².

Answer: From A = πr(r + s): 33264 = (22/7) × 84 × (84 + s) = 264 × (84 + s), so 84 + s = 126, s = 42 cm.

But s < r, which is impossible (from s² = r² + h², s is always greater than r).

So no such cone exists — the given numbers are mutually inconsistent.

[note: as printed, the row is inconsistent: it gives s = 42 cm < r = 84 cm, which is impossible for a cone.]

Question: Exercise 9.6, Question 2: In the following chart, find the unknown quantities about a cone: (i) r = 3 cm, h = 7 cm.

Answer: $s = \sqrt{(7^2 + 3^2)} = \sqrt{58} \approx 7.62$ cm.

Volume $V = (1/3)\pi r^2 h = (1/3) \times (22/7) \times 9 \times 7 = 66$ cm³.

Question: Exercise 9.6, Question 2: (ii) r = 3.5 cm, h = 6 cm.

Answer: $s = \sqrt{(6^2 + 3.5^2)} = \sqrt{48.25} \approx 6.95$ cm.

Volume $V = (1/3) \times (22/7) \times 12.25 \times 6 = 77$ cm³.

Question: Exercise 9.6, Question 2: (iii) h = 28 cm, s = 35 cm.

Answer: $r = \sqrt{(35^2 - 28^2)} = \sqrt{(1225 - 784)} = \sqrt{441} = 21$ cm.

Volume $V = (1/3) \times (22/7) \times 441 \times 28 = 12936$ cm³.

Question: Exercise 9.6, Question 2: (iv) h = 14 cm, volume V = 154 cm³.

Answer: $r^2 = 3V \div (\pi h) = (3 \times 154) \div ((22/7) \times 14) = 462 \div 44 = 10.5$, $r = \sqrt{10.5} \approx 3.24$ cm.

$s = \sqrt{(14^2 + 10.5)} = \sqrt{206.5} \approx 14.37$ cm.

Question: Exercise 9.6, Question 2: (v) r = 6.3 cm, s = 105 cm.

Answer: $h = \sqrt{(105^2 - 6.3^2)} = \sqrt{(11025 - 39.69)} = \sqrt{10985.31} \approx 104.81$ cm.

Volume V = (1/3) × (22/7) × 39.69 × 104.81 ≈ 4358 cm³.

Question: Exercise 9.6, Question 3: A cone's base circle area is 2464 cm². If its height is three times the radius of the base circle, find its volume.

Answer: From πr² = 2464, r² = 2464 × 7 ÷ 22 = 784, r = 28 cm.

Height h = 3r = 84 cm.

Volume V = (1/3) × (base area) × h = (1/3) × 2464 × 84 = 68992 cm³.

Question: Exercise 9.6, Question 4: A cone's volume is 352 cm³. If the radius of its base circle is 4 cm, find the cone's vertical height.

Answer: From V = (1/3)πr²h, h = 3V ÷ (πr²) = (3 × 352) ÷ ((22/7) × 16) = 1056 ÷ (352/7) = 21 cm.

[note: the book prints "چورس سينٽي ميٹر" (square cm) for the volume here;

it should be cubic cm.]

Question: Exercise 9.6, Question 5: If a cone's radius is 3.5 m and its height is 6 m, find its volume.

Answer: V = (1/3) × (22/7) × 3.5² × 6 = (22/7) × 12.25 × 2 = 22 × 3.5 = 77 m³.

Question: Exercise 9.7, Question 1: A sphere has radius 7 cm. Tell what its volume is.

Answer: V = (4/3) × (22/7) × 7³ = (88/21) × 343 = 1437.33 cm³.

Question: Exercise 9.7, Question 2: A conical cup is like a Chinese cup. If the cup's radius and height are respectively 6 cm and 10.5 cm, tell how much ice cream it holds.

Answer: $V = (1/3) \times (22/7) \times 6^2 \times 10.5 = (22/7) \times 12 \times 10.5 = 396\ cm^3$.

So the cup holds 396 cm^3 of ice cream.

Question: Exercise 9.7, Question 3: A conical tent has height 4.2 m and its base circle radius is 5.6 m. Find the volume of air inside the conical tent.

Answer: $V = (1/3) \times (22/7) \times 5.6^2 \times 4.2 = (22/7) \times 31.36 \times 1.4 = 137.984\ m^3 \approx 137.98\ m^3$ of air.

Question: Exercise 9.7, Question 4: A conical tank has height 7 m and radius 6 m. Tell how much wheat it will hold, if 1 cubic metre of volume holds 100 kg of wheat.

Answer: $V = (1/3) \times (22/7) \times 6^2 \times 7 = 264\ m^3$.

Wheat = 264 × 100 = 26400 kg.

Question: Exercise 9.7, Question 5: A conical water tank has height 7 m and its base circle radius is 3 m. Tell how many litres of water can be collected in it?

Answer: $V = (1/3) \times (22/7) \times 3^2 \times 7 = 66\ m^3$.

1 m^3 = 1000 litres, so water = 66 × 1000 = 66000 litres.

Question: Exercise 9.7, Question 6: A conical water tank is 6.3 m high and the radius of its base circle is 8.4 m. Tell what the volume of the tank is.

Answer: V = (1/3) × (22/7) × 8.4² × 6.3 = (22/7) × 70.56 × 2.1 = 465.696 m³ ≈ 465.70 m³.

Question: Exercise 9.7, Question 7: A football is like a balloon filled with air. If its volume is 9π/16 cm³, what is its radius?

Answer: From (4/3)πr³ = 9π/16, r³ = (9/16) × (3/4) = 27/64, so r = 3/4 cm.

Review Exercise 9, Question 1: In the given figure, the rectangle's length is 8 cm and width is 6 cm. Its diagonal length is _____. (i) 12 cm (ii) 16 cm (iii) 10 cm (iv) 14 cm.

Review Exercise 9, Question 2: In the given figure, right triangle ABC has base 9 cm and hypotenuse 15 cm. The perpendicular's length is _____. (i) 10 cm (ii) 12 cm (iii) 14 cm (iv) 16 cm.

Question: Review Exercise 9, Question 3: According to Heron's formula: (i) ▲ = √S(S + a)(S + b)(S + c) (ii) ▲ = √S(S − a)(S − b)(S − c) (iii) ▲ = √S + S(S − a) + (S − b) + (S − c) (iv) ▲ = √S(S − a − b − c).

Answer: (ii) ▲ = √S(S − a)(S − b)(S − c).

Review Exercise 9, Question 4: The value of S used in Heron's formula is _____: (i) S = (a − b − c)/2 (ii) S = (a + b − c)/2 (iii) S = (a + b + c)/2 (iv) S = (a + b + c)/3.

Review Exercise 9, Question 5: The formula for the surface area of a sphere is _____: (i) A = πr² (ii) A = 3πr² (iii) A = 2πr² (iv) A = 4πr².

Review Exercise 9, Question 6: The formula for the volume of a sphere is

_____: (i) V = (2/3)πr² (ii) V = (4/3)πr² (iii) V = (3/4)πr³ (iv) V = (4/3)πr³.

Review Exercise 9, Question 7: The value of π is approximately _____. (i) 3 1/7 (ii) 3 2/7 (iii) 3 3/7 (iv) 3 4/7.

Review Exercise 9, Question 8: If a cone has r = 1 and h = 3, its volume is _____: (i) π m³ (ii) π m² (iii) π m (iv) π.

Review Exercise 9, Question 9: A cone has radius 1 m; the area of its circular part is _____: (i) π m (ii) π m² (iii) π m³ (iv) π.

Review Exercise 9, Question 10: A cone with slant height 5 m and radius 2 m has total surface area _____: (i) 11 m³ (ii) 22 m³ (iii) 33 m³ (iv) 44 m³.

10. Demonstrative Geometry

Exercise 10.1, Question 1 (a): Fill in the blanks: Demonstrative geometry is a _____ of mathematics, in which _____ are proved by _____.

Answer: branch, propositions (statements), logical reasoning.

Exercise 10.1, Question 1 (b): Fill in the blanks: The _____ method is the one in which a general conclusion is drawn from particular examples, and the _____ method is the one in which a particular conclusion is drawn from general principles.

Answer: inductive, deductive

Exercise 10.1, Question 1 (c): Fill in the blank: In the _____ method, a general conclusion is drawn from particular examples.

Answer: inductive

Exercise 10.1, Question 1 (d): Fill in the blank: In the _____ method, a particular conclusion is drawn from general principles.

Answer: deductive

Exercise 10.1, Question 1 (e): Fill in the blank: The statement which is accepted _____ is called an axiom or a postulate.

Answer: without proof

Exercise 10.1, Question 1 (f): Fill in the blanks: Fundamental assumptions are of two kinds: (1) _____ and (2) _____.

Answer: axioms, postulates.

Exercise 10.1, Question 1 (g): Fill in the blank: The fundamental assumption which concerns geometry is called a _____.

Answer: postulate.

Exercise 10.1, Question 1 (h): Fill in the blanks: The four important elements of the deductive method are: (1) _____, (2) _____, (3) _____, (4) _____.

Answer: method.

Incorrect: Exercise 10.1, Question 2 (a): State whether this statement is true or false: More than one line can be drawn through two different points.

By Postulate 1, exactly one line can be drawn through two different points.

Incorrect: Exercise 10.1, Question 2 (b): State whether this statement is true or false: If P and Q do not coincide, then symbolically we write them as P = Q.

If P and Q do not coincide (that is, they are distinct points), we write them as P ≠ Q, not P = Q. [note: The book's answer key lists this item as correct, but as printed the statement is false; either the key is misprinted or the exercise intends "coincide" without "not".]

Incorrect: Exercise 10.1, Question 2 (c): State whether this statement is true or false: Joining two points by a line gives a line.

Joining two points gives a line segment, not a whole line (Postulate 6).

Correct: Exercise 10.1, Question 2 (d): State whether this statement is

true or false: A plane surface is smooth and unlimited on all sides.

By Postulate 4, a plane surface is smooth and unlimited on all sides.

Correct: Exercise 10.1, Question 2 (e): State whether this statement is true or false: The measure of the complementary angle of x° is (90° − x°).

The measure of the complementary angle of any angle x° is (90° − x°).

Correct: Exercise 10.1, Question 2 (f): State whether this statement is true or false: If a = b and b = c, then a = c.

This is the transitive property of equality (Axiom 3).

Question: Exercise 10.1, Question 3: Write the basic parts of a proof.

Answer: The basic parts of a proof are six: (1) Statement — what is to be proved, (2) Given — what is given, (3) Required to prove — stating clearly what is to be proved, (4) Construction — an addition to the figure if needed, (5) Proof — proving by logical reasoning, (6) Q.E.D.

(proved).

Question: Exercise 10.2, Question 1 (i): In the figure, name the supplementary angle of ∠BOC.

Answer: ∠AOC.

Because ∠BOC and ∠AOC are a pair of adjacent angles on a straight line, they are supplementary (Theorem 1).

Question: Exercise 10.2, Question 1 (ii): In the figure, name the complementary angle of ∠BOC.

Answer: ∠POC.

In the figure, line OP is perpendicular to AB, so ∠BOC and ∠POC together make 90°, that is, they are complementary.

Question: Exercise 10.2, Question 1 (iii): If m∠BOC = 50°, write the measures of its complementary and supplementary angles.

Answer: Complementary angle = 90° − 50° = 40°.

Supplementary angle = 180° − 50° = 130°.

Question: Exercise 10.2, Question 2: In the figure, m∠BOC = 45°, m∠COD = 3x and m∠AOC = 2x. Find the measure of x.

Answer: The three angles are on the same side of a straight line, so their sum is 180° (Corollary 1 of Theorem 1): 2x + 3x + 45° = 180° → 5x = 135° → x = 27°.

Hence m∠COD = 3x = 81° and m∠AOC = 2x = 54°.

Question: Exercise 10.2, Question 3 (a): In the figure, m∠BOC = x = 40°. Write the measures of the other three angles.

Answer: ∠AOC and ∠BOC are supplementary: m∠AOC = 180° − 40° = 140°.

∠BOD is the vertical angle of ∠AOC: m∠BOD = 140°.

∠AOD is the vertical angle of ∠BOC: m∠AOD = 40°.

Question: Exercise 10.2, Question 3 (b): In the figure, m∠BOC = 45°. Write the measures of the other three angles.

Answer: ∠AOC and ∠BOC are supplementary: m∠AOC = 180° − 45° = 135°.

∠BOD is the vertical angle of ∠AOC: m∠BOD = 135°.

∠AOD is the vertical angle of ∠BOC: m∠AOD = 45°.

Question: Exercise 10.2, Question 4: Prove that the bisectors of two adjacent supplementary angles are perpendicular to each other.

Answer: Let ∠AOC and ∠BOC be two adjacent supplementary angles.

Let OE bisect ∠AOC and OD bisect ∠BOC.

By Theorem 1, m∠AOC + m∠BOC = 180°.

Dividing both sides by 2: (1/2)m∠AOC + (1/2)m∠BOC = 90°, that is, m∠EOC + m∠DOC = 90°.

But m∠EOC + m∠DOC = m∠EOD, so m∠EOD = 90°, that is, OE ⊥ OD.

Q.E.D.

Question: Exercise 10.2, Question 5: Prove that if the bisectors of two adjacent angles are perpendicular to each other, then the two angles are

supplementary.

Answer: Let OE and OD be the bisectors of ∠AOC and ∠BOC, and OE ⊥ OD, that is, m∠EOD = 90°.

Now m∠EOD = m∠EOC + m∠DOC = (1/2)m∠AOC + (1/2)m∠BOC = 90°.

Multiplying both sides by 2: m∠AOC + m∠BOC = 180°, that is, ∠AOC and ∠BOC are supplementary.

Q.E.D.

Question: Exercise 10.2, Question 6: In the figure of Question 4, write the pairs of complementary angles.

Answer: In the figure of Question 4, OE bisects ∠AOC and OD bisects ∠BOC.

Let m∠AOE = m∠EOC = a and m∠BOD = m∠DOC = b;

by Question 4, a + b = 90°.

Hence the complementary pairs are: (∠AOE, ∠BOD), (∠BOD, ∠COE), (∠DOC, ∠COE), (∠DOC, ∠AOE) — each pair sums to a + b = 90°.

Question: Exercise 10.3, Question 1: In the figure, m∠BOC = 70°. Write the measures of the other three angles.

Answer: ∠AOD is the vertical angle of ∠BOC: m∠AOD = 70° (Theorem 3).

∠AOC and ∠BOC are supplementary: m∠AOC = 180° − 70° = 110°.

∠BOD is the vertical angle of ∠AOC: m∠BOD = 110°.

Question: Exercise 10.3, Question 2: Two lines cut each other at an angle of 30°. Write the measures of the other three angles.

Answer: One angle is 30°.

Its vertical angle is also 30° (Theorem 3).

The other two angles are each 180° − 30° = 150° (supplementary, Theorem 1).

Hence the other three angles are 30°, 150°, 150°.

Question: Exercise 10.3, Question 3: In the figure, prove that ∠1 ≅ ∠3.

Answer: In the figure, ∠1 + ∠2 = 180° (angles on a straight line, Theorem 1) and ∠3 + ∠2 = 180° (for the same reason).

Hence ∠1 + ∠2 = ∠3 + ∠2.

Subtracting ∠2 from both sides (Axiom 5) gives ∠1 = ∠3, that is, ∠1 ≅ ∠3.

Q.E.D.

Question: Exercise 10.3, Question 4: Prove that the supplements of the same angle are equal to each other.

Answer: Let ∠C be the supplementary angle of both ∠A and ∠B.

Then m∠A + m∠C = 180° and m∠B + m∠C = 180°.

Hence m∠A + m∠C = m∠B + m∠C.

Subtracting m∠C from both sides (Axiom 5) gives m∠A = m∠B, that is, ∠A ≅ ∠B.

Q.E.D.

Question: Exercise 10.3, Question 5, figure (i): In the figure, write the pairs of complementary, supplementary and vertical angles.

Answer: (a) Complementary pairs: (∠c, ∠d), (∠g, ∠h).

(b) Supplementary pairs: (∠a, ∠b), (∠e, ∠f), (∠f, ∠a).

(c) Vertical pairs: (∠a, ∠e), (∠b, ∠f).

Question: Exercise 10.3, Question 5, figure (ii): In the figure, write the pairs of complementary, supplementary and vertical angles.

Answer: (a) Complementary pairs: (∠c, ∠e), (∠d, ∠f), (∠c, ∠f), (∠d, ∠e).

(b) Supplementary pairs: (∠a, ∠b), (∠g, ∠h).

(c) Vertical pairs: (∠c, ∠f), (∠d, ∠e).

Question: Exercise 10.3, Question 5, figure (iii): In the figure, write the pairs of complementary, supplementary and vertical angles.

Answer: (a) Complementary pairs: (∠c, ∠d).

(b) Supplementary pairs: (∠e, ∠h), (∠h, ∠g), (∠g, ∠f), (∠f, ∠e).

(c) Vertical pairs: (∠e, ∠g), (∠h, ∠f).

Question: Exercise 10.3, Question 6, figure (i): In the figure, find the measure of x.

Answer: In the figure, one line is perpendicular to the other, so x + 90° + 20° = 180° (angles on a straight line).

Hence x = 180° − 90° − 20° = 70°.

Question: Exercise 10.3, Question 6, figure (ii): In the figure, find the measures of x and y.

Answer: Two slanting lines are perpendicular to each other.

y + 90° + 35° = 180° (angles on a straight line) → y = 55°.

x and y are complementary (two parts of the same right angle): x = 90° − y = 90° − 55° = 35°.

Question: Exercise 10.3, Question 6, figure (iii): In the figure, find the measures of x, y and z.

Answer: The two upper slanting lines are perpendicular to each other, and one line is perpendicular to the horizontal line.

90° − x = 30° (with the given 30° angle) → x = 60°.

y = 90° − x = 90° − 60° = 30°.

z = 90° − y = 90° − 30° = 60°.

Question: Exercise 10.3, Question 6, figure (iv): In the figure, find the measures of x, y and z.

Answer: The right-angle mark lies in the x wedge, so x = 90°.

The upper-right ray is 15° above the horizontal, so y = 15° (vertical angle).

The bottom angle z = 90°.

[note: The figure is not drawn to scale;

x = 90° comes from the right-angle mark.]

Question: Exercise 10.4, Question 1: AB and CD are two lines cutting each other at point O. Prove that AC ≅ BD.

Answer: Join A to C and B to D.

In ΔAOC and ΔBOD: OA ≅ OB and OC ≅ OD (O bisects both lines), and ∠AOC ≅ ∠BOD (vertical angles, Theorem 3).

Hence ΔAOC ≅ ΔBOD (SAS, Theorem 4).

Hence AC ≅ BD (corresponding sides of congruent triangles).

[note: The SAS step needs OA ≅ OB and OC ≅ OD;

this is read from the figure and is implied by the book's hint.]

Question: Exercise 10.4, Question 2: Prove that if two lines cut each other, their vertical angles are equal to each other.

Answer: Let two lines cut each other, forming four angles: ∠1, ∠2, ∠3, ∠4.

∠1 + ∠2 = 180° and ∠3 + ∠2 = 180° (Theorem 1).

Hence ∠1 + ∠2 = ∠3 + ∠2, and subtracting ∠2 gives ∠1 = ∠3 (Axiom 5).

Similarly ∠2 = ∠4.

Hence the vertical angles are equal.

Q.E.D.

[note: The printed question appears to use the word for "alternate", but the angles meant are the vertical (opposite) angles;

this is the general statement of Exercise 10.3 Question 3.]

Question: Exercise 10.4, Question 3: Prove that all four angles of a rectangle are equal to each other.

Answer: Let ABCD be a rectangle, with ∠A = 90° (definition).

AD ∥ BC, so ∠A + ∠B = 180° (same-side interior angles, Theorem 1) → ∠B = 90°.

AB ∥ CD, so ∠B + ∠C = 180° → ∠C = 90°, and ∠C + ∠D = 180° → ∠D = 90°.

Hence all four angles are equal (each 90°).

Q.E.D.

Question: Exercise 10.4, Question 4: Prove that in an isosceles triangle, the bisector of the vertex angle is perpendicular to the base.

Answer: Let ΔABC have AB ≅ AC, and let AD bisect ∠A, meeting BC at D.

In ΔABD and ΔACD: AB ≅ AC (given), ∠BAD ≅ ∠CAD (AD is the bisector), AD is common.

Hence ΔABD ≅ ΔACD (SAS, Theorem 4).

Hence ∠ADB = ∠ADC.

But ∠ADB + ∠ADC = 180° (angles on a straight line), so each is 90°, that is, AD ⊥ BC.

Q.E.D.

Question: Exercise 10.4, Question 5: Prove that every point on the perpendicular bisector of a line is equidistant from the ends of the line.

Answer: Let OP be the perpendicular bisector of AB, that is, OP ⊥ AB and O is the midpoint of AB.

Let P be any point on OP.

In ΔAOP and ΔBOP: AO ≅ BO (O is the midpoint), ∠AOP = ∠BOP = 90° (OP ⊥ AB), OP is common.

Hence ΔAOP ≅ ΔBOP (SAS).

Hence AP ≅ BP.

Q.E.D.

Question: Exercise 10.4, Question 6: Prove that the medians of the equal sides of an isosceles triangle are equal to each other.

Answer: Let ΔABC have AB ≅ AC.

Let BD and CE be the medians of the two equal sides (D is the midpoint of AC;

E is the midpoint of AB).

In ΔABD and ΔACE: AB ≅ AC (given), ∠A is common, AD ≅ AE (halves of equal sides: AD = (1/2)AC, AE = (1/2)AB).

Hence ΔABD ≅ ΔACE (SAS, Theorem 4).

Hence BD ≅ CE.

Q.E.D.

Question: Exercise 10.4, Question 7: Prove that from an exterior point, more than one perpendicular cannot be drawn to a line.

Answer: Suppose the contrary: from an exterior point P, two perpendiculars PA and PB can be drawn to line l.

Then ΔPAB would have two right angles (∠PAB and ∠PBA), which contradicts Corollary 1 of Theorem 6 (a triangle has at most one right angle).

Hence the supposition is false, and only one perpendicular can be drawn from P to l.

Q.E.D.

Question: Exercise 10.4, Question 8 (i): In the figure, say about the pairs (∠1, ∠2) and (∠3, ∠4): are they supplementary?

Answer: Yes.

(∠1, ∠2) is a pair of adjacent angles on a straight line, and (∠3, ∠4) is likewise a pair of adjacent angles on a straight line.

Hence each pair sums to 180°, that is, they are supplementary (Theorem 1).

Question: Exercise 10.4, Question 8 (ii): In the figure, say about ∠1 and ∠4, and ∠3 and ∠2: are they equal?

Answer: No. ∠1 and ∠4 are alternate interior angles, and ∠3 and ∠2 are also alternate interior angles, but the two lines are not parallel.

Alternate interior angles are equal only when the lines are parallel (Theorem 7).

Hence they are not equal.

Question: Exercise 10.4, Question 8 (iii): In the figure, say about ∠2 and ∠3: are they equal?

Answer: No. ∠2 and ∠3 are alternate interior angles (the slanting line cuts the other two lines), but the two lines are not parallel, so they are not equal (by Theorem 7, equality requires the lines to be parallel).

Question: Exercise 10.4, Question 8 (iv): In the figure, say which pairs are supplementary and why.

Answer: (∠1, ∠2) and (∠3, ∠4) are supplementary, because they are pairs of adjacent angles on a straight line (Theorem 1).

(∠1, ∠3) and (∠2, ∠4) are supplementary only when the two lines are parallel: then ∠1 = ∠4 (alternate, Theorem 7) and ∠3 + ∠4 = 180° (straight line) → ∠1 + ∠3 = 180°, and similarly ∠2 + ∠4 = 180°.

Question: Exercise 10.4, Question 8 (v): In the figure, say when the same-side interior angles are supplementary.

Answer: The same-side interior angles are supplementary when the two lines cut by the transversal are parallel (Corollary 2 of Theorem 7).

Question: Exercise 10.4, Question 8 (vi): Discuss the figure: where do two non-parallel lines meet when extended, and what does Theorem 6 say about it?

Answer: Two non-parallel lines meet when extended on the side where the same-side interior angles together are less than 180°, and they form a triangle with the transversal.

By Theorem 6, the exterior angle of that triangle is greater than each remote interior angle (in particular greater than m∠2 and m∠4).

The full answer comes from Theorems 7 and 8.

Question: Exercise 10.5, Question 1, figure (a): In the figure, find the measure of x.

Answer: Draw a line through the bend parallel to the two parallel lines.

The upper part is (180° − 130°) = 50° (alternate interior angle), and the lower part is 40° (alternate interior angle).

Hence x = 50° + 40° = 90°.

Question: Exercise 10.5, Question 1, figure (b): In the figure, find the measure of x.

Answer: Draw a line through the bend parallel to the parallel lines.

The 135° bend splits into 75° (alternate interior angle) and 60°.

The lower segment makes 60° with the lower line, and x is its supplement: x = 180° − 60° = 120°.

(Or: x = 180° − (135° − 75°) = 120°.)

Question: Exercise 10.5, Question 1, figure (c): In the figure, find the measure of x.

Answer: Draw a line through the dent (notch) point parallel to the upper and lower horizontal edges.

The alternate interior angles are 25° and 30°, so the dent angle x = 25° + 30° = 55°.

[note: The figure also marks the right-hand point as x, but with the given 115° and 105° that point's interior angle is 40° (180° − 65° − 75°), not 55°;

the two x-marks in the printed figure do not agree.

The book's key gives 55° as the dent angle.]

Question: Exercise 10.5, Question 1, figure (d): In the figure, find the measure of x.

Answer: The figure has a triangle whose angles are 40° (the vertical angle of the given 40° at the crossing), 75° and x.

The angles of a triangle sum to 180° (Theorem 9): x = 180° − 40° − 75° = 65°.

[note: The book's key lists an extra (d) 140° entry which matches no figure in Question 1 (its (e) 65° matches this figure).]

Question: Exercise 10.5, Question 2: If two lines AB and CD cut each other

at point O, prove that AC and BD are parallel and AD and BC are parallel.

Answer: O is the midpoint of both lines (OA ≅ OB, OC ≅ OD — the setup of Exercise 10.4 Question 1).

In ΔAOC and ΔBOD: OA ≅ OB, OC ≅ OD, ∠AOC ≅ ∠BOD (vertical).

Hence ΔAOC ≅ ΔBOD (SAS) → ∠OAC = ∠OBD (alternate interior angles) → AC ∥ BD (Theorem 8).

Similarly ΔAOD ≅ ΔBOC → ∠OAD = ∠OBC → AD ∥ BC.

Q.E.D.

Question: Exercise 10.5, Question 3: D and E are the midpoints of the two sides AB and AC of triangle ΔABC respectively. If DE is extended to point F so that EF ≅ DE, prove that CF ∥ AB and DE ∥ BC.

Answer: In ΔADE and ΔCFE: AE ≅ CE (E is the midpoint of AC), ∠AED ≅ ∠CEF (vertical angles), DE ≅ EF (given).

Hence ΔADE ≅ ΔCFE (SAS).

Hence CF ≅ AD and ∠A ≅ ∠ECF.

∠A and ∠ECF are alternate interior angles, so CF ∥ AB (Theorem 8).

Moreover, CF ≅ AD ≅ BD (D is the midpoint of AB), so DBCF is a parallelogram (one pair of opposite sides equal and parallel), so DF ∥ BC, that is, DE ∥ BC.

Q.E.D.

Question: Exercise 10.5, Question 4: Prove that the opposite angles of a parallelogram are equal to each other.

Answer: Let ABCD be a parallelogram (AB ∥ CD, AD ∥ BC).

AB is a transversal cutting AD and BC, so m∠A + m∠B = 180° (same-side interior angles, Corollary 2 of Theorem 8).

Similarly BC is a transversal cutting AB and CD, so m∠B + m∠C = 180°.

Hence m∠A + m∠B = m∠B + m∠C → m∠A = m∠C.

Similarly m∠B = m∠D.

Hence the opposite angles are equal.

Q.E.D.

Question: Exercise 10.5, Question 5: In an isosceles triangle, what can be the measure of each angle?

Answer: In an isosceles triangle the two base angles are equal (Theorem 5) and each is less than 90°, because the angles of a triangle sum to 180° (Theorem 9): if one base angle were 90° or more, the sum would exceed 180°.

The vertex angle = 180° − 2×(base angle).

Hence the vertex angle can be acute, right (90°, in which case each base angle is 45°), or obtuse (less than 180°).

Question: Exercise 10.5, Question 6: One angle of a triangle equals the sum of the other two angles. In which case can this happen?

Answer: This can happen exactly when the triangle is right-angled.

Let one angle be A and the other two B and C, with A = B + C.

The angles of a triangle sum to 180° (Theorem 9): A + B + C = 180° → A + A = 180° → A = 90°.

Hence that angle is a right angle and the other two sum to 90° (Corollary 3 of Theorem 9).

Question: Exercise 10.5, Question 7: Prove that the sum of the interior angles of any quadrilateral is the sum of four right angles.

Answer: Draw any one diagonal of the quadrilateral.

The diagonal divides the quadrilateral into two triangles.

Each triangle's angles sum to 180° = 2 right angles (Theorem 9).

Hence the sum of the quadrilateral's interior angles = 2 right angles + 2 right angles = 4 right angles = 360°.

Q.E.D.

Question: Review Exercise 10, Question 1: Demonstrative geometry is a branch of mathematics. What is the important work of this branch?

Answer: The important work of this branch is to prove geometrical statements (propositions) by logical reasoning and proof — that is, to establish facts about figures by reasoning, not merely by looking or measuring.

Question: Review Exercise 10, Question 2 (a): What do you know about the following term: undefined terms.

Answer: Undefined terms are the basic terms which are not defined, but are understood directly, such as point, line and plane.

All of geometry rests on them.

Question: Review Exercise 10, Question 2 (b): What do you know about the following term: assumptions.

Answer: Assumptions are statements accepted without proof, on which proofs rest.

They are of two kinds: axioms and postulates.

Question: Review Exercise 10, Question 2 (c): What do you know about the following term: propositions.

Answer: Propositions are statements that are to be proved or that are proved, such as a theorem, a corollary and a rider/problem.

Every proposition is proved by reasoning.

Question: Review Exercise 10, Question 2 (d): What do you know about the following term: postulate.

Answer: A postulate is the fundamental assumption which concerns geometry and is accepted without proof, such as "exactly one line can be drawn through two different points".

Question: Review Exercise 10, Question 2 (e): What do you know about the following term: axiom.

Answer: An axiom is the fundamental assumption which applies generally to all of mathematics and is accepted without proof, such as "if a = b and b = c, then a = c" (the transitive property of equality).

Question: Review Exercise 10, Question 2 (f): What do you know about the following term: corollary.

Answer: A corollary is a proposition which follows directly from another proposition and needs little separate proof, such as the corollaries of Theorem 1.

Question: Review Exercise 10, Question 3: State the four properties of Reasoning.

Answer: The four important elements of deductive reasoning are: (1) defined terms, (2) undefined terms, (3) propositions, (4) assumptions.

In the working form of a proof there are also four parts: given, required to prove, construction, proof.

Question: Review Exercise 10, Question 4 (a): Write the statement of Theorem 1 and prove it.

Answer: Theorem 1: The sum of the angles formed at any point on a straight line is 180° (2 right angles).

Proof: Let AB be a straight line and O a point on it, from which ray OP emerges.

Draw OP perpendicular to AB (construction).

Now m∠AOP + m∠POB = 180°, because together they make the straight line AB.

Q.E.D.

Question: Review Exercise 10, Question 4 (b): Prove that the sum of the angles formed on the same side of any line is 2 right angles.

Answer: Let AB be a line and O a point on it, and let several rays on the same side form angles ∠1, ∠2, ∠3, ...

By Theorem 1, the whole angle between the first and last rays is 180°, and that whole angle is the sum of all these small angles.

Hence ∠1 + ∠2 + ∠3 + ... = 180° = 2 right angles.

Q.E.D.

Question: Review Exercise 10, Question 5 (a): Prove: if two lines cut each other, their vertical angles are equal to each other.

Answer: Let two lines cut each other, forming four angles.

Any two adjacent angles sum to 180° (Theorem 1).

Hence ∠1 + ∠2 = 180° and ∠3 + ∠2 = 180° → ∠1 = ∠3.

Similarly ∠2 = ∠4.

Hence the vertical angles are equal (Theorem 3).

Q.E.D.

Question: Review Exercise 10, Question 5 (b): The base angles of an isosceles triangle are equal to each other. Prove it.

Answer: Let ΔABC have AB ≅ AC.

Draw the bisector AD of ∠A, meeting BC at D.

In ΔABD and ΔACD: AB ≅ AC, ∠BAD ≅ ∠CAD, AD is common.

Hence ΔABD ≅ ΔACD (SAS, Theorem 4).

Hence ∠B = ∠C, that is, the base angles are equal (Theorem 5).

Q.E.D.

Question: Review Exercise 10, Question 6: Prove that from a point, only one perpendicular can be drawn to a line.

Answer: Suppose the contrary: from an exterior point P, two perpendiculars PA and PB can be drawn to line l.

Then ΔPAB would have two right angles, which contradicts Corollary 1 of Theorem 6 (a triangle has at most one right angle).

Hence the supposition is false, and only one perpendicular can be drawn from P to l (Corollary 2 of Theorem 6).

Q.E.D.

Question: Review Exercise 10, Question 7 (a): Prove that the exterior angle of any triangle equals the sum of the remote interior angles.

Answer: Let ΔABC have BC extended to D, so ∠ACD is the exterior angle.

The angles of a triangle sum to 180° (Theorem 9): ∠A + ∠B + ∠ACB = 180°.

And ∠ACD + ∠ACB = 180° (angles on a straight line).

Hence ∠ACD + ∠ACB = ∠A + ∠B + ∠ACB → ∠ACD = ∠A + ∠B.

Q.E.D.

Question: Review Exercise 10, Question 7 (b): Write the statement of Theorem 5 and prove it.

Answer: Theorem 5: The base angles of an isosceles triangle are equal to each other.

Proof: As in Review Exercise 10 Question 5 (b): draw the bisector AD of ∠A, prove ΔABD ≅ ΔACD (SAS), hence ∠B = ∠C.

Q.E.D.

Correct: Review Exercise 10, Question 8 (a) (i): State whether this statement is true or false: If two lines cut each other, their vertical angles are equal to each other.
Vertical angles are always equal (Theorem 3). [note: The book's answer key lists this item as false; as printed the statement is true, because vertical angles are equal. Either the key is misprinted or a negation is missing from the printed statement.]

Correct: Review Exercise 10, Question 8 (a) (ii): State whether this statement is true or false: From any point outside a line, only one line parallel to that line can be drawn.
From any exterior point, only one line parallel to the given line can be drawn (the parallel postulate).

Correct: Review Exercise 10, Question 8 (a) (iii): State whether this statement is true or false: The sum of the three angles of a triangle is 180°.
The sum of the three angles of a triangle is 180° (Theorem 9). [note: The book's answer key lists this item as false; as printed the statement is true, because the angle sum is 180°. Either the key is misprinted or a negation is missing from the printed statement.]

Incorrect: Review Exercise 10, Question 8 (a) (iv): State whether this statement is true or false: If a ray emerges from a point on a line, the two angles formed with it can be complementary.
The two angles formed by a ray emerging from a point on a line are

adjacent angles on a straight line, so their sum is 180° (supplementary, Theorem 1), not 90° (complementary). Hence they cannot be complementary.

Correct: Review Exercise 10, Question 8 (a) (v): State whether this statement is true or false: An exterior angle of a triangle equals the sum of its remote interior angles.
This is Corollary 5 of Theorem 9 (the exterior angle theorem).

Review Exercise 10, Question 8 (b) (i): Fill in the blank: On a _____, infinitely many lines are determined from every point.
Answer: plane,

Review Exercise 10, Question 8 (b) (ii): Fill in the blank: In a plane, exactly one _____ is determined by two different points.
Answer: line

Review Exercise 10, Question 8 (b) (iii): Fill in the blank: From a point not on a line, more than one _____ cannot be drawn.
Answer: perpendicular

Review Exercise 10, Question 8 (b) (iv): Fill in the blanks: If two lines are parallel to each other, their _____ angles and _____ angles are equal to each other.
Answer: alternate, corresponding

Review Exercise 10, Question 8 (b) (v): Fill in the blank: The measure of an exterior angle of a triangle equals the _____ of its remote interior

angles.

Answer: sum

Question: Review Exercise 10, Question 9 (i): In the given figure, prove that m∠a = m∠b = m∠c = m∠d, and write the pairs of parallel lines.

Answer: The pairs of parallel lines in the figure are: AB ∥ CD, AB ∥ FG, FG ∥ CD, EF ∥ GC.

AB ∥ FG and EF is the transversal, so a = b (alternate interior angles, Theorem 7).

EF ∥ GC and FG is the transversal, so b = c (alternate interior angles).

FG ∥ CD and GC is the transversal, so c = d (alternate interior angles).

Hence m∠a = m∠b = m∠c = m∠d.

Q.E.D.

Question: Review Exercise 10, Question 9 (ii) (a): In the following figure, find the measure of the angle shown by x.

Answer: Draw a line through the bend parallel to the parallel lines.

The bend angle 42° is the sum of two alternate interior angles: 42° = 20° + x → x = 42° − 20° = 22°.

Question: Review Exercise 10, Question 9 (ii) (b): In the following figure, find the measure of the angle shown by x.

Answer: At the bend, the angle between the slanting line and the horizontal is 180° − 130° = 50° (supplementary).

x is the exterior angle of a triangle whose remote interior angles are 75° and 50°, so x = 75° + 50° = 125° (Corollary 5 of Theorem 9).

Question: Review Exercise 10, Question 9 (ii) (c): In the following figure, find the measure of the angle shown by x.

Answer: The slanting line makes 120° with the lower horizontal line, so it makes 180° − 120° = 60° with the upper horizontal ray (corresponding angles).

At the upper point, this 60° angle splits into 40° and x: x = 60° − 40° = 20°.

11. Trigonometry

Question: Exercise 11.1, Question 1 (a): In the given right triangle PQR, find the six trigonometric ratios of ∠P. (mPQ = 15 cm, mPR = 12 cm, mQR = 9 cm)

Answer: For ∠P: opposite side = QR = 9 cm, adjacent side = PR = 12 cm, hypotenuse = PQ = 15 cm.

Hence: Sin P = 9/15 = 3/5, Cos P = 12/15 = 4/5, Tan P = 9/12 = 3/4, Cot P = 12/9 = 4/3, Sec P = 15/12 = 5/4, Cosec P = 15/9 = 5/3.

Question: Exercise 11.1, Question 1 (b): In the given right triangle PQR, find the six trigonometric ratios of ∠Q. (mPQ = 15 cm, mPR = 12 cm, mQR = 9 cm)

Answer: For ∠Q: opposite side = PR = 12 cm, adjacent side = QR = 9 cm, hypotenuse = PQ = 15 cm.

Hence: Sin Q = 12/15 = 4/5, Cos Q = 9/15 = 3/5, Tan Q = 12/9 = 4/3, Cot Q = 9/12 = 3/4, Sec Q = 15/9 = 5/3, Cosec Q = 15/12 = 5/4.

Question: Exercise 11.1, Question 2 (i): For the given right triangle LMN, find the value of Sin x. (x = m∠M, y = m∠N, l = 20 cm, m = 16 cm, n = 12 cm)

Answer: For x: opposite side = m = 16 cm, hypotenuse = l = 20 cm.

Sin x = 16/20 = 4/5.

Question: Exercise 11.1, Question 2 (ii): For the given right triangle LMN, find the value of Cos y.

Answer: For y: adjacent side = m = 16 cm, hypotenuse = l = 20 cm.

Cos y = 16/20 = 4/5.

Question: Exercise 11.1, Question 2 (iii): For the given right triangle LMN, find the value of Tan x.

Answer: For x: opposite side = 16 cm, adjacent side = n = 12 cm.

Tan x = 16/12 = 4/3.

Question: Exercise 11.1, Question 2 (iv): For the given right triangle LMN, find the value of Cosec y.

Answer: For y: hypotenuse = 20 cm, opposite side = n = 12 cm.

Cosec y = 20/12 = 5/3.

Question: Exercise 11.1, Question 2 (v): For the given right triangle LMN, find the value of Sec x.

Answer: For x: hypotenuse = 20 cm, adjacent side = n = 12 cm.

Sec x = 20/12 = 5/3.

Question: Exercise 11.1, Question 2 (vi): For the given right triangle LMN, find the value of Cot y.

Answer: For y: adjacent side = m = 16 cm, opposite side = n = 12 cm.

Cot y = 16/12 = 4/3.

Question: Exercise 11.1, Question 2 (vii): For the given right triangle LMN, find the value of Cos x.

Answer: For x: adjacent side = n = 12 cm, hypotenuse = 20 cm.

Cos x = 12/20 = 3/5.

Question: Exercise 11.1, Question 2 (viii): For the given right triangle LMN, find the value of Sin y.

Answer: For y: opposite side = n = 12 cm, hypotenuse = 20 cm.

Sin y = 12/20 = 3/5.

Question: Exercise 11.1, Question 2 (ix): For the given right triangle LMN, find the value of Cot x.

Answer: For x: adjacent side = 12 cm, opposite side = 16 cm.

Cot x = 12/16 = 3/4.

Question: Exercise 11.1, Question 2 (x): For the given right triangle LMN, find the value of Sec y.

Answer: For y: hypotenuse = 20 cm, adjacent side = m = 16 cm.

Sec y = 20/16 = 5/4.

Question: Exercise 11.1, Question 2 (xi): For the given right triangle LMN, find the value of Cosec x.

Answer: For x: hypotenuse = 20 cm, opposite side = m = 16 cm.

Cosec x = 20/16 = 5/4.

Question: Exercise 11.1, Question 2 (xii): For the given right triangle LMN, find the value of Tan y.

Answer: For y: opposite side = n = 12 cm, adjacent side = m = 16 cm.

Tan y = 12/16 = 3/4.

Question: Exercise 11.1, Question 2: State which of the above ratios are equal to each other.

Answer: The equal pairs are: Sin x = Cos y = 4/5;

Tan x = Cot y = 4/3;

Cosec y = Sec x = 5/3;

Cos x = Sin y = 3/5;

Cot x = Tan y = 3/4;

Sec y = Cosec x = 5/4.

(In each pair, one angle's ratio equals the other angle's corresponding complementary ratio, because x + y = 90°.)

Question: Exercise 11.1, Question 3 (i): For the given right triangle UVW,

verify that Sin α · Cosec α = 1. (m∠W = 90°, m∠U = α, m∠V = β, UW = 8 cm, WV = 15 cm, UV = 17 cm)

Answer: For α: Sin α = 15/17, Cosec α = 17/15.

Hence Sin α · Cosec α = (15/17) × (17/15) = 1.

Verified.

Question: Exercise 11.1, Question 3 (ii): For the given right triangle UVW, verify that Sec β · Cos β = 1.

Answer: For β: Sec β = 17/15, Cos β = 15/17.

Hence Sec β · Cos β = (17/15) × (15/17) = 1.

Verified.

Question: Exercise 11.1, Question 3 (iii): For the given right triangle UVW, verify that Tan α · Cot α = 1.

Answer: For α: Tan α = 15/8, Cot α = 8/15.

Hence Tan α · Cot α = (15/8) × (8/15) = 1.

Verified.

Question: Exercise 11.1, Question 3 (iv): For the given right triangle UVW, verify that Sin β / Cos β = Tan β.

Answer: For β: Sin β = 8/17, Cos β = 15/17.

Hence Sin β / Cos β = (8/17) ÷ (15/17) = 8/15.

And Tan β = 8/15.

Hence Sin β / Cos β = Tan β.

Verified.

Question: Exercise 11.1, Question 3 (v): For the given right triangle UVW, verify that Cos α / Sin α = Cot α.

Answer: For α: Cos α = 8/17, Sin α = 15/17.

Hence Cos α / Sin α = (8/17) ÷ (15/17) = 8/15.

And Cot α = 8/15.

Hence Cos α / Sin α = Cot α.

Verified.

Question: Exercise 11.1, Question 3 (vi): For the given right triangle UVW, verify that Cos α · Cosec α = Cot α.

Answer: For α: Cos α = 8/17, Cosec α = 17/15.

Hence Cos α · Cosec α = (8/17) × (17/15) = 8/15.

And Cot α = 8/15.

Hence Cos α · Cosec α = Cot α.

Verified.

Question: Exercise 11.1, Question 3 (vii): For the given right triangle UVW, verify that Sec β / Cosec β = Tan β.

Answer: For β: Sec β = 17/15, Cosec β = 17/8.

Hence Sec β / Cosec β = (17/15) ÷ (17/8) = 8/15.

And Tan β = 8/15.

Hence Sec β / Cosec β = Tan β.

Verified.

Question: Exercise 11.1, Question 3 (viii): For the given right triangle UVW, verify that Tan α / Cot β = 1.

Answer: Tan α = 15/8, Cot β = 15/8 (for β: adjacent side = 15, opposite side = 8).

Hence Tan α / Cot β = (15/8) ÷ (15/8) = 1.

Verified.

Question: Exercise 11.1, Question 4 (i): In right triangle ABC, ∠C is the right angle, with m∠A = x and m∠B = y. Prove that Sin x = Cos y.

Answer: x + y = 90°, so y = 90° − x.

In terms of sides: Sin x = (side opposite x)/(hypotenuse) = a/c, and Cos y = (side adjacent to y)/(hypotenuse) = a/c (the side adjacent to y is also a).

Hence Sin x = Cos y.

[note: The side measures given in the figure (16, 24, 30) do not satisfy Pythagoras, so the proof uses general sides a, b, c instead of the particular measures;

the identities follow from x + y = 90° alone.]

Question: Exercise 11.1, Question 4 (ii): In right triangle ABC, prove that Tan y = Cot x.

Answer: Tan y = (side opposite y)/(side adjacent to y) = b/a, and Cot x = (side adjacent to x)/(side opposite x) = b/a.

Hence Tan y = Cot x.

Question: Exercise 11.1, Question 4 (iii): In right triangle ABC, prove that Sec x = Cosec y.

Answer: Sec x = (hypotenuse)/(side adjacent to x) = c/b, and Cosec y = (hypotenuse)/(side opposite y) = c/b.

Hence Sec x = Cosec y.

Question: Exercise 11.1, Question 4 (iv): In right triangle ABC, prove that Cos x = Sin y.

Answer: Cos x = (side adjacent to x)/(hypotenuse) = b/c, and Sin y = (side

opposite y)/(hypotenuse) = b/c.

Hence Cos x = Sin y.

Question: Exercise 11.1, Question 4 (v): In right triangle ABC, prove that Cot y = Tan x.

Answer: Cot y = (side adjacent to y)/(side opposite y) = a/b, and Tan x = (side opposite x)/(side adjacent to x) = a/b.

Hence Cot y = Tan x.

Question: Exercise 11.1, Question 4 (vi): In right triangle ABC, prove that Cosec x = Sec y.

Answer: Cosec x = (hypotenuse)/(side opposite x) = c/a, and Sec y = (hypotenuse)/(side adjacent to y) = c/a.

Hence Cosec x = Sec y.

Question: Exercise 11.2, Question 1 (i): Find the value of x when Cos x = √3/2.

Answer: Cos 30° = √3/2 (table).

Hence Cos x = Cos 30° → x = 30°.

Question: Exercise 11.2, Question 1 (ii): Find the value of x when Cos x = 1/√2.

Answer: Cos 45° = 1/√2 (table).

Hence Cos x = Cos 45° → x = 45°.

Question: Exercise 11.2, Question 1 (iii): Find the value of x when Cos x = 1/2.

Answer: Cos 60° = 1/2 (table).

Hence Cos x = Cos 60° → x = 60°.

Question: Exercise 11.2, Question 2 (i): Find the value of y when Tan y = √3.

Answer: Tan 60° = √3 (table).

Hence Tan y = Tan 60° → y = 60°.

Question: Exercise 11.2, Question 2 (ii): Find the value of y when Tan y = 1.

Answer: Tan 45° = 1 (table).

Hence Tan y = Tan 45° → y = 45°.

Question: Exercise 11.2, Question 2 (iii): Find the value of y when Tan y = 1/√3.

Answer: Tan 30° = 1/√3 (table).

Hence Tan y = Tan 30° → y = 30°.

Question: Exercise 11.2, Question 3 (i): Find the value of ∠A when Cosec m∠A = 2/1.

Answer: Cosec 30° = 2 (table).

Hence Cosec m∠A = Cosec 30° → m∠A = 30°.

Question: Exercise 11.2, Question 3 (ii): Find the value of ∠A when Cosec m∠A = √2.

Answer: Cosec 45° = √2 (table).

Hence Cosec m∠A = Cosec 45° → m∠A = 45°.

Question: Exercise 11.2, Question 3 (iii): Find the value of ∠A when Cosec m∠A = 2/√3.

Answer: Cosec 60° = 2/√3 (table).

Hence Cosec m∠A = Cosec 60° → m∠A = 60°.

Question: Exercise 11.2, Question 4 (i): Find the value of: Sin 30° × Sin 60° + Cos 30° × Cos 60°.

Answer: Sin 30° = 1/2, Sin 60° = √3/2, Cos 30° = √3/2, Cos 60° = 1/2.

Hence = (1/2)(√3/2) + (√3/2)(1/2) = √3/4 + √3/4 = 2√3/4 = √3/2.

Question: Exercise 11.2, Question 4 (ii): Find the value of: Sec 30° × Cosec 60° + Tan 60° × Cot 30°.

Answer: Sec 30° = 2/√3, Cosec 60° = 2/√3, Tan 60° = √3, Cot 30° = √3.

Hence = (2/√3)(2/√3) + (√3)(√3) = 4/3 + 3 = 13/3 = 4 1/3.

Question: Exercise 11.2, Question 4 (iii): Find the value of: Cos 30° × Sin 30° + Cosec 30° × Sec 30°.

Answer: Cos 30° = √3/2, Sin 30° = 1/2, Cosec 30° = 2, Sec 30° = 2/√3.

Hence = (√3/2)(1/2) + (2)(2/√3) = √3/4 + 4/√3 = √3/4 + 4√3/3 = (3√3 + 16√3)/12 = 19√3/12.

Question: Exercise 11.2, Question 4 (iv): Find the value of: Sin 60° × Sin 45° − Cos 30° × Cos 45°.

Answer: Sin 60° = √3/2, Sin 45° = 1/√2, Cos 30° = √3/2, Cos 45° = 1/√2.

Hence = (√3/2)(1/√2) − (√3/2)(1/√2) = 0.

Question: Exercise 11.2, Question 4 (v): Find the value of: Tan 30° × Cot 60° − Sin 45° × Cos 45°.

Answer: Tan 30° = 1/√3, Cot 60° = 1/√3, Sin 45° = 1/√2, Cos 45° = 1/√2.

Hence = (1/√3)(1/√3) − (1/√2)(1/√2) = 1/3 − 1/2 = (2 − 3)/6 = −1/6.

Question: Exercise 11.2, Question 4 (vi): Find the value of: Sec 45° × Cosec 45° + Sec 30° × Cosec 60°.

Answer: Sec 45° = √2, Cosec 45° = √2, Sec 30° = 2/√3, Cosec 60° = 2/√3.

Hence = (√2)(√2) + (2/√3)(2/√3) = 2 + 4/3 = 10/3 = 3 1/3.

Question: Exercise 11.2, Question 4 (vii): Find the value of: Cot 60° × Sec 30° − Cosec 60° × Tan 30°.

Answer: Cot 60° = 1/√3, Sec 30° = 2/√3, Cosec 60° = 2/√3, Tan 30° =

1/√3.

Hence = (1/√3)(2/√3) − (2/√3)(1/√3) = 2/3 − 2/3 = 0.

Question: Exercise 11.2, Question 5 (i): Find x from the equation: Sin 30° = Cos x.

Answer: Sin 30° = 1/2, and Cos 60° = 1/2.

Hence Cos x = Cos 60° → x = 60°.

(Or: Sin 30° = Cos (90° − 30°) = Cos 60° → x = 60°.)

Question: Exercise 11.2, Question 5 (ii): Find x from the equation: Tan 60° = Cot x.

Answer: Tan 60° = √3, and Cot 30° = √3.

Hence Cot x = Cot 30° → x = 30°.

(Or: Tan 60° = Cot (90° − 60°) = Cot 30° → x = 30°.)

Question: Exercise 11.2, Question 5 (iii): Find x from the equation: Sec x = Cosec 45°.

Answer: Cosec 45° = √2, and Sec 45° = √2.

Hence Sec x = Sec 45° → x = 45°.

(Or: Cosec 45° = Sec (90° − 45°) = Sec 45° → x = 45°.)

Question: Exercise 11.2, Question 5 (iv): Find x from the equation: Cos x = Sin 70°.

Answer: Sin 70° = Cos (90° − 70°) = Cos 20°.

Hence Cos x = Cos 20° → x = 20°.

Question: Exercise 11.2, Question 5 (v): Find x from the equation: Sec 40° = Cosec x.

Answer: Sec 40° = Cosec (90° − 40°) = Cosec 50°.

Hence Cosec x = Cosec 50° → x = 50°.

Question: Exercise 11.2, Question 5 (vi): Find x from the equation: Cot x = Tan 55°.

Answer: Tan 55° = Cot (90° − 55°) = Cot 35°.

Hence Cot x = Cot 35° → x = 35°.

Question: Exercise 11.2, Question 5 (vii): Find x from the equation: Cosec 10° = Sec x.

Answer: Cosec 10° = Sec (90° − 10°) = Sec 80°.

Hence Sec x = Sec 80° → x = 80°.

Question: Exercise 11.2, Question 5 (viii): Find x from the equation: Tan x = Cot 65°.

Answer: Cot 65° = Tan (90° − 65°) = Tan 25°.

Hence Tan x = Tan 25° → x = 25°.

Question: Exercise 11.2, Question 5 (ix): Find x from the equation: Sin x = Cos 35°.

Answer: Cos 35° = Sin (90° − 35°) = Sin 55°.

Hence Sin x = Sin 55° → x = 55°.

Exercise 11.2, Question 6 (i): Fill in the blank: Sin 35° = Cos _____.

Answer: (90° − 35°) = Cos 55°.

Exercise 11.2, Question 6 (ii): Fill in the blank: Tan 65° = Cot _____.

Answer: (90° − 65°) = Cot 25°.

Exercise 11.2, Question 6 (iii): Fill in the blank: Sec 20° = Cosec _____.

Answer: (90° − 20°) = Cosec 70°.

Exercise 11.2, Question 6 (iv): Fill in the blank: Cot 53° = Tan _____.

Answer: (90° − 53°) = Tan 37°.

Exercise 11.2, Question 6 (v): Fill in the blank: Cos 56° = Sin _____.

Answer: (90° − 56°) = Sin 34°.

Exercise 11.2, Question 6 (vi): Fill in the blank: Cosec 75° = Sec _____.

Answer: (90° − 75°) = Sec 15°.

Question: Exercise 11.3, Part A, Question 1: Solve the right triangle ΔABC. (m∠C = 90°, m∠A = 30°, mBC = 3 cm)

Answer: m∠B = 90° − m∠A = 90° − 30° = 60°.

Sin 30° = mBC/mAB → 1/2 = 3/mAB → mAB = 6 cm.

Tan 30° = mBC/mAC → 1/√3 = 3/mAC → mAC = 3√3 cm.

Question: Exercise 11.3, Part A, Question 2: Solve the right triangle ΔDEF. (m∠E = 90°, m∠D = 60°, mDE = 4 cm)

Answer: m∠F = 90° − m∠D = 90° − 60° = 30°.

Tan 60° = mEF/mDE → √3 = mEF/4 → mEF = 4√3 cm.

Cos 60° = mDE/mDF → 1/2 = 4/mDF → mDF = 8 cm.

Question: Exercise 11.3, Part A, Question 3: Solve the right triangle ΔLMN. (m∠M = 90°, m∠N = 45°, mML = 8 cm)

Answer: m∠L = 90° − m∠N = 90° − 45° = 45°.

Tan 45° = mML/mMN → 1 = 8/mMN → mMN = 8 cm.

Hypotenuse mLN = √(8² + 8²) = √128 = 8√2 cm.

Question: Exercise 11.3, Part B, figure 1: Find the unknown parts of the right triangle in the figure. (∠A = 30°, ∠C = 90°, hypotenuse AB = 10 cm)

Answer: m∠B = 90° − 30° = 60°.

Sin 30° = mBC/mAB → 1/2 = mBC/10 → mBC = 5 cm.

Cos 30° = mAC/mAB → √3/2 = mAC/10 → mAC = 5√3 cm.

Question: Exercise 11.3, Part B, figure 2: Find the unknown parts of the right triangle in the figure. (∠B = 60°, ∠C = 90°, hypotenuse DB = 14 cm)

Answer: m∠D = 90° − 60° = 30°.

Sin 60° = mDC/mDB → √3/2 = mDC/14 → mDC = 7√3 cm.

Cos 60° = mBC/mDB → 1/2 = mBC/14 → mBC = 7 cm.

Question: Exercise 11.3, Part B, figure 3: Find the unknown parts of the right triangle in the figure. (∠D = 45°, ∠C = 90°, hypotenuse DG = 8√2 cm)

Answer: m∠G = 90° − 45° = 45°.

The triangle is isosceles right, so mDC = mGC.

Cos 45° = mDC/mDG → 1/√2 = mDC/(8√2) → mDC = 8 cm.

Hence mGC = 8 cm.

Question: Exercise 11.3, Part B, figure 4: Find the unknown parts of the right triangle in the figure. (∠F = 30°, ∠C = 90°, hypotenuse EF = 9 cm)

Answer: m∠E = 90° − 30° = 60°.

Sin 30° = mEC/mEF → 1/2 = mEC/9 → mEC = 4.5 cm.

Cos 30° = mCF/mEF → √3/2 = mCF/9 → mCF = 9√3/2 cm.

Question: Exercise 11.3, Part B, figure 5: Find the unknown parts of the right triangle in the figure. (∠N = 60°, ∠O = 90°, hypotenuse MN = 16 cm)

Answer: m∠M = 90° − 60° = 30°.

Sin 60° = mMO/mMN → √3/2 = mMO/16 → mMO = 8√3 cm.

Cos 60° = mON/mMN → 1/2 = mON/16 → mON = 8 cm.

Question: Exercise 11.3, Part B, figure 6: Find the unknown parts of the right triangle in the figure. (∠P = 60°, ∠Q = 90°, QR = 3√3 cm)

Answer: m∠R = 90° − 60° = 30°.

Tan 60° = mQR/mPQ → √3 = 3√3/mPQ → mPQ = 3 cm.

Sin 60° = mQR/mPR → √3/2 = 3√3/mPR → mPR = 6 cm.

Question: Exercise 11.3, Part C, Question 1: Solve the right triangle ΔLMN. (m∠M = 90°, l = 4 cm, n = 4√3 cm)

Answer: Tan m∠L = l/n = 4/(4√3) = 1/√3 → m∠L = 30°.

m∠N = 90° − 30° = 60°.

Hypotenuse m = √(l² + n²) = √(16 + 48) = √64 = 8 cm.

Question: Exercise 11.3, Part C, Question 2: Solve the right triangle ΔPQR. (m∠Q = 90°, p = 6√3 cm, r = 6 cm)

Answer: Tan m∠P = p/r = 6√3/6 = √3 → m∠P = 60°.

m∠R = 90° − 60° = 30°.

Hypotenuse q = √(p² + r²) = √(108 + 36) = √144 = 12 cm.

Question: Exercise 11.3, Part C, Question 3: Solve the right triangle ΔXYZ.

(m∠Y = 90°, x = z = 7 cm)

Answer: Tan m∠X = x/z = 7/7 = 1 → m∠X = 45°.

m∠Z = 90° − 45° = 45°.

Hypotenuse y = √(x² + z²) = √(49 + 49) = √98 = 7√2 cm.

Question: Exercise 11.3, Part D, Question 1: Solve the right triangle ΔACB.

(m∠C = 90°, a = 5 cm, c = 10 cm)

Answer: Sin m∠A = a/c = 5/10 = 1/2 → m∠A = 30°.

m∠B = 90° − 30° = 60°.

b = √(c² − a²) = √(100 − 25) = √75 = 5√3 cm.

Question: Exercise 11.3, Part D, Question 2: Solve the right triangle ΔDEF.

(m∠F = 90°, d = 7√3 cm, f = 14 cm)

Answer: Sin m∠D = d/f = 7√3/14 = √3/2 → m∠D = 60°.

m∠E = 90° − 60° = 30°.

e = √(f² − d²) = √(196 − 147) = √49 = 7 cm.

Question: Exercise 11.3, Part D, Question 3: Solve the right triangle ΔLMN. (m∠L = 90°, m = 8 cm, l = 8√2 cm)

Answer: Sin m∠M = m/l = 8/(8√2) = 1/√2 → m∠M = 45°.

m∠N = 90° − 45° = 45°.

n = √(l² − m²) = √(128 − 64) = √64 = 8 cm.

Question: Exercise 11.4, Question 1: A ladder is placed against a wall as shown in the figure. The ladder makes a 60° angle with the floor. If its lower end is 3 m from the wall, find the length of the ladder.

Answer: In the right triangle: Cos 60° = (adjacent side)/(hypotenuse) = 3/(length of the ladder).

Cos 60° = 1/2, so 1/2 = 3/(length of the ladder) → length of the ladder = 6 m.

Question: Exercise 11.4, Question 2: The base of a hill is 200 m from a point on the ground. From that point, the angle of elevation of the top of the hill is 45°. Find the height of the hill.

Answer: In the right triangle: Tan 45° = (height of the hill)/(200).

Tan 45° = 1, so height of the hill = 200 × 1 = 200 m.

Question: Exercise 11.4, Question 3: On the seashore there is a 170 m high lighthouse. A ship in the sea makes a 30° angle of depression as shown in the figure. Find the distance of the ship from the base of the lighthouse.

Answer: The angle of depression is 30°, so the angle of elevation of the top of the lighthouse from the ship is also 30° (alternate angles).

In the right triangle: Tan 30° = 170/(distance) → 1/√3 = 170/(distance) → distance = 170√3 ≈ 294.4 m (about 294 m).

Question: Exercise 11.4, Question 4: A flag is shown in the figure. Find its height. (In the figure, base = 173 m, angle = 30°)

Answer: In the right triangle: Tan 30° = (height of the flag)/(173) → 1/√3 = (height)/173 → height = 173/√3 = 173√3/3 ≈ 99.88 m (about 100 m).

[note: The book's answer key lists 11.55 m for this question, which does not match the printed figure (base 173 m);

11.55 m is the answer for a 20 m base (20 × Tan 30° = 11.55).]

Question: Exercise 11.4, Question 5: A truck climbs a 100 m slope whose angle is 30°. Tell how much the truck has climbed vertically.

Answer: In the right triangle: Sin 30° = (vertical climb)/(100) → 1/2 = (vertical climb)/100 → vertical climb = 50 m.

Question: Review Exercise 11, Question 1 (i): Verify that: 2 Sin 45° + (1/2) Cosec 45° = 3/√2.

Answer: L.H.S.

= 2 × (1/√2) + (1/2) × (√2) = 2/√2 + √2/2 = √2 + √2/2 = 3√2/2.

And 3/√2 = 3√2/2 (multiplying numerator and denominator by √2).

Hence L.H.S.

= R.H.S.

Verified.

Question: Review Exercise 11, Question 1 (ii): Verify that: Cos 60° · Cos 30° − Sin 60° · Sin 30° = 0.

Answer: L.H.S.

= (1/2)(√3/2) − (√3/2)(1/2) = √3/4 − √3/4 = 0 = R.H.S.

Verified.

Question: Review Exercise 11, Question 1 (iii): Verify that: Sin 60° Cos 30° − Cos 60° Sin 30° = 1/2.

Answer: L.H.S.

= (√3/2)(√3/2) − (1/2)(1/2) = 3/4 − 1/4 = 2/4 = 1/2 = R.H.S.

Verified.

Question: Review Exercise 11, Question 1 (iv): Verify that: (Cos 30° − Sin 30°) × (Cos 30° + Sin 30°) = 1/2.

Answer: L.H.S.

= Cos^2 30° − Sin^2 30° (difference-of-squares formula: (a − b)(a + b) = a^2 − b^2) = $(\sqrt{3}/2)^2 - (1/2)^2$ = 3/4 − 1/4 = 1/2 = R.H.S.

Verified.

Question: Review Exercise 11, Question 1 (v): Verify that: Sin 45° × Cos 45° + Sin 30° × Cos 60° = 3/4.

Answer: L.H.S.

= (1/√2)(1/√2) + (1/2)(1/2) = 1/2 + 1/4 = 3/4 = R.H.S.

Verified.

Question: Review Exercise 11, Question 2: On the bank of a canal there is a 9 m high banner. It appears at a 45° angle from the place opposite on the other bank of the canal, as shown in the figure. Find the width of the canal.

Answer: In the right triangle: Tan 45° = 9/(width of the canal).

Tan 45° = 1, so width of the canal = 9/1 = 9 m.

Question: Review Exercise 11, Question 3: A kite, with a string, is at a 30° angle of elevation with the ground, as shown in the figure. The kite is 60 m above the ground. Find the length of the string.

Answer: In the right triangle: Sin 30° = 60/(length of the string).

Sin 30° = 1/2, so 1/2 = 60/(length of the string) → length of the string = 120 m.

Review Exercise 11, Question 4 (i): Fill in the blank: Cosec 30° = _____.

Answer: 1/Sin 30° = 1/(1/2) = 2. (The book's key gives Sec 60°, which is the same value.)

Review Exercise 11, Question 4 (ii): Fill in the blank: Cot 60° = _____.

Answer: 1/Tan 60° = 1/√3. (The book's key gives Tan 30°, which is the same value.)

Review Exercise 11, Question 4 (iii): Fill in the blank: Cos 45° = _____.

Answer: 1/√2. (The book's key gives Sin 45°, which is the same value.)

Review Exercise 11, Question 4 (iv): Fill in the blank: Sec 30° = _____.

Answer: 1/Cos 30° = 1/(√3/2) = 2/√3. (The book's key gives Cosec 60°, which is the same value.)

Review Exercise 11, Question 4 (v): Fill in the blank: Tan m∠C is the inverse of _____.

Answer: Cot m∠C, because Tan = 1/Cot.

Review Exercise 11, Question 4 (vi): Fill in the blank: Sin m∠A = (side opposite ∠A) / _____.

Answer: ∠A)/(hypotenuse). [note: The book's answer key lists (90° − m∠A) for this item, which does not fit the printed fraction; the correct denominator is the hypotenuse.]

Review Exercise 11, Question 4 (vii): Fill in the blank: Cos (90° − m∠A) = Sin _____.

Answer: m∠A (complementary-angles identity).

Review Exercise 11, Question 4 (viii): Fill in the blank: Tan 30° = Cot _____.

Answer: (90° − 30°) = Cot 60°.

Review Exercise 11, Question 4 (ix): Fill in the blank: Cosec 10° = Sec _____.

Answer: (90° − 10°) = Sec 80°.

Review Exercise 11, Question 4 (x): Fill in the blank: Sec (90° − θ) = Cosec _____.

Answer: θ (complementary-angles identity).

Review Exercise 11, Question 5 (i): Choose the correct answer: Cos 60° = _____.

(a) √3/2

✓ (b) 1/2

(c) 1/√2

Review Exercise 11, Question 5 (ii): Choose the correct answer: Cos 30° = _____. (a) 1/√3 (b) 1 (c) √3

Answer: √3/2, not √3; the question's options/key appear misprinted.]

Review Exercise 11, Question 5 (iii): Choose the correct answer: Tan 45° = _____.

(a) 2

(b) √2

✓ (c) 1

Review Exercise 11, Question 5 (iv): Choose the correct answer: Cos² 30° = _____.

✓ (a) 3/4

(b) √3/2

(c) 1/√2

Review Exercise 11, Question 5 (v): Choose the correct answer: Cosec 60° = _____.

(a) √3/2

✓ (b) 2/√3

(c) 1/2

Review Exercise 11, Question 5 (vi): Choose the correct answer: Sin² 45° = _____.

(a) 1/√2

(b) 2

✓ (c) 1/2

Review Exercise 11, Question 5 (vii): Choose the correct answer: Cosec² 60° = _____.

(a) 2

✓ (b) 4/3

(c) 2/√3

Incorrect: Review Exercise 11, Question 6 (i): State whether this statement is true or false: Cos θ = 1/Sin θ.

Cos θ = 1/Sec θ, not 1/Sin θ. (1/Sin θ = Cosec θ.)

Incorrect: Review Exercise 11, Question 6 (ii): State whether this statement is true or false: Cosec θ = 1/Sec θ.

Cosec θ = 1/Sin θ, not 1/Sec θ. (1/Sec θ = Cos θ.)

Incorrect: Review Exercise 11, Question 6 (iii): State whether this statement is true or false: Sec β = 1/Cosec β.

Sec β = 1/Cos β, not 1/Cosec β. (1/Cosec β = Sin β.)

Correct: Review Exercise 11, Question 6 (iv): State whether this statement is true or false: Tan θ · Cot θ = 1.

Cot θ = 1/Tan θ, so Tan θ · Cot θ = 1.

Correct: Review Exercise 11, Question 6 (v): State whether this statement is true or false: Tan x · Cot x = 1.

Cot x = 1/Tan x, so Tan x · Cot x = 1.

Incorrect: Review Exercise 11, Question 6 (vi): State whether this statement is true or false: (side adjacent to ∠A)/(hypotenuse) = Sin m∠A.

(Side adjacent to ∠A)/(hypotenuse) = Cos m∠A, not Sin m∠A. (Sin m∠A = (side opposite ∠A)/(hypotenuse).)

Correct: Review Exercise 11, Question 6 (vii): State whether this statement is true or false: Cot x = Cos x / Sin x.

Cot x = 1/Tan x = 1/(Sin x/Cos x) = Cos x/Sin x.

Incorrect: Review Exercise 11, Question 6 (viii): State whether this statement is true or false: Sin 20° = Cosec 70°.

Sin 20° = Cos 70° (complementary-angles identity), not Cosec 70°. (Cosec 70° = 1/Sin 70°.)

Correct: Review Exercise 11, Question 6 (ix): State whether this statement is true or false: The inverse of Sin x = Cosec x.
Cosec x = 1/Sin x, so Cosec x is the inverse (reciprocal) of Sin x.
[note: The book's answer key lists this item as false, but by the book's own usage (Review Exercise 11 Question 4 (v): Tan is the inverse of Cot), "inverse" means reciprocal, which makes this statement true.]

Incorrect: Review Exercise 11, Question 6 (x): State whether this statement is true or false: Cos (90° − 30°) = Sec 60°.
Cos (90° − 30°) = Cos 60° = 1/2, whereas Sec 60° = 2.

12. Information Handling

Question: Review Exercise, Question 1: From the table below, find the class frequency, class limits, true class limits, upper class limits, and class width. (Classes: 24–28, 29–33, 34–38, 39–43, 44–48; frequencies: 03, 16, 12, 23, 16)

Answer: Table: Class 24–28: frequency 03, class limits 24–28, true class limits 23.5–28.5, upper class limit 28.5.

Class 29–33: frequency 16, class limits 29–33, true class limits 28.5–33.5, upper class limit 33.5.

Class 34–38: frequency 12, class limits 34–38, true class limits 33.5–38.5, upper class limit 38.5.

Class 39–43: frequency 23, class limits 39–43, true class limits 38.5–43.5, upper class limit 43.5.

Class 44–48: frequency 16, class limits 44–48, true class limits 43.5–48.5, upper class limit 48.5.

Class width = 5 (for every class).

[note: The book prints the third class as "38–34", which is wrong;

the correct class is 34–38.]

Question: Review Exercise, Question 2 (i): 30 students of a class sat in an exam. Their obtained marks out of 10 are given below. Make a frequency chart from the given data. (Marks: 4, 3, 3, 0, 2, 2, 4, 3, 3, 5, 2, 2, 4, 3, 5, 1, 0, 3, 5, 4, 3, 4, 4, 0, 3, 2, 3, 6, 0, 1)

Answer: Frequency chart: Obtained marks 0: number of students 4.

1: 2.

2: 5.

3: 9.

4: 6.

5: 3.

6: 1.

Total = 30 students.

[note: The book's answer key lists frequency 5 for 4 marks and 4 for 5 marks, which does not match the printed table;

from the printed table, 4 marks occur 6 times and 5 marks occur 3 times.]

Question: Review Exercise, Question 2 (ii): Find the range of the given data.

Answer: Range = largest value − smallest value = 6 − 0 = 6.

Question: Review Exercise, Question 2 (iii): How many marks did the most students obtain?

Answer: 3 marks — obtained by 9 students (the highest frequency).

Question: Review Exercise, Question 2 (iv): How many marks did the fewest students obtain?

Answer: 6 marks — obtained by only 1 student (the lowest frequency).

Question: Review Exercise, Question 3 (i): The teacher members of a school were asked how much time they spend daily with their children. The following answers were obtained. Make a frequency chart from the given data. (Time in hours: 4, 3, 1, 6, 2, 2, 3, 1, 4, 5, 3, 4, 1, 2, 5, 3, 2, 2, 3, 2, 2, 3, 1, 1, 4, 2, 3)

Answer: Frequency chart: Time 1 hour: number of members 5.

2 hours: 8.

3 hours: 7.

4 hours: 4.

5 hours: 2.

6 hours: 1.

Total = 27 members.

Question: Review Exercise, Question 3 (ii): How many members in all took

part in this survey?

Answer: Total members = 5 + 8 + 7 + 4 + 2 + 1 = 27 members.

Question: Review Exercise, Question 3 (iii): What is the maximum time spent with the children?

Answer: The maximum time = 6 hours.

Question: Review Exercise, Question 3 (iv): How many members spent the maximum time with the children?

Answer: Only 1 member spent 6 hours (the frequency of the maximum time is 1).

[note: The book's answer key also carries an extra line "8 people spent 2 hours";

that is the frequency of the 2-hour value (8), not the answer to (iv).]

Question: Review Exercise, Question 4 (i): 150 people of an institution took a 10-mark test. The result is given in the table. How many people got the highest marks? (Marks 1–10; frequencies: 6, 12, 15, 21, 35, 24, 20, 10, 6, 1)

Answer: The highest marks = 10 marks, obtained by only 1 person.

Question: Review Exercise, Question 4 (ii): How many people got the lowest marks?

Answer: The lowest marks = 1 mark, obtained by 6 people.

Question: Review Exercise, Question 4 (iii): How many marks did the most

people obtain?

Answer: 5 marks — obtained by 35 people (the highest frequency).

Question: Exercise 12.1, Question 1: The weights (in kg) of 30 boy scouts selected from different schools are as follows: 40, 39, 36, 32, 38, 40, 42, 35, 41, 29, 40, 42, 43, 44, 42, 47, 45, 43, 45, 42, 36, 34, 32, 31, 33, 35, 37, 39, 40, 46. Make a frequency distribution with 5 classes and draw a histogram.

Answer: Range = 47 − 29 = 18.

Class width = 18 ÷ 5 = 3.6 ≈ 4.

Frequency table: Class 29–32: frequency 4 (true limits 28.5–32.5).

Class 33–36: frequency 6 (true limits 32.5–36.5).

Class 37–40: frequency 8 (true limits 36.5–40.5).

Class 41–44: frequency 8 (true limits 40.5–44.5).

Class 45–48: frequency 4 (true limits 44.5–48.5).

Histogram: true class limits on the X-axis, frequency on the Y-axis;

the bar heights are 4, 6, 8, 8, 4 in order.

Question: Exercise 12.1, Question 2: The units of electricity used in 25 houses of a colony are given in order below: 700, 720, 750, 740, 750, 730,

690, 695, 690, 695, 700, 710, 705, 690, 685, 695, 690, 705, 710, 700, 680, 677, 730, 725, 720. Make a frequency distribution with 5 classes and draw a histogram.

Answer: Range = 750 − 677 = 73.

Class width = 73 ÷ 5 = 14.6 ≈ 15.

Frequency table: Class 677–691: frequency 7 (true limits 676.5–691.5).

Class 692–706: frequency 8 (true limits 691.5–706.5).

Class 707–721: frequency 4 (true limits 706.5–721.5).

Class 722–736: frequency 3 (true limits 721.5–736.5).

Class 737–751: frequency 3 (true limits 736.5–751.5).

Histogram: true class limits on the X-axis, frequency on the Y-axis;

the bar heights are 7, 8, 4, 3, 3 in order.

Question: Exercise 12.1, Question 3: 28 students of a class sat in the monthly math test. They got marks as follows: 60, 58, 62, 61, 59, 65, 67, 67, 64, 60, 63, 70, 67, 69, 60, 70, 80, 79, 78, 71, 82, 50, 45, 55, 56, 52, 54, 59. Make a frequency distribution with 4 classes and draw a histogram.

Answer: Range = 82 − 45 = 37.

Class width = 37 ÷ 4 = 9.25 ≈ 10.

Frequency table: Class 45–54: frequency 4 (true limits 44.5–54.5).

Class 55–64: frequency 12 (true limits 54.5–64.5).

Class 65–74: frequency 8 (true limits 64.5–74.5).

Class 75–84: frequency 4 (true limits 74.5–84.5).

Histogram: true class limits on the X-axis, frequency on the Y-axis;

the bar heights are 4, 12, 8, 4 in order.

Question: Exercise 12.1, Question 4: The record of rainfall (in cm) during the last 25 years is as follows: 15, 25, 16, 18, 9, 5, 7, 14, 21, 23, 4, 11, 22, 3, 5, 5, 12, 25, 13, 17, 5, 26, 7, 9, 24. Make a frequency distribution of rainfall with 5 classes and draw its histogram.

Answer: Range = 26 − 3 = 23.

Class width = 23 ÷ 5 = 4.6 ≈ 5.

Frequency table: Class 3–7: frequency 8 (true limits 2.5–7.5).

Class 8–12: frequency 4 (true limits 7.5–12.5).

Class 13–17: frequency 5 (true limits 12.5–17.5).

Class 18–22: frequency 3 (true limits 17.5–22.5).

Class 23–27: frequency 5 (true limits 22.5–27.5).

Histogram: true class limits on the X-axis, frequency on the Y-axis;

the bar heights are 8, 4, 5, 3, 5 in order.

Question: Exercise 12.1, Question 5: The frequency distribution of marks obtained on the computer by the students of a class is: classes 26–30, 31–35, 36–40, 41–45, 46–50, 51–55; frequencies 05, 17, 13, 24, 15, 14. Draw a histogram for the given frequency distribution.

Answer: True class limits: 25.5–30.5, 30.5–35.5, 35.5–40.5, 40.5–45.5, 45.5–50.5, 50.5–55.5.

Histogram: true class limits on the X-axis, frequency on the Y-axis;

the bar heights are 5, 17, 13, 24, 15, 14 in order.

Question: Exercise 12.1, Question 6: Draw a histogram for the frequency distribution of the daily earnings (in rupees) of labourers. (Classes: 501–600, 601–700, 701–800, 801–900, 901–1000, 1001–1100; frequencies: 10, 15, 20, 25, 30, 35)

Answer: True class limits: 500.5–600.5, 600.5–700.5, 700.5–800.5, 800.5–900.5, 900.5–1000.5, 1000.5–1100.5.

Histogram: true class limits on the X-axis, frequency on the Y-axis;

the bar heights are 10, 15, 20, 25, 30, 35 in order.

Question: Exercise 12.2, Question 1 (i): Find the arithmetic mean of the following data: 3, 8, 5, 4, 6, 0, 7, 1, 2.

Answer: Sum = 3 + 8 + 5 + 4 + 6 + 0 + 7 + 1 + 2 = 36.

n = 9.

Arithmetic mean = 36 ÷ 9 = 4.

Question: Exercise 12.2, Question 1 (ii): Find the arithmetic mean of the following data: 11, 13, 15, 7, 9, 1, 3, 5.

Answer: Sum = 11 + 13 + 15 + 7 + 9 + 1 + 3 + 5 = 64.

n = 8.

Arithmetic mean = 64 ÷ 8 = 8.

Question: Exercise 12.2, Question 1 (iii): Find the arithmetic mean of the following data: 11, 16, 13, 17, 10, 15, 18, 14, 12.

Answer: Sum = 11 + 16 + 13 + 17 + 10 + 15 + 18 + 14 + 12 = 126.

n = 9.

Arithmetic mean = 126 ÷ 9 = 14.

Question: Exercise 12.2, Question 1 (iv): Find the arithmetic mean of the following data: 4, 16, 32, 8, 40, 12, 20, 28, 24, 36.

Answer: Sum = 4 + 16 + 32 + 8 + 40 + 12 + 20 + 28 + 24 + 36 = 220.

n = 10.

Arithmetic mean = 220 ÷ 10 = 22.

Question: Exercise 12.2, Question 1 (v): Find the arithmetic mean of the following data: 1.2, 2.4, 3.6, 4.8, 5.1, 6.3, 7.5, 8.7.

Answer: Sum = 1.2 + 2.4 + 3.6 + 4.8 + 5.1 + 6.3 + 7.5 + 8.7 = 39.6.

n = 8.

Arithmetic mean = 39.6 ÷ 8 = 4.95.

Question: Exercise 12.2, Question 1.B (i): Find the value of the unknown term: $\bar{X} = 12$, $\Sigma x = 132$, $n = ?$

Answer: $n = \Sigma x \div \bar{X} = 132 \div 12 = 11$.

Question: Exercise 12.2, Question 1.B (ii): Find the value of the unknown term: $n = 15$, $\Sigma x = 75$, $\bar{X} = ?$

Answer: $\bar{X} = \Sigma x \div n = 75 \div 15 = 5$.

Question: Exercise 12.2, Question 1.B (iii): Find the value of the unknown term: $\Sigma x = 96$, $\bar{X} = 8$, $n = ?$

Answer: $n = \Sigma x \div \bar{X} = 96 \div 8 = 12$.

Question: Exercise 12.2, Question 1.B (iv): Find the value of the unknown term: $\bar{X} = 13$, $n = 11$, $\Sigma x = ?$

Answer: $\Sigma x = n \times \bar{X} = 11 \times 13 = 143$.

[note: The book's answer key lists 141, which is wrong;

11 × 13 = 143.]

Question: Exercise 12.2, Question 2 (i): Find the weighted mean from the following table: values (x): 10, 20, 30, 40, 50, 60, 70; weights (w): 1, 2, 1, 1, 2, 2, 1.

Answer: Σxw = (10×1) + (20×2) + (30×1) + (40×1) + (50×2) + (60×2) + (70×1) = 10 + 40 + 30 + 40 + 100 + 120 + 70 = 410.

Σw = 1 + 2 + 1 + 1 + 2 + 2 + 1 = 10.

Weighted mean = 410 ÷ 10 = 41.

[note: The book's answer key writes 410/7, which divides by the count of values (7) instead of the sum of weights (10);

the correct weighted mean is 410/10 = 41.]

Question: Exercise 12.2, Question 2 (ii): Find the weighted mean from the following table: values (x): 0, 2, 4, 6, 8, 10, 12; weights (w): 1, 2, 3, 4, 5, 3, 2.

Answer: Σxw = (0×1) + (2×2) + (4×3) + (6×4) + (8×5) + (10×3) + (12×2) = 0 + 4 + 12 + 24 + 40 + 30 + 24 = 134.

Σw = 1 + 2 + 3 + 4 + 5 + 3 + 2 = 20.

Weighted mean = 134 ÷ 20 = 6.7.

[note: The book's answer key writes 134/7;

the correct divisor is the sum of weights (20): 134/20 = 6.7.]

Question: Exercise 12.2, Question 2 (iii): Find the weighted mean from the following table: values (x): 1, 3, 5, 7, 9, 11; weights (w): 3, 2, 1, 1, 2, 1.

Answer: Σxw = (1×3) + (3×2) + (5×1) + (7×1) + (9×2) + (11×1) = 3 + 6 + 5 + 7 + 18 + 11 = 50.

Σw = 3 + 2 + 1 + 1 + 2 + 1 = 10.

Weighted mean = 50 ÷ 10 = 5.

[note: The book's answer key writes 50/6;

the correct divisor is the sum of weights (10): 50/10 = 5.]

Question: Exercise 12.2, Question 2 (iv): Find the weighted mean from the following table: values (x): 5, 15, 25, 15, 35, 45; weights (w): 2, 1, 2, 1, 2, 2.

Answer: Σxw = (5×2) + (15×1) + (25×2) + (15×1) + (35×2) + (45×2) = 10 + 15 + 50 + 15 + 70 + 90 = 250.

Σw = 2 + 1 + 2 + 1 + 2 + 2 = 10.

Weighted mean = 250 ÷ 10 = 25.

[note: The book's answer key writes 170/6, which does not match the

printed table;

from the printed table Σxw = 250 and Σw = 10, so the weighted mean is 25.]

Question: Exercise 12.2, Question 3 (i): Find the median of the following data: 17, 19, 11, 3, 5, 7, 9, 13, 15.

Answer: In order: 3, 5, 7, 9, 11, 13, 15, 17, 19.

n = 9 (odd).

Median = 5th value = 11.

Question: Exercise 12.2, Question 3 (ii): Find the median of the following data: 45, 5, 15, 20, 35, 40, 10, 25, 30.

Answer: In order: 5, 10, 15, 20, 25, 30, 35, 40, 45.

n = 9 (odd).

Median = 5th value = 25.

Question: Exercise 12.2, Question 3 (iii): Find the median of the following data: 22, 2, 20, 4, 18, 6, 16, 8, 14, 10, 12.

Answer: In order: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22.

n = 11 (odd).

Median = 6th value = 12.

Question: Exercise 12.2, Question 3 (iv): Find the median of the following data: 35, 21, 49, 77, 45, 27, 33, 55, 63, 81, 99.

Answer: In order: 21, 27, 33, 35, 45, 49, 55, 63, 77, 81, 99.

n = 11 (odd).

Median = 6th value = 49.

Question: Exercise 12.2, Question 3 (v): Find the median of the following data: 39, 78, 91, 75, 104, 105, 19, 29, 99, 79, 89.

Answer: In order: 19, 29, 39, 75, 78, 79, 89, 91, 99, 104, 105.

n = 11 (odd).

Median = 6th value = 79.

Question: Exercise 12.2, Question 3 (vi): Find the median of the following data: 2.5, 3.5, 1.5, 4.5, 5.5, 5.0, 9.9, 9.5, 10.5, 11.5, 6.5, 7.5, 5.5.

Answer: In order: 1.5, 2.5, 3.5, 4.5, 5.0, 5.5, 5.5, 6.5, 7.5, 9.5, 9.9, 10.5, 11.5.

n = 13 (odd).

Median = 7th value = 5.5.

Question: Exercise 12.2, Question 4.A (i): Find the mode of the following data: 14, 21, 27, 14, 28, 29.

Answer: 14 occurs twice, all others once.

Mode = 14.

Question: Exercise 12.2, Question 4.A (ii): Find the mode of the following data: 24, 31, 37, 24, 38, 39.

Answer: 24 occurs twice, all others once.

Mode = 24.

Question: Exercise 12.2, Question 4.A (iii): Find the mode of the following data: 23, 29, 41, 29, 23, 77, 55, 29.

Answer: 29 occurs three times (the most), 23 twice.

Mode = 29.

Question: Exercise 12.2, Question 4.A (iv): Find the mode of the following data: 20, 26, 38, 20, 26, 74, 52, 26.

Answer: 26 occurs three times (the most), 20 twice.

Mode = 26.

Question: Exercise 12.2, Question 4.A (v): Find the mode of the following data: 71, 51, 93, 81, 71, 57, 51, 71, 81, 51, 71.

Answer: 71 occurs four times (the most), 51 three times, 81 twice.

Mode = 71.

Question: Exercise 12.2, Question 4.A (vi): Find the mode of the following data: 93, 75, 85, 89, 77, 55, 33.

Answer: No value occurs twice.

There is no mode.

Question: Exercise 12.2, Question 4.A (vii): Find the mode of the following data: 930, 750, 850, 890, 770, 550, 330.

Answer: No value occurs twice.

There is no mode.

Question: Exercise 12.2, Question 4.B (i): Fill in the blank: 3, 3, 4, 4, 5, 8, 7, 8, ___, 9 if the mode is 8.

Answer: 8.

8 must occur three times so that it is the unique mode (3 and 4 already occur twice each).

Question: Exercise 12.2, Question 4.B (ii): Fill in the blank: 4, 5, 8, 10, 20, ___, 15 if the median is 8.

Answer: 8.

[note: For the median to be 8, any value ≤ 8 can fill the blank;

8 is the natural fill.]

Question: Exercise 12.2, Question 4.B (iii): Fill in the blank: 6, 9, 7, 10, ___, 4 if the arithmetic mean is 8.

Answer: 12.

(6 + 9 + 7 + 10 + 4 + x) ÷ 6 = 8 → 36 + x = 48 → x = 12.

Question: Exercise 12.2, Question 5: The sum of some numbers is 30 and the arithmetic mean is 5; tell how many numbers there are.

Answer: n = Σx ÷ $\bar{X}$ = 30 ÷ 5 = 6 numbers.

Question: Exercise 12.2, Question 6: The arithmetic mean of some numbers is 45. Two of the numbers are 64 and 36. The last 2 numbers are equal to x. Find the value of x.

Answer: There are 4 numbers in all: 64, 36, x, x.

(64 + 36 + x + x) ÷ 4 = 45 → (100 + 2x) ÷ 4 = 45 → 100 + 2x = 180 → 2x = 80 → x = 40.

Question: Exercise 12.2, Question 7 (i): If the median is 5, fill in the blank: 1, 2, 5, ___, 11.

Answer: 5 (or any value ≥ 5).

n = 5, median = 3rd value.

In order 1, 2, 5, x, 11, the 3rd value is 5 when x ≥ 5.

Question: Exercise 12.2, Question 7 (ii): If the median is 5, fill in the blank: 11, 8, 5, ___, 2.

Answer: 5 (or any value ≤ 5).

n = 5, median = 3rd value.

In order 2, 5, 8, 11, x, the 3rd value is 5 when $x \leq 5$.

Question: Exercise 12.2, Question 7 (iii): If the median is 5, fill in the blank: 20, 11, ___, 5, 2, 1.

Answer: 5.

n = 6, median = mean of the 3rd and 4th values.

In order 1, 2, 5, x, 11, 20, $(5 + x) \div 2 = 5 \rightarrow x = 5$.

Question: Exercise 12.2, Question 7 (iv): If the median is 5, fill in the blank: 20, 11, ___, 4, 2, 1.

Answer: 6.

n = 6, median = mean of the 3rd and 4th values.

In order 1, 2, 4, x, 11, 20, $(4 + x) \div 2 = 5 \rightarrow x = 6$.

Question: Exercise 12.2, Question 8 (i): If the mode is 5, fill in the blank: 11, ___, 5, 2, 1.

Answer: 5.

5 must occur twice so that it is the unique mode.

Question: Exercise 12.2, Question 8 (ii): If the mode is 5, fill in the blank: 11, 8, 5, ___, 2.

Answer: 5.

5 must occur twice so that it is the unique mode.

Question: Exercise 12.2, Question 8 (iii): If the mode is 5, fill in the blank: 20, 11, ___, 5, 2, 1.

Answer: 5.

5 must occur twice so that it is the unique mode.

Question: Exercise 12.2, Question 8 (iv): If the mode is 5, fill in the blank: 11, ___, 5, 5, 2, 1.

Answer: Any value except 11, 2, or 1 (such as 5).

5 already occurs twice;

putting 11, 2, or 1 in the blank would give another value frequency 2 as well, so the mode would not be unique.

Question: Exercise 12.2, Question 8 (v): If the mode is 5, fill in the blank: 11, ___, 5, 5, 2, 2, 1.

Answer: 5.

5 must occur three times so that it is the unique mode (2 already occurs twice).

Question: Exercise 12.3, Question 1: A customer bought vegetables: types (potatoes, onions, tomatoes, carrots, pinibon), weights (in kg): 12, 8, 5, 3, 4; price per kg (in rupees): 25, 30, 60, 40, 80. Find the average price per kg.

Answer: Average with weights (weighted mean): Σxw = (12×25) + (8×30) + (5×60) + (3×40) + (4×80) = 300 + 240 + 300 + 120 + 320 = 1280.

Σw = 12 + 8 + 5 + 3 + 4 = 32.

Average price per kg = 1280 ÷ 32 = 40 rupees per kg.

Question: Exercise 12.3, Question 2: The heights (in cm) of 20 students of a class are given below: 130, 135, 140, 145, 142, 141, 140, 135, 136, 137, 132, 134, 136, 138, 140, 142, 144, 143, 139, 138. Find the arithmetic mean, median, and mode of the students' heights. Is the inequality Mode ≥ Median ≥ Arithmetic Mean correct?

Answer: Sum = 2767, n = 20.

Arithmetic mean = 2767 ÷ 20 = 138.35.

In order: 130, 132, 134, 135, 135, 136, 136, 137, 138, 138, 139, 140, 140, 140, 141, 142, 142, 143, 144, 145.

Median = (10th + 11th) ÷ 2 = (138 + 139) ÷ 2 = 138.5.

Mode = 140 (three times, the most).

Check: 140 ≥ 138.5 ≥ 138.35 — the inequality is correct for this data.

[note: The book's answer key lists median 138 and answer "No";

138 is the lower middle value, not the mean of the two middle values

(138.5).

With the correct median 138.5, the inequality is correct.]

Question: Exercise 12.3, Question 3: 20 crates of Sindhi mangoes were sent from Hyderabad to Sukkur. The number of rotten mangoes in the crates is given in order below: 0, 1, 2, 3, 0, 1, 2, 4, 3, 0, 2, 1, 4, 3, 0, 2, 3, 1, 2, 3. Find the arithmetic mean, mode, and median of the number of rotten mangoes. Check whether the inequality Mode ≥ Median ≥ Arithmetic Mean is correct.

Answer: Sum = 37, n = 20.

Arithmetic mean = 37 ÷ 20 = 1.85.

In order: 0, 0, 0, 0, 1, 1, 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 4, 4.

Median = (10th + 11th) ÷ 2 = (2 + 2) ÷ 2 = 2.

Mode = 2 and 3 (each five times — two modes).

Check: 2 ≥ 2 ≥ 1.85 and 3 ≥ 2 ≥ 1.85 — the inequality is correct for this data.

[note: The book's answer key lists arithmetic mean 1.75 and answer "No";

the printed data sum to 37 (1.85), not 35 (1.75).

With the correct numbers, the inequality is correct.]

Question: Exercise 12.3, Question 4: 45 students took the Science and Math tests. The frequency distribution of their obtained marks is given: marks 10, 20, 30, 40, 50, 60, 70, 80, 90; Math students 5, 4, 7, 8, 6, 6, 4, 5, 3; Science students 4, 4, 8, 8, 5, 3, 2, 7, 4. Find the average marks in each subject. On this basis, tell in which subject the students performed better.

Answer: Math: $\Sigma xw = (10\times5) + (20\times4) + (30\times7) + (40\times8) + (50\times6) + (60\times6) + (70\times4) + (80\times5) + (90\times3) = 50 + 80 + 210 + 320 + 300 + 360 + 280 + 400 + 270 = 2270$.

Number of students = 48.

Average = $2270 \div 48 \approx 47.29$.

Science: $\Sigma xw = (10\times4) + (20\times4) + (30\times8) + (40\times8) + (50\times5) + (60\times3) + (70\times2) + (80\times7) + (90\times4) = 40 + 80 + 240 + 320 + 250 + 180 + 140 + 560 + 360 = 2170$.

Number of students = 45.

Average = $2170 \div 45 \approx 48.22$.

The Science average (48.22) is higher than the Math average (47.29), so the students performed better in Science.

[note: The printed table's Math frequencies sum to 48, not 45 as stated in the question;

the book's answer-key figures (2090/45 and 2170/45) do not match the printed table.]

Question: Exercise 12.3, Question 5: The table of the number of پائٹ of 100 families is given below: پائٹ per family: 2, 3, 4, 5, 6, 7, 8, 9, 10; number of families: 24, 32, 20, 10, 5, 4, 2, 2, 1. Find the average of the number of پائٹ.

Answer: Σxw = (2×24) + (3×32) + (4×20) + (5×10) + (6×5) + (7×4) + (8×2) + (9×2) + (10×1) = 48 + 96 + 80 + 50 + 30 + 28 + 16 + 18 + 10 = 376.

Number of families = 100.

Average پائٹ per family = 376 ÷ 100 = 3.76.

Review Exercise 12, Question 1 (i): The arithmetic mean of 1, 0, 5, 5, 6, 7 is:

(a) 3

✓ (b) 4

(c) 5

(d) 6.

Review Exercise 12, Question 1 (ii): The mode of the data 3, 5, 3, 4, 3, 2 is:

(a) 2

(b) 5

✓ (c) 3

(d) 4.

Review Exercise 12, Question 1 (iii): The median of the data 2, 3, 4, 5, 10 is:

✓ (a) 4

(b) 5

(c) 3

(d) 10.

Review Exercise 12, Question 1 (iv): The median of the data 0, 2, 4, 2, 6, 8 is:

(a) 2

(b) 4

(c) 6

✓ (d) 3.

Review Exercise 12, Question 1 (v): The number of modes in the data 2, 4, 8, 6, 6, 4, 2 is:

(a) 1

(b) 2

✓ (c) 3

(d) 0.

Review Exercise 12, Question 1 (vi): The formula for the arithmetic mean is:

✓ (a) $\bar{X} = \Sigma x \div n$

(b) $\bar{X} = \Sigma x$

(c) $\bar{X} = \Sigma x \cdot n$

(d) $n = \Sigma x$.

Review Exercise 12, Question 1 (vii): The value that occurs more often than other values is called:

(a) arithmetic mean

(b) median

(c) frequency distribution

✓ (d) mode.

Review Exercise 12, Question 1 (viii): If n = 5 and $\bar{X}$ = 6, then Σx is equal to:

(a) 5

(b) 6

✓ (c) 30

(d) 5/6.

Review Exercise 12, Question 1 (ix): The method of presenting raw data in a short and organized manner is called:

✓ (a) frequency distribution

(b) arithmetic mean

(c) median

(d) mode.

Question: Review Exercise 12, Question 1 (x): The characteristic that indicates the central tendency of the data: (a) frequency distribution (b) arithmetic mean (c) mode (d) median.

Answer: (b), (c), and (d) are all measures of central tendency;

only (a) frequency distribution is not a measure of central tendency.

[note: As printed, the question has no unique answer;

it most likely intends "which is NOT a measure of central tendency", whose answer is (a).]

Question: Review Exercise 12, Question 2 (i): 110 families live in a plaza. The histogram opposite shows the water (in litres) used by each family. Looking at the graph, tell what is the Model number of the drinking water usage shown in litres. (Histogram: classes 0–5, 5–10, 10–15, 15–20, 20–25, 25–30; frequencies 5, 15, 25, 35, 35, 35)

Answer: Three classes — 15–20, 20–25, and 25–30 litres — each have frequency 35 (the highest, and equal), so the mode is threefold: 15–20, 20–25, and 25–30 litres.

[note: The printed histogram's bars sum to 150, not 110 as stated in the question;

the book's answer-key answers (20 families, 5 families, 20–25 litres) do not match the printed graph.]

Question: Review Exercise 12, Question 2 (ii): Find the arithmetic mean per family from the histogram above.

Answer: Class midpoints: 2.5, 7.5, 12.5, 17.5, 22.5, 27.5.

$\Sigma xw = (2.5\times5) + (7.5\times15) + (12.5\times25) + (17.5\times35) + (22.5\times35) + (27.5\times35) = 12.5 + 112.5 + 312.5 + 612.5 + 787.5 + 962.5 = 2800.$

Total families = 150.

Arithmetic mean = 2800 ÷ 150 ≈ 18.67 litres per family.

Question: Review Exercise 12, Question 2 (iii): Find the median (Median) class in the histogram above.

Answer: Total = 150, median position = 75th family.

Cumulative frequencies: 5, 20, 45, 80, 115, 150.

The 75th family falls in the 15–20 litres class.

Median class = 15–20 litres.

Question: Review Exercise 12, Question 3 (i): 30 students of a class sat in an exam. Their obtained marks out of 40 are: 18, 24, 10, 3, 14, 26, 22, 18, 25, 11, 29, 17, 16, 11, 25, 11, 14, 15, 4, 22, 13, 23, 29, 10, 27, 18, 10, 25, 13, 14. Make a frequency distribution with 5 classes from the raw data.

Answer: Range = 29 − 3 = 26.

Class width = 26 ÷ 5 = 5.2 ≈ 6 (rounded up so that 5 classes cover the whole range).

Frequency table: Class 3–8: frequency 2 (true limits 2.5–8.5).

Class 9–14: frequency 11 (true limits 8.5–14.5).

Class 15–20: frequency 6 (true limits 14.5–20.5).

Class 21–26: frequency 8 (true limits 20.5–26.5).

Class 27–32: frequency 3 (true limits 26.5–32.5).

Total = 30.

Question: Review Exercise 12, Question 3 (ii): In the above data, how many marks did the most students obtain?

Answer: 10, 11, 14, 18, and 25 — each obtained by three students (five modes).

Question: Review Exercise 12, Question 3 (iii): Draw a histogram for the above distribution.

Answer: Histogram: true class limits on the X-axis, frequency on the Y-axis;

the bar heights are 2, 11, 6, 8, 3 in order.

Question: Review Exercise 12, Question 4: In a good way, construct a frequency distribution with four classes for the following data and also draw its histogram. (Data: 64, 63, 60, 50, 55, 60, 60, 62, 66, 67, 68, 66, 60, 73, 67, 65, 70, 72, 69, 59, 59, 50, 60, 70, 69, 70, 63)

Answer: Range = 73 − 50 = 23.

Class width = 23 ÷ 4 = 5.75 ≈ 6.

Frequency table: Class 50–55: frequency 3 (true limits 49.5–55.5).

Class 56–61: frequency 7 (true limits 55.5–61.5).

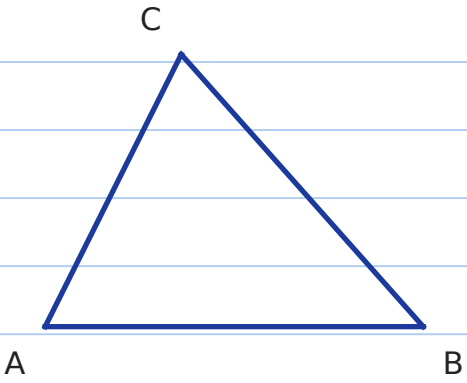
Class 62–67: frequency 9 (true limits 61.5–67.5).

Class 68–73: frequency 8 (true limits 67.5–73.5).

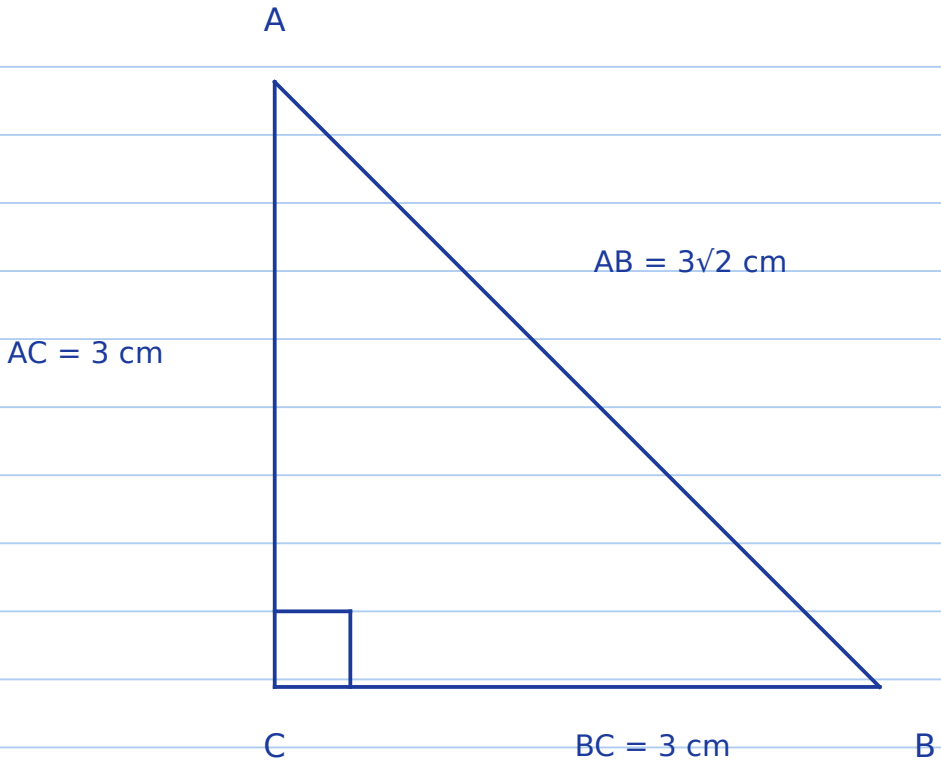
Total = 27.

Histogram: true class limits on the X-axis, frequency on the Y-axis;

the bar heights are 3, 7, 9, 8 in order.



9. Areas and Volumes



11. Trigonometry

